Chapter 8
Types of Nanomaterials and Their Properties

8.1 INTRODUCTION

It is an interesting question as to, starting with a few atoms, how the bulk materials reach their structure and related properties. Do they undergo any structural changes or even their smallest unit cell is similar to that in bulk material? This question has been addressed by many. There are reasons to believe that the small clusters or nanoparticles are not just the fragments of bulk materials. There can be entirely different structures as well as bonds and bond strengths in clusters which can even differ from nanomaterials.

It has been well established now that all the materials, may be metals, semiconductors or insulator clusters or nanomaterials, have size dependent physico-chemical properties. In most of the cases a cluster size is below 1 nm and that of nanoparticles is in the 1 to 100 nm range. Interestingly at such a small size even the shape of the material and interactions between clusters or nanomaterials decide the properties of the material. This opens up a huge possibility of tailor making the materials, which have different properties just due to their size, shape and/or assembly. In this chapter we will discuss to some extent clusters, semiconductor metal nanomaterials and magnetic nanomaterials. Nanomaterials would mean 0-D, 1-D or 2-D materials with the relevant dimensions smaller than ~100 nm to qualify them as nanomaterials. The general analysis methods for all the nanomaterials are as mentioned in Chapter 7. In this chapter some additional relevant analysis is discussed for semiconductor and metal nanoparticles. The chapter closes with some additional properties like mechanical, structural, electrical and thermal properties of the nanomaterials in general.



8.2 CLUSTERS

Clusters are aggregates of small number of atoms and can be considered as intermediates of atoms and nanomaterials. Depending upon the number of atoms they can be roughly divided into following:

1. Microclusters – number of atoms between 2 and 10-13 atoms
2. Small clusters – number of atoms between 13 to about 100 atoms
3. Large clusters – number of atoms between about 100 and 1000 atoms.

With number of atoms larger than 10^3 and below 10^6 atoms it is a nano-particle. Although the classification of clusters and nanomaterials cannot be very precise, usually clusters are smaller than ~1 nm.

One may also ask the question, what is the difference between a large atom, small molecule with couple of atoms and a cluster? For example a 92-atom sodium atom cluster and uranium atom with atomic number 92. A sodium atom cluster with one valence electron and 92 free electrons of sodium atom would make it appear like a uranium atom. However, in uranium atom the positive charge is concentrated in the nucleus in a very small volume and 92 electrons have some specific electron configuration. On the other hand, in a sodium atom cluster, each atom's positive charge is spatially separated from the other atoms and each one would have its ionic shell configuration; only 92 electrons would be free to form an electron gas of the cluster. As far as molecules are concerned they have stable configuration of certain types and number of atoms. They have fixed composition, structure, bond lengths, bond angles and, hence, fixed properties. The clusters on the other hand are the aggregates of atoms and may or may not be stable. The structure, bonds etc. vary with number of atoms and so do the properties. In clusters large fraction of atoms is on the surface. For example in a cluster of 55 atoms like that of argon or sodium 32 atoms are on the surface and only 23 are inside. Often we are only able to understand their presence in some experiments but may not be able to collect them. However, their theoretical as well as experimental investigations can lead to the understanding of the basic processes that take place while the clusters are formed and also lead to applications in catalysis, epitaxial materials and so on.

Types of Clusters

Besides the classification of clusters depending upon the number of atoms they contain, it is possible to realize various types of clusters as follows:

1. Homogeneous clusters – containing only single type of elements, e.g. Ar, Na, K, Si, Au, Pb etc.
2. Heterogeneous clusters – 3Na + 3K, 2Na + 3K etc. $(NaCl)_n$ or Na_nCl_{n+y}, $(Na_{n+1}Cl_n)^+$, $(Na_nCl_{n+1})^-$ and many other kinds and compositions.
3. Oppositely charged clusters may be attracted and stay together.
4. Covalently bonded clusters like carbon, silicon etc. Fullerenes or C60 atom clusters are examples of such clusters and will be discussed in Chapter 9.



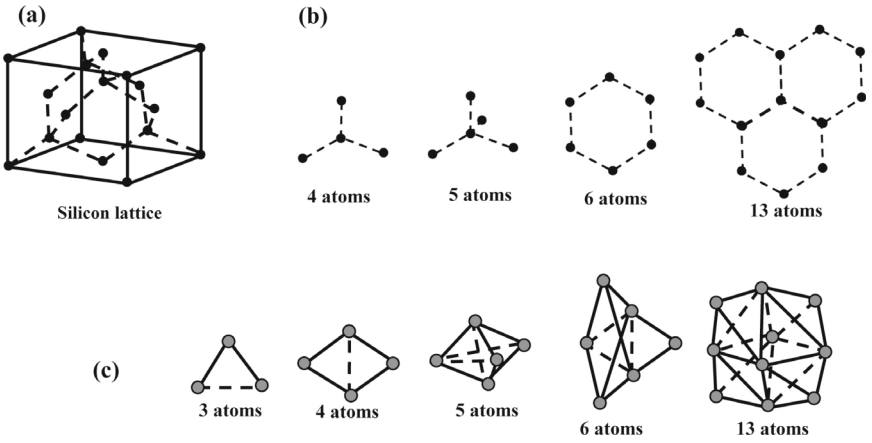


Fig. 8.1: (a) Silicon atoms forming a unit cell, (b) fragments of silicon unit cell, and (c) stable clusters of silicon.

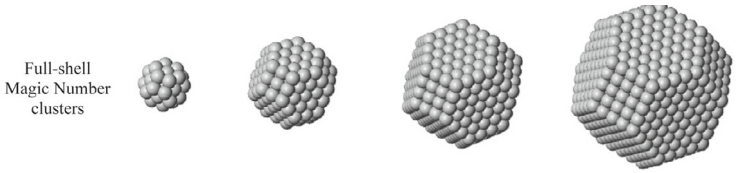
In a silicon cluster, the stable configurations are not the fragments of bulk lattice and instead of having 4, 5, 6 atoms in clusters, stable clusters having 3, 4 atoms with different structure rather than bulk fragmentation are energetically favoured (Fig. 8.1).

Box 8.1
Clusters and Platonic Solids

A cluster may be considered as being formed shell by shell around a central atom. As the number of shells increase, total number of atoms goes on increasing and percentage of surface atoms compared to interior goes on decreasing. The magic number of atoms in a cluster can be obtained by first considering some basic types which were formulated by ancient mathematicians like Pythagoras and Plato around 450 B.C. The basic shapes considered were cube, octahedron, icosahedrons, tetrahedron and dodecahedron related to earth, air, water, fire and heavenly constellation respectively. One of the common metal cluster is 13-atom icosahedrons. It has one central atom and 12 surrounding atoms, 20 faces, 12 vertices and 30 edges. Number of atoms in the nth shell of an icoshedron can be obtained from $10\,n^2+2$.

Figure 8.2 shows few icosahedrons clusters.





Number of shells	1	2	3	4
Number of atoms in cluster	13	55	147	309
Surface atoms (%)	92	76	63	52

Fig. 8.2: Icosahedron clusters with complete shells.

5. Metal atom clusters i.e. a phase in which they would ultimately develop if the cluster and nanoparticle size is exceeded. In fact even before a material reaches a bulk size with some critical number of atoms characteristic of the material it undergoes various transitions assuming local stable sizes. This has been very well exemplified for potassium clusters (see Fig. 8.3). Here the ionization energy is plotted as a function of the number of atoms in a cluster. More the ionization energy more stable is the cluster. One can see that 8, 18, 20 … atoms clusters are locally stable compared to the other neighbouring clusters. After having about 92 atoms, the cluster reaches the ionization potential (or work function value) of a bulk potassium metal. The clusters also can have different structures compared to the bulk material. One can get more insight about the metal clusters by using a jellium model.
6. Clusters of rare gases – close shell atom clusters of Ar, Kr etc. are held together by weak van der Waals forces and formed at low temperatures.
7. Clusters with hydrogen bonds, for example a $(H_2O)_n$ cluster. It is a strongly held molecular cluster, stronger than cluster of rare gases but much weaker than metallic, covalent or ionic clusters.
8. Magnetic clusters – clusters of atoms with resultant magnetic moment. The magnetic moments on clusters like Co, Fe and Ni may show the moment close to magnetic moment/atom.

Usually shell theory applies to atoms, nuclei as well as clusters. In all the three cases fermions are confined in ~10^{-12} cm for nuclei, 10^{-8} cm for atoms and ~10^{-7} to 10^{-6} cm in case of clusters. Using appropriate angular momentum conditions for the spherical harmonic, square well or spherical potential shells are developed which are a direct result of application of quantum mechanics. In all these cases interacting fermions obeying Pauli exclusion principle result ultimately to shells with magic numbers or in case of clusters some stable clusters are referred to as 'magic clusters'.

In order to obtain very small clusters, starting with 2-3 atoms upto few tens of atoms, special, very sophisticated machines are built. Usually an intense beam of



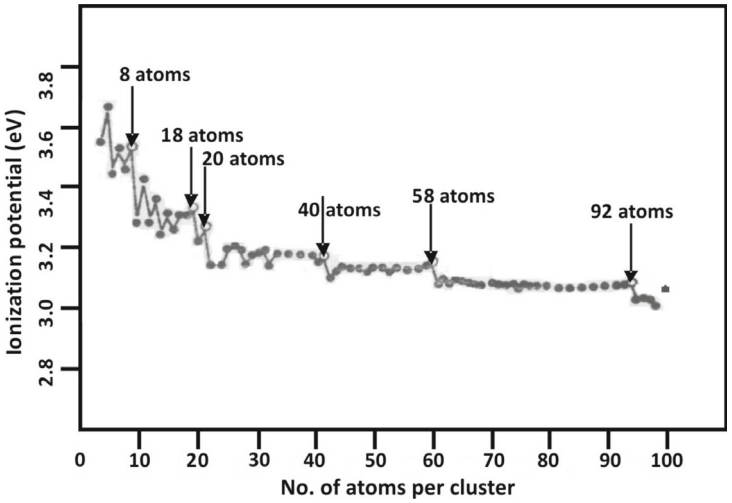


Fig. 8.3: Variation of ionization energies of different potassium clusters having different number of atoms in them.

laser is made incident on a bulk piece of material whose clusters are to be investigated. The evaporation normally contains clusters of different sizes which need to be mass selected using a mass spectrometer. The mass selected cluster would then have abundance of clusters of uniform size (same number of atoms per cluster). Such a cluster is then fragmented by another beam of laser which is then analyzed with another mass spectrometer. This analysis would then lead to the understanding of stability of the clusters. For example, if a four-atom cluster is less stable than a three-atom cluster, the fragmentation should show the abundance of three-atom clusters than four-atom clusters.

In some cases where ionization of the clusters is investigated, an oven is used as a source of clusters, where the source material from the hot oven is expanded through a nozzle, ionized in a chamber using photons or electron beam and then analyzed using a mass spectrometer. Of course in both the types of apparatus discussed here, adequate vacuum system is a must in order to avoid any contaminations.

8.3 SEMICONDUCTOR NANOPARTICLES (QUANTUM DOTS)

Every material has a characteristic size below which size dependent properties are realized. In semiconductors this size is nothing but the size of the exciton. Understanding the concept of exciton and estimating its size for different semiconductor materials is the first step towards understanding the semiconductor nanoparticles.



8.3.1 Excitons

In semiconductor or insulator, valence and conduction bands are separated by some finite energy gap characteristic of the material. When an electron from the valence band gets sufficient energy to overcome the energy gap, may be by thermal excitation or absorption of photon, and go to conduction band, a hole is left behind. The electron–hole pair so formed is a quasi-particle called exciton (see Fig. 8.4). An exciton, whose centre of mass motion is quantized, can move in the crystal.

Different kinds of excitons are identified in a variety of materials. Note that the spins of electron and hole in an exciton pair can be either parallel or antiparallel to each other. If the spins of hole and electron in an exciton are parallel to each other the 'dark exciton' results and 'bright exciton' otherwise. The properties of dark and bright excitons are different. As its name suggests dark exciton is optically inactive but the bright exciton is optically active. The exciton is useful in optical sensing and emission. Dark exciton, one may think, would not be useful. However, dark excitons and bright excitons can be interchanged by spin flip-flop and this has potential application in spin storage and qbits.

Depending upon the intensity of the light used to excite the excitons, it is also possible under intense illumination that a large number of excitons (exciton liquid) are formed which start interacting with each other. Two interacting excitons can attract each other forming a 'biexciton'. One can write

$$\delta E_{\text{exc}} = E_{\text{biexc}} - 2E_{\text{exc}} \tag{8.1}$$

Here E_{exc} is the energy of a single exciton, E_{biexc} is the energy of the two interacting excitons and δE_{exc} is the difference in the energies given by equation 8.1. Obviously the bound pair will be formed if the energy $\delta E_{\text{exc}} < 0$. Although there is a lot of interesting research being carried out to study the lifetime, dynamics and optical properties of biexcitons, we will not go into more depth here as it is beyond the scope of this book.

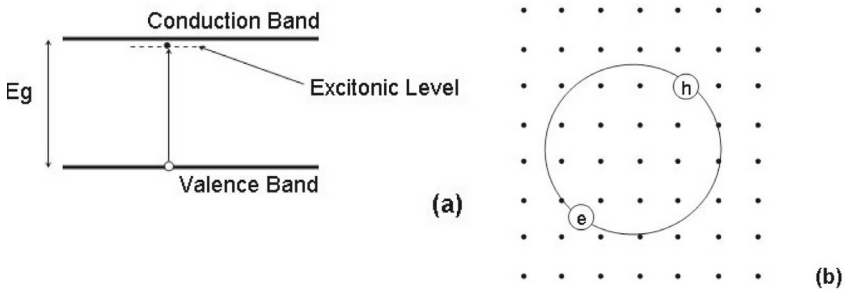


Fig. 8.4: (a) Energy level diagram for a typical semiconductor along with the excitation energy level. (b) Mott-Wannier exciton in a typical lattice.

We shall consider here single (or widely isolated), optically active excitons. When the electron-hole pair is tightly bound with distance between electron and hole comparable to the lattice constant of the material, it is called Frenkel exciton. At the other extreme, one may have an exciton with electron-hole separation much larger compared to the lattice constant. Such a weakly bound electron-hole pair is called as Mott-Wannier exciton. The energy of such an exciton is slightly less than the energy gap (E_g) between valence and conduction band. In fact the energy gap E_g can be also defined as the energy necessary to create a free electron and a free hole. The Hamiltonian for Mott-Wannier exciton is given as

$$H = \frac{P_e^2}{2m_e} + \frac{P_h^2}{2m_h} - \frac{e^2}{\varepsilon|r_e - r_h|} \tag{8.2}$$

where m_e and m_h are effective electron and hole mass respectively, and ε is the dielectric constant of semiconductor material. It should be remembered that the mass of an electron and hole in a solid material and even a nanomaterial is not same as that of a 'free electron' mass. It may be smaller or larger than the free electron mass depending upon the curvature of a band. For more details one should refer to a text book on Solid State Physics. First two terms on the right hand side of Eq. (8.2) are kinetic energies of electron and hole respectively and the third term is the Coulomb energy of electron-hole attraction or potential energy.

The Bohr radius of such an exciton is given as

$$r_B = \frac{\hbar^2 \varepsilon}{e^2}\left[\frac{1}{m_e} + \frac{1}{m_h}\right] \tag{8.3}$$

In Table 8.1 radii for few semiconductor spherical nanoparticles are illustrated.

It may be noticed that these values of r_B are of the order of a few nanometres. We observe in a nanoparticle and other nanostructures a situation in which exciton is confined in the nanostructure. The sizes of excitons are often comparable to the sizes of nanoparticles or nanostructures.

Table 8.1: Bohr radius (r_B) for exciton in some semiconductors

Semiconductor	$E_g(eV)$	m_e	m_h	ε	$r_B(nm)$
GaAs	1.52	0.067	0.082	13.1	18.8
InP	1.42	0.077	0.64	12.4	9.6
InSb	0.24	0.0145	0.39	17.6	66.7



8.3.2 *Effective Mass Approximation*

There are various theories like Effective Mass Approximation (EMA) and Tight Binding (TB) developed to explain the size dependent energy gap variation (being most important property of semiconductor materials on which other properties depend) in nanostructures. Depending upon the material or group of materials one theory may be more applicable than the other. Nonetheless, EMA works out to be reasonably well in most of the cases. It nicely explains the dependence of energy gap opening with the reduction of size and will be discussed here. Historically, Efros and Efros first formulated this theory which was extensively developed by L.E. Brus. Briefly it is outlined below.

Consider a dielectric sphere with dielectric constant ε_2 embedded in a medium of dielectric constant ε_1. Let the radius of the sphere be R (see Fig. 8.5).

If $\mathbf{r}_1$ and $\mathbf{r}_2$ are the positions of the charges inside the sphere, the potential energy can be written as

$$V(\mathbf{S}_1, \mathbf{S}_2) = \pm \frac{e^2}{\varepsilon_2 |\mathbf{r}_1 - \mathbf{r}_2|} + \mathbf{P}(\mathbf{r}_1) \pm \mathbf{P}(\mathbf{r}_2) \pm \mathbf{P}_M(\mathbf{r}_1, \mathbf{r}_2) \tag{8.4}$$

where polarization $\mathbf{P}$ is given as

$$\mathbf{P}(\mathbf{r}) = \sum_{n=0}^{\infty} \alpha_n \left(\frac{r}{R}\right)^{2n} \frac{e^2}{2R} \tag{8.5}$$

and

$$\alpha_n = \frac{(\varepsilon - 1)(n+1)}{\varepsilon_2(\varepsilon + n + 1)} \text{ with } \varepsilon = \frac{\varepsilon_2}{\varepsilon_1}$$

$$\mathbf{P}_M(\mathbf{r}_1, \mathbf{r}_2) = \sum_{n=0}^{\infty} \alpha_n \frac{\varepsilon_2 + r_1^n r_2^n}{R^{2n+1}} P_n(\cos\theta) \tag{8.6}$$

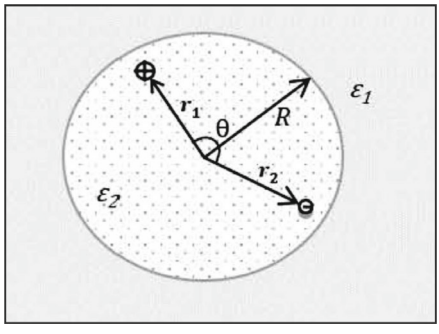


Fig. 8.5: Spherical semiconductor nano-particle embedded in a medium with different dielectric medium.

Here, θ is the angle between r_1 and r_2 and P_n is a Legendre polynomial.

When R is larger than r_1 and r_2

$$V \to \frac{\varepsilon^2}{\varepsilon_2 |\boldsymbol{r}_1 - \boldsymbol{r}_2|} \tag{8.7}$$

This is the potential energy (third) term given in equation (8.2) while writing the Hamiltonian term for the Mott-Wannier exciton.

We can now write the Schrödinger equation as

$$\left[-\frac{\hbar^2}{2m}\nabla_e^2 - \frac{\hbar^2}{2m}\nabla_h^2 + V_0(\mathbf{r}_e, \mathbf{r}_h) \right] \Phi(\mathbf{r}_e, \mathbf{r}_h) = E\Phi(\mathbf{r}_e, \mathbf{r}_h) \tag{8.8}$$

Consider $V_0 = \infty$ outside the sphere.

For small values of R (nanoparticles) we consider wave function $\psi_n(\mathbf{r})$ of the type

$$\Psi_n(\mathbf{r}) = \frac{C_n}{\mathbf{r}} \sin\left(\frac{n\pi r}{R}\right) \tag{8.9}$$

with

$$E_n = \frac{n^2\pi^2\hbar^2}{2mR^2} \tag{8.10}$$

Here m is the effective mass of the charge (electron or hole).

Consider

$$\Phi_0 = \psi(\mathbf{r}_e)\psi(\mathbf{r}) \tag{8.11}$$

as the wave function for Eq. (8.8).

In such a case the energy of the first excited state would be

$$\Delta E = \frac{\hbar^2\pi^2}{2R^2}\left[\frac{1}{m_e} + \frac{1}{m_h}\right] - \frac{1.8e^2}{\varepsilon_2 R} + \text{polarization energy} \tag{8.12}$$

The first term on the right hand side of the equation is localization energy of electron and hole whereas the second term is due to Coulomb interaction energy between electrons and hole. The third term is due to polarization of the cluster but is usually very small and size independent, hence neglected. Details of the formulations are not given here but can be found in the original paper by L.E. Brus. The first two terms have signs opposite for each value of R. But as the particle size decreases, the first term starts dominating and ΔE starts increasing rapidly. This would effectively mean that the energy gap in a semiconductor which is characteristic of a material (e.g. CdS $E_g = 2.42$ eV, ZnS $Eg = 3.6$ eV, GaAs $Eg = 1.42$ eV) would increase



Table 8.2: Kinetic energy, Coulomb energy, energy shift (ΔE) and total energy gap calculated using Eq. (8.11) for spherical particles of GaAs of various radii

Energies (eV)	*Particle radius (nm)*			
	20	10	5	2
Kinetic energy	0.015	0.059	0.24	1.48
Coulomb energy	-0.012	-0.024	-0.489	-0.119
Energy shift (ΔE)	0.003	0.039	0.19	1.361
Total energy gap (E_g)	1.42	1.46	1.61	2.78

with decreasing particle size or in other words we obtain size dependent energy gap which is

$$\text{Total energy gap } = E_{g(\text{bulk})} + \Delta E(\text{exciton}) \tag{8.13}$$

Obviously, ΔE becomes important when cluster size is comparable to the excitation size. This can be seen from the example given in Table 8.2, where ΔE for various values of R for GaAs are shown.

8.3.3 *Optical Properties of Quantum Dots*

The simplest experiment to determine the size dependence in semiconductor nanoparticles, i.e. Quantum Dots is to study absorption spectrum of the material as a function of wavelength of incident photons. When photons are incident on semiconductor material they will be absorbed only when the minimum energy of photons is enough to excite an electron from the valence band to conduction band, i.e. when the photon energy equals the energy gap of the semiconductor. If lower energy photons are incident, there cannot be any absorption. Therefore there is a sudden rise in absorption when the photon energy is same as the energy gap. This is the onset of absorption. If the energy gap increases there should be a shift in the onset of absorption towards the shorter wavelength. As shown in Fig. 8.6, one expects a blue shift with absorption in smaller and smaller particles, which is indicative of increasing energy gap. For every small particle (see Fig. 8.7) one often uses the terminology borrowed from chemistry, viz. Highest Occupied Molecular Orbital (HOMO) amd Lowest Unoccupied Molecular Orbital (LUMO) instead of top of valence band and bottom of conduction band used in case of extended solid.

This change in absorption would have an interesting effect on the originally coloured materials. It is known now from literature that Cd_3P_2 is a dark brown semiconductor with energy gap of approximately 0.5 eV. When its nanoparticles are made, it progressively passes through a series of colours like brown, red, yellow and white with particle size changing from ~30 Å to ~15 Å. For ~15 Å particles the band gap increases to 4 eV.



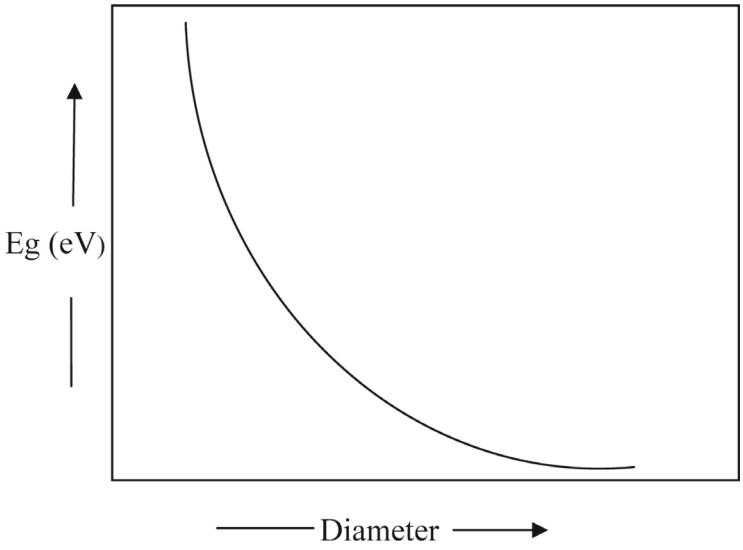


Fig. 8.6: Schematic representation of variation of energy gap with particle size.

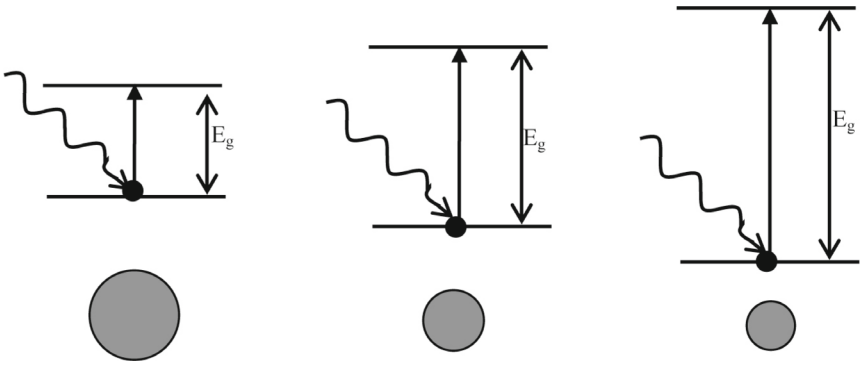


Fig. 8.7: With decreasing particle size, energy gap increases. This would give rise to blue shift in the absorption spectra.

The same is true for CdS (Figure in the preface of this book). The bulk semiconductor with energy gap of 2.42 eV is orange in colour. As the particles become smaller and energy gap increases it becomes yellowish and ultimately white. It is quite easy to show by chemical analysis techniques that this white material is CdS and nothing else. In fact, observation of different colours due to CdSe in glass matrix led scientists to think that CdSe nanoparticles of different sizes might have been formed.

Many nanomaterials exhibit enhanced luminescence as compared to their bulk counterparts. Some materials like silicon which are not luminescent in their bulk



form become luminescent in the nano form. Therefore luminescence investigations of nanomaterials are often carried out and are quite interesting.

We shall discuss here the principles of (a) photoluminescence, (b) electroluminescence, (c) cathodoluminescence and (d) thermoluminescence.

(a) ***Photoluminescence***

When the external stimulus is electromagnetic radiation or photons, the observed luminescence is known as photoluminescence. An electron from the valence band can be excited to a level in the conduction band if photon of sufficient energy to make a transition is available. This process leaves a hole in the valence band. The excited electron can loose energy by emission of phonon in a relatively shorter time ($\sim 10^{-12}$-10^{-13} sec) before it can relax and make a radiative transition (Fig. 8.8b). The life time of radiative process is much longer 10^{-7}– 10^{-9} up to few milliseconds (fluorescence) or even few seconds (phosphorescence) than the life time of nonradiative transition.

Various possibilities of luminescence mechanisms are illustrated in Fig. 8.8. When emission of light due to the transition of an electron from the valence band maximum to the conduction band minimum, as shown in Fig. 8.8(a), takes place it is called the band edge emission. The energy of emitted light or photoluminescence is the energy difference between the conduction band minimum and the valence band maximum. It is also possible as in Fig. 8.8(c) that the electron is first excited to the conduction band from where it moves in the material and gets trapped in some localized level. Subsequently it makes a radiative transition to valence band emitting the photon of a longer wavelength compared to that emitted by band edge luminescence. If there are localized levels introduced by impurities in the energy gap then non-radiative transition from conduction band minimum to impurity level occurs, followed by radiative transition to the lower levels (Fig. 8.8(d)) and then another non-radiative transition to valence band maximum.

The wavelength of luminescence, luminescence efficiency and radiative lifetime depend upon the material. In case of nanomaterials additionally the size of the material plays an important role. This can be readily understood by considering the schematics in Fig. 8.9 shown for a spherical nanoparticle.

In a particle with dimension comparable to Bohr diameter of electron, electron and hole wavefunctions can overlap. Therefore the probability of efficient recombination

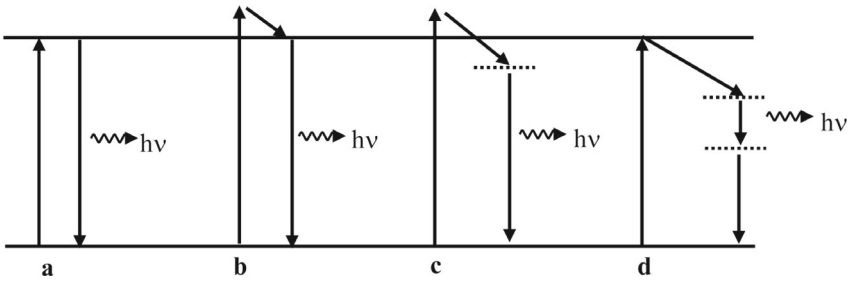


Fig. 8.8: Various processes of luminescence.

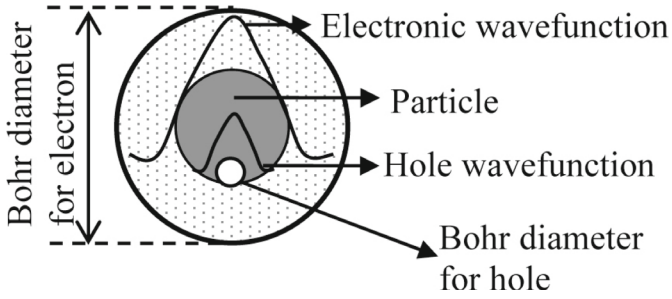


Fig. 8.9: Localization of electron and hole wavefunction inside a nanoparticle.

increases. Due to altered electronic structure in nanomaterials, the transition rules also relax in certain cases and some transitions which are forbidden in bulk material become possible. The fluorescent efficiencies in general are found to be enhanced in nanoparticles. Interestingly, doping of nanoparticles also is possible. Dopants can introduce some energy levels in the band gap of the host material as discussed above. Such levels can be responsible for altering the luminescence properties of the host material. As an example, consider the doping of ZnS nanoparticles with manganese. ZnS is a wide direct band gap material. The band gap luminescence of bulk ZnS is at ~344 nm. If Mn is introduced in small quantity (~0.1 at. %) it produces discrete energy levels in the band gap. A photoexcited electron can make a transition from conduction band minimum to upper level and then combine with a trapped hole in lower level to emit luminescence at ~590 nm. In nanoparticles also such luminescence is observable but with more probability or higher luminescence efficiency, due to increased overlap, arising because of small size in nanoparticles, of wavefunctions. The process can be accompanied by band edge luminescence and/or defect emission.

(b) ***Electroluminescence***

Luminescence observed by the application of an electric field to a material is known as electroluminescence. It can be observed by applying either low or high ($>10^4$ V/cm) field; accordingly it is classified as 'injection luminescence' and 'high field electroluminescence' respectively.

Injection luminescence: Light emitting diodes are based on the principle of minority carrier injection in a diode. When a p-n diode is formed, a depletion region is set up with V_d as the diffusion potential between the p-n regions (see Fig. 8.10).

In a p-type semiconductor, holes are the majority carriers and electrons are minority carriers. On the other hand, in an n-type semiconductor, electrons are majority carriers and holes constitute minority carriers.

By forward biasing the diode, the diffusion barrier can be reduced and potential becomes $V_d - V$, where V is the applied voltage. Under this condition, the electrons from n-region find it easier to travel to p-region and holes can easily migrate from p-region to n-region. Thus minority carriers are injected on each side. The injected

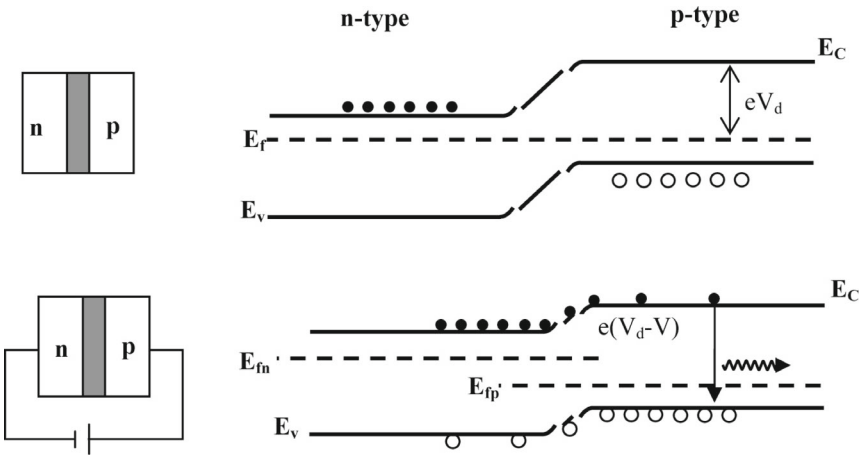


Fig. 8.10: p–n junction at thermal equilibrium (top) and with forward bias (bottom).

carriers can combine with easily available opposite charge carriers in each region to produce luminescence.

It can be observed that the injection luminescence is a 'band edge' type luminescence. Therefore for a fixed semiconductor, the wavelength of luminescence is fixed. However in nanomaterials we know that energy gap can be tuned with particle size. Hence the luminescence also can be tuned to desired wavelength in nanomaterials.

High field electroluminescence: This type of electroluminescence is used in 'display panels'. Emission of electrons by application of very high electric field (~10^5 to 10^7 V/cm) is known as *field emission*. As illustrated in Fig. 8.11 an energy barrier existing for a material has to be overcome by an electron in order to leave it. The barrier, however, reduces on application of the electric field, the width of which depends on the applied field. As can be seen from the figure, more the applied field, smaller is the width of the barrier. In general, an electron needs energy sufficient to overcome the barrier in general. However quantum mechanically, if the barrier width is reduced sufficiently, the electrons can tunnel through the barrier. This type of field emission requires very high field as mentioned above.

In a metal semiconductor contact, the barrier height is reduced due to transfer of electrons either from metal to semiconductor or semiconductor to metal depending upon the type (n or p) and value of work functions of both metal and semiconductor. A metal–semiconductor contact is known as Schottky barrier. Electrons can be injected from metal to p-type semiconductor by high electric field. Note that here electrons are emitted from metal into the semiconductor by a different phenomenon as compared to that in injection luminescence. In injection luminescence, charge carriers overcome the potential barrier and penetrate the other material whereas in high field luminescence it is through tunneling as the electrons pass from metal to semiconductor. After passing from metal to semiconductor, if the electron comes in the vicinity of a hole it combines radiatively.

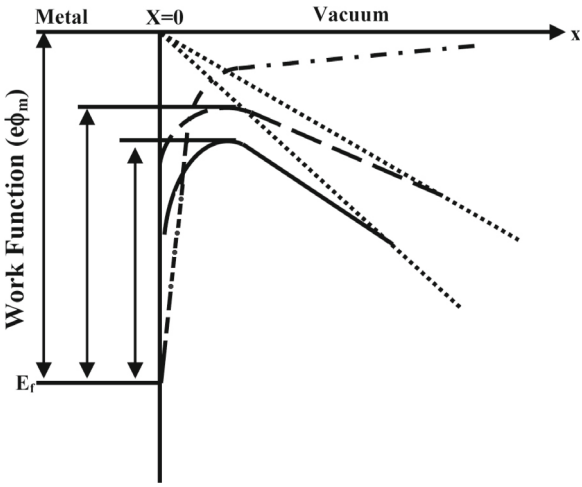


Fig. 8.11: Field emission from a metal.

High field luminescence in nanomaterials: Nanomaterials are useful in display panels using high field. The fine particles resolve the images better as compared to microparticles. However due to high quantum efficiency of luminescence, electroluminescence also is an efficient process.

(c) ***Cathodoluminescence***

Electrons of very high energy striking a semiconductor material produce luminescence known as 'cathodoluminescence'. The incident electrons here are from some filament or field emission cathode. They strike the luminescent material in vacuum at high energy. The electrons on impact with the material are able to loose their energy by various processes. Phenomenon of cathodoluminescence is used in oscilloscope and old television display screens.

Nanomaterials are also useful to produce high efficiency cathodo-luminescence due to the same reasons as for high field luminescent materials. Cathodoluminescence in ZnS doped with Mn and ZnO nanoparticles has been investigated. It has been observed that nanoparticles are better luminescent materials than microparticles of the respective material.

(d) ***Thermoluminescence***

In semiconductors with large band gaps it is found that if they are excited at very low temperatures with photons in the UV range, on heating to some temperature which depends upon the dopant ions, light is emitted even in the absence of any other stimulus. The phenomenon is known as 'thermoluminescence' or 'after glow'.

As illustrated in Fig. 8.12, luminescence is due to trapped electrons. The trapped electrons are activated by thermal energy and can combine with hole making radiative transition or causing luminescence. Figure 8.12 also gives some examples in which luminescence due to different co-activation is observed for ZnS doped with Cu.

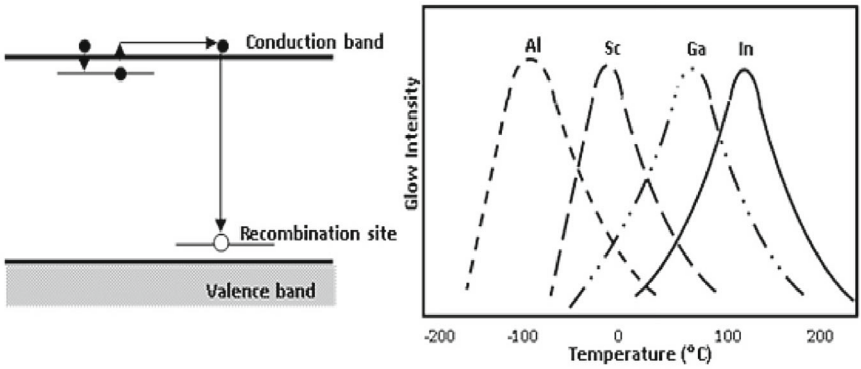


Fig. 8.12: Thermoluminescence energy level diagram.

Observation of thermoluminescence is an effective method of studying the trap levels in materials.

Nanomaterials also have defect levels. The surface atoms can act as efficient traps for electrons/holes. Therefore thermoluminescence can be quite strong in nanomaterials. Thermoluminescence has been reported for ZnS nanoparticles doped with copper and other co-dopants (or co-activators) illustrated in Fig. 8.12.

8.4 METAL NANOPARTICLES (PLASMONIC MATERIALS)

Windows in old churches, palaces, houses etc. are designed with beautiful tinted glasses. Such glasses are made by dissolving small amount (<5%) of metal particles like gold, silver, cobalt, iron and nickel. Such glasses are transparent but have different colours like red, pink, blue, green or other shades depending upon the dissolved metal particles. Indeed the colour of glasses is due to metal nanoparticles. Note that the colours produced by dissolving the metal powders are not same as their bulk metal colour. For example gold metal has yellow 'golden' colour in the bulk form but the colour produced by dissolving small amount of gold in glass may be reddish, pink or even blue. Although the art of making tinted glasses is more than 2000 years old, scientific interest in metal nanoparticles started with M. Faraday's synthesis of gold nanoparticles in 1857. He reduced chloroauric acid ($HAuCl_4$) using citric acid [$CH_2(COO)_2H_2O$] and showed that Au metal nanoparticles produced intense magenta-red colour as against yellow appearance of bulk gold metal. Attempts were since then made to explain the observed intense colours due to metal nanoparticles. Apart from their beautiful appearance, it is now realized that metal nanoparticles have applications in sensors, solar cells, cancer therapy, cloaking devices and photonic

devices for communication or optical circuits. Here we will try to understand the theories developed to explain the observed behaviour of metal nanostructures. It should be remembered that metal nanoparticles differ from the semiconductor particles. Metals have a large free electron density of valence electrons. ~10^{22} to 10^{23}/cm^3. On the other hand semiconductors have much lower electron density ~10^{16} to 10^{18}/cm^3 depending upon the level of doping. Doping is a process of introducing electron donor or acceptor atoms in small quantities in order to make the n or p type semiconductors. Therefore in semiconductor nanomaterials the observed properties are explained based on the electron confinement in a potential well of appropriate shape depending on the nanomaterial dimensionality (0-D, 1-D or 2-D). In metal nanostructures a large number of electrons are considered together and the confinement is often known as 'dielectric confinement' rather than 'electron confinement'. We now proceed to discuss how to describe observed behaviour of metal nanoparticles and nanostructures.

The small particles of different shapes like spherical particles, wires, rods, cubes and thin films or surfaces of metals need to be treated differently. The small particles exhibit *localized surface plasmon resonance* (SPR) and surfaces of metals or thin films have propagating surface waves viz. *surface plasmon polariton* as separately discussed below and schematically illustrated in Fig. 8.13.

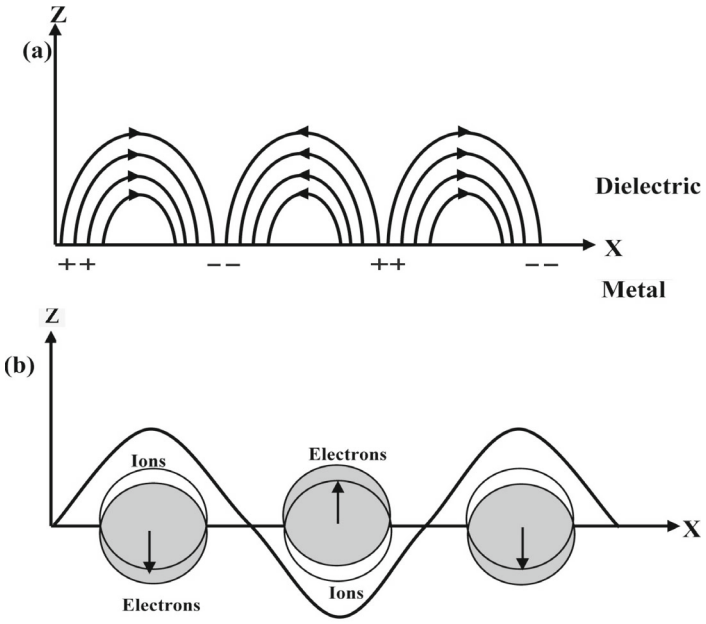


Fig. 8.13: Schematic of SPR: (a) Propagating SPR and (b) Localized SPR.

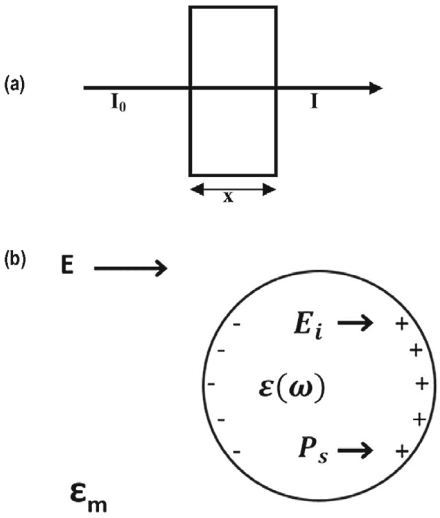


Fig. 8.14: (a) Incident radiation I_0 passes through a medium of thickness x and (b) A particle embedded in a dielectric medium (Mie theory).

8.4.1 Localized Surface Plasmon Resonance

In 1908, G. Mie explained by using Maxwell's equations the scattering (and absorption) of light from very small particles. When electromagnetic radiation is incident on the spherical particles of uniform size, embedded in a medium, reduction in intensity is observed. It is necessary to consider mainly the dielectric constant of medium in which the particles float and the dielectric constant of the particles. Interaction between the particles is neglected in Mie theory. His theory can be briefly described as follows. The details can be found in some standard text books on optics.

A beam of electromagnetic radiation of intensity I_0 and wavelength λ passes through a medium having dielectric constant ε_m. If the particles are embedded uniformly in the medium and multiple reflections do not take place, the transmitted intensity would be given by D'Lambert equation (Fig. 8.14)

$$I = I_0 e^{-\mu x} \tag{8.14}$$

where μ is extinction coefficient.

$$\mu = \frac{N}{VC_{ext}} \tag{8.15}$$

N is the number of particles in medium, V – the volume of colloidal particles and

C_{ext} is extinction cross section of a particle.

$$C_{\text{ext}} = C_{\text{abs}} + C_{\text{scatt}} \tag{8.16}$$

In the Mie theory

$$C_{\text{ext}} = \frac{2\pi}{k^2} \sum_{n=1}^{\infty} (2n+1) R_e (a_n + b_n) \tag{8.17}$$

where a_{n} and b_{n} are scattering coefficients. They are functions of a and λ in terms of Ricatti-Bessel functions. The details of the Mie theory can be found in the book by Born and Wolf (see the complete reference at the end of this chapter).

$$k = \frac{2\pi\sqrt{\varepsilon_m}}{\lambda} \tag{8.18}$$

For very small particles ($kR<<1$) having radius R, extinction is mainly due to absorption. The extinction coefficient is given by

$$C_{\text{ext}} = \frac{24\pi^2 R^3 \varepsilon_m^{\frac{3}{2}}}{\lambda} \frac{\varepsilon_2}{(\varepsilon_1 + 2\varepsilon_m)^2 + \varepsilon_2^2} \tag{8.19}$$

where

$$\varepsilon(\omega) = \varepsilon_1(\omega) + i\varepsilon_2(\omega) \tag{8.20}$$

$\varepsilon(\omega)$ is the dielectric constant of the particles. As can be seen from above equation, C_{ext} depends on R^3. Absorption coefficient μ is inversely proportional to $1/V$ i.e. $1/R^3$. Thus absorption is independent of particle size. From the Eq. (8.19), it is also seen that extinction would be maximum when

$$\varepsilon_1 + 2\varepsilon_{\text{m}} = 0 \text{ or } \varepsilon_1 = -2\varepsilon_{\text{m}} \tag{8.21}$$

if ε_2 is small. This gives rise to strong resonance band. Bandwidth and peak height is mainly determined by $\varepsilon_2(\omega)$ (interestingly C_{ext} vanishes if $\varepsilon_2 = \infty$ as well as $\varepsilon_2 = 0$).

Although Mie theory successfully explains the observation of an absorption band for metal nanoparticles in the visible range of wavelengths, for particle size less than ~10 nm, it hardly explains the observed shifts in the absorption peaks with change in the particle size. One, therefore, should consider dielectric constant which depends not only on the wavelength (or frequency) but also on the particle size. For metals the dielectric response of electrons is better described by Drude model. The theory of LSPR is based on the Drude model of dielectric response of free electrons in a metal.

Consider that the electric field vector of an electromagnetic wave with frequency ω is written as

$$E = E_0 \exp(-i\omega t) \tag{8.22}$$

on a metal particle. The electrons of mass m are then accelerated. The equation of motion is given as

$$m\frac{d^2x}{dt^2} + \gamma\frac{dx}{dt} + m\omega_0^2 x = eEe^{-i\omega t} \tag{8.23}$$

An electron in a metal experiences collisions (which give rise to resistivity in materials) with lattice and impurities. The relaxation time of an electron is $\tau = 1/\gamma$. Solving the equation (8.23) we get

$$x = \frac{eE/m}{\omega_0^2 - \omega^2 - i\omega_d\omega} \tag{8.24}$$

where ${\omega_0}^2 = f/m$, in which 'f' is the restoring force constant, and $\omega_d = \gamma/m$ is the damping. The resonance frequency here is ω_0 and it may or may not be equal to the frequency of the incident radiation.

The electron displacement 'x' is in the same direction of applied field when the resonance frequency ω_0 is more than the incident frequency ω. The displacement can be negative if $\omega > \omega_0$.

Consider that the displacement x is not just for a single electron but for N electrons/cm^3 in a particle, then they would create polarization $\mathbf{P}$

$$\mathbf{P} = N\,e\mathbf{x} \tag{8.25}$$

Polarization can also be written in terms of the dielectric function as

$$\mathbf{P} = \varepsilon_0(1 - \varepsilon)\mathbf{E} \tag{8.26}$$

Using the equations

$$\varepsilon(\omega) = 1 + \frac{Ne^2/m\varepsilon_0}{\omega_0^2 - \omega^2 - i\omega_d\omega} \tag{8.27}$$

and

$$\varepsilon(\omega) = \varepsilon'(\omega) + i\varepsilon''(\omega) \tag{8.28}$$

where ε' and ε'' are the real and imaginary parts of the dielectric functions respectively. Therefore,

$$\varepsilon'(\omega) = 1 + \frac{\left(\omega_0^2 - \omega^2\right)\omega_p^2}{\left(\omega_0^2 - \omega^2\right)^2 + \omega^2\omega_d^2} \tag{8.29}$$

When $\omega_0 = 0$ and

$$\varepsilon(\infty) = 1,\ \omega_p^2 = \frac{4\pi Ne^2}{m} \text{ is the volume plasma,} \tag{8.30}$$

$$\varepsilon'(\omega) = \varepsilon(\infty) - \frac{\omega_p^2}{\omega^2 + \omega_d^2} \tag{8.31}$$

and

$$\varepsilon''(\omega) = \frac{\omega_d\omega_p^2}{\omega\left(\omega^2 + \omega_d^2\right)} \tag{8.32}$$

$\omega_0 = 0$ is possible when the electrons do not have any restoring force. $\varepsilon(\infty)$ is the dielectric constant at very high frequency.

When $\omega_0 = 0$, the electrons are out of phase with the electric field **E**. This implies that the real part of the dielectric constant would always be negative. This is schematically shown in Fig. 8.15.

Additionally, as shown schematically in Fig. 8.16, the electrons from the conduction band (*s*) and the filled band (*d*) to empty states above the conduction band also get photo-excited.

Thus the dielectric constant may get further modified by interband transitions. In silver and gold, the real part of the dielectric constant gets modified due to contribution from the 'd' and 'sp' bands. In the visible and ultra-violet regions, interband

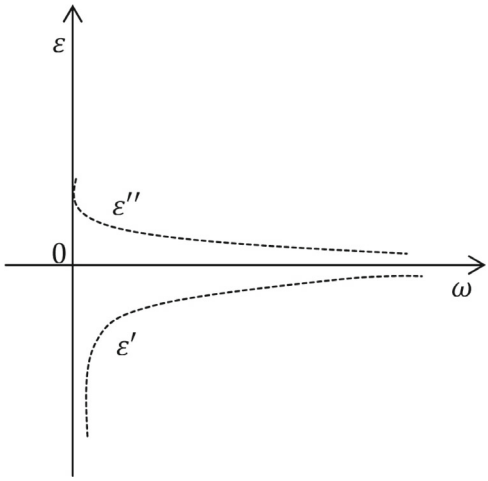


Fig. 8.15: Real and imaginary parts of the dielectric function with respect to the frequency.

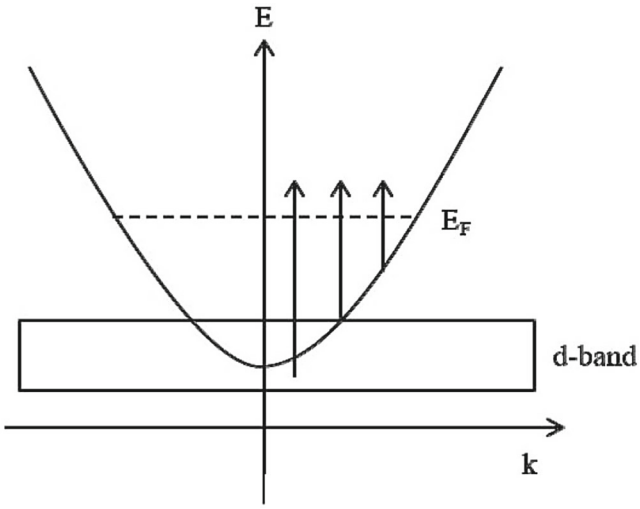


Fig. 8.16: Schematic diagram of various transitions from filled inter and intra bands in a metal.

transitions contribute substantially whereas the 'red' or the long wavelengths derive contributions from intraband transitions.

Further, one needs to take into account that at small sizes of metal particles, the collisions of electrons with surface of the particle would increase. Therefore, the damping term γ would become

$$\gamma = \gamma_0 + g\frac{v_F}{D} \tag{8.33}$$

where g is a proportionality factor (~1), v_F – the velocity of electrons at the Fermi surface, and D is the diameter of the particle. If D is much smaller than the wavelength of incident radiation and the resonance condition given by $\varepsilon' + 2\varepsilon_m = 0$, is satisfied, the resonance will occur at

$$\Omega(r) = \frac{\omega_p}{\sqrt{\varepsilon'_{\text{interband}} + 2\varepsilon_m}} \tag{8.34}$$

One can see from the above equation and the discussion in this section, how the dielectric constant of the metal particles (alongwith the dielectric constant of the medium in which they are dispersed) determines the position of the SPR peak. Therefore the phenomenon of SPR is aptly referred to as the 'dielectric confinement'. This equation works well for silver, but complications arise for gold and copper due to the overlap of interband transitions with the surface plasmon band. Thus in order to explain the observed difference in the surface plasmon resonance positions of gold and silver nanoparticles in spite of almost same lattice constants and electron density (hence ω_p which is 8.98 eV for silver and 9.01 eV for gold), it is necessary

to remember that interband transitions need to be taken into account. In Table 8.3 values for some metals in vacuum ($\varepsilon_m = 1$) are given.

For the silver nanoparticles, surface plasmon resonance peaks are much sharper compared to those due to gold nanoparticles. This is due to the occurrence of silver plasmon resonance peak where imaginary part of dielectric constant (ε'') has nearly zero value. This is not the case for gold, for which absorption due to interband transition for ε'' has finite value.

Interestingly only few metals like gold, silver, copper and palladium nanoparticles show surface plasmon resonance in the visible range or close to visible range. There are many metals which surprisingly do not exhibit strong resonance peaks in the visible range. This is because they have interband transitions spread over UV–VIS-IR region which overlap with their resonance energies making them difficult to observe.

For a given metal nanoparticle surface plasmon resonance energy is affected by the medium (dielectric constant) in which they are embedded. The surface plasmon resonance peak position also depends on the particle size (Fig. 8.17).

Table 8.3: Values for some metals: $\hbar\Omega$(ib) – interband energy and $\hbar\Omega$(r) – surface plasmon energy

m_e *(effective mass of electron)*		N *(e/m^3)*	*l (nm)*	v_F *(10^6 m/s)*	ω_p *(eV)*	$\hbar\Omega\,(ib)$ *(eV)*	$\hbar\Omega\,(r)$ *(eV)*
Ag	1.03	5.85	55	1.4	8.98	3.9	3.5
Au	1.01	5.9	42	1.4	9.01	2.3	2.4
Cu	1.17	8.48	28	1.05	9.05	2.1	2.0

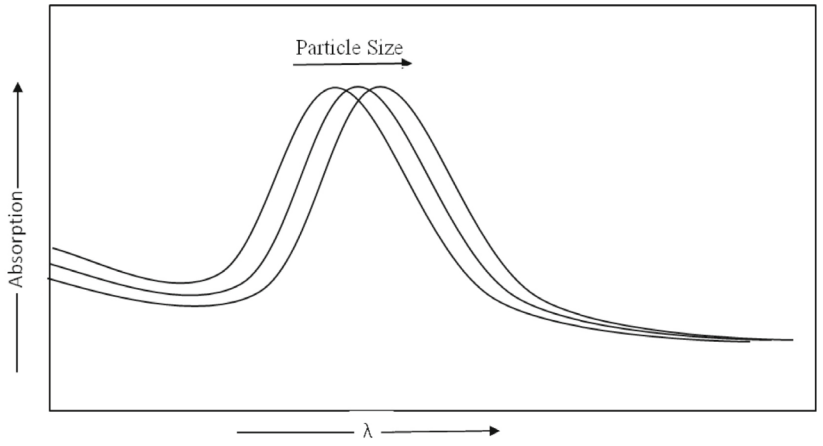


Fig. 8.17: The surface plasmon resonance absorption peak position red-shifts with increase of the particle size.

Usually small particles are dominated by absorption and large particles by scattering. Gold and silver nanoparticles also show effects on surface plasmon resonance due to shape. This is easily attributed to the differences in their polarization/depolarization effects. The orientation of nanoparticles with respect to the incident radiation also would get modified (see Fig. 8.18). For example a nanorod standing parallel or transverse to the incident beam would have different value of C_{ext}.

A nanorod also readily shows two surface plasmon resonance peaks as shown in Fig. 8.19.

The two peaks appear due to the cross section (diameter) and length of the particle. If the diameter of the nanorod is kept constant and length changed, it is also possible to keep the resonance peak due to cross section at same energy and vary the longitudinal peak (due to length). The length/diameter or aspect ratio variation is often made in order to meet the needs of some applications.

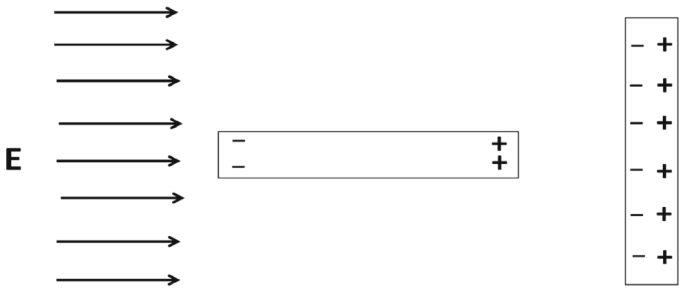


Fig. 8.18: Illustration of the change in the magnitude of the polarization due to the orientation of nanorods.

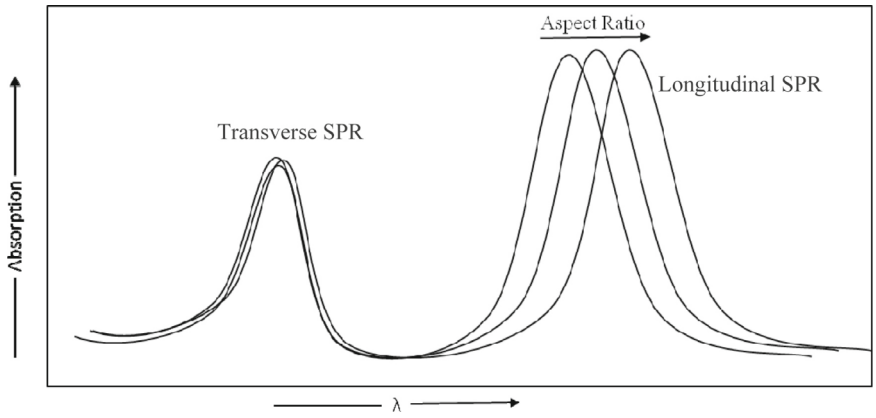


Fig. 8.19: Transverse and longitudinal surface plasmon peaks of nanorods. With increasing rod length the longitudinal plasmon resonance peak keeps on shifting to the longer wavelength but transverse resonance peak remains at the same position if the diameter of the rod remains unchanged.

Numerous other shapes like cubes, stars, tetra pods, flowers and so on are observed by tuning the synthesis conditions. They produce tunable surface plasmon resonance peaks.

Temperature is another parameter which can affect the surface plasmon resonance. With increase in temperature, volume expansion takes place shifting the electron density. This causes a red shift in the resonance peak position. The peak broadening also can occur as a result of increased scattering of electrons. The solvent temperature decreases its refractive index which shifts the resonance peak to blue side.

8.4.2 Surface Plasmon Polariton

Surface plasmon polaritons have become very important in nanotechnology because the scientists are quite hopeful of developing a branch called *nanophotonics* using them. Using some metallic nanostructures or thin films, unlike in localized surface plasmon resonance, it is possible to let the waves of surface plasmon polaritons to propagate along the interface between metal surface and a dielectric (vacuum, air, glass or any other dielectric medium) medium over large distances. This is because of the coupling of the electromagnetic radiation and the surface plasmons. Coupling of a photon and surface plasmon is known as *surface plasmon polariton.* It should be remembered that polariton is a general term used to refer to any coupling between a photon and an elementary excitation (exciton, phonon and plasmon). Thus surface plasmon polaritons are confined and can travel over large distances of even few micrometres which makes them very useful to transfer the light from one end to the other end of the nanostructure. This becomes useful to design nano optical devices and circuits.

As shown in Fig. 8.20, there are two components of the surface plasmon polariton viz. longitudinal (along X-axis) and the transverse (along Z-axis).

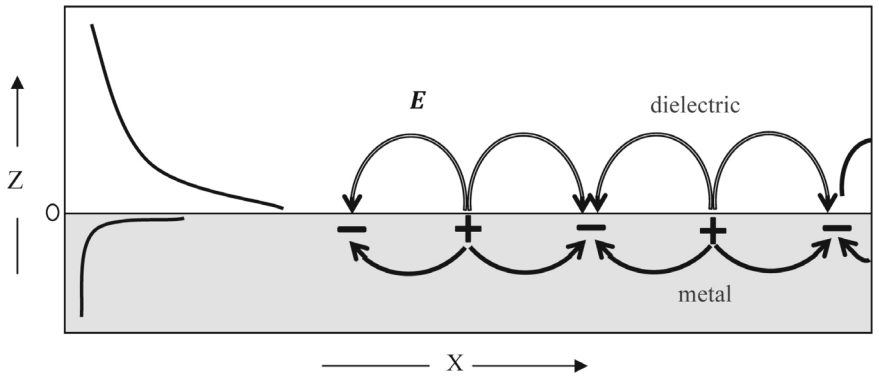


Fig. 8.20: Surface plasmon polariton. Metal shown in grey and white portion of the figure is dielectric.

The electric field inside the metal can be written as

$$E_{\mathrm{m}}(x, z, t) = E_{\mathrm{m},0} e^{i(k_x x - k_z z - \omega t)} \tag{8.35}$$

and in the dielectric as

$$E_{\mathrm{d}}(x, z, t) = E_{\mathrm{m},0} \frac{k_x}{k_z} e^{i(k_x x + k_z z - \omega t)} \tag{8.36}$$

Note that *m* and *d* denote metal and dielectric. The electric field has both x and z components, although the amplitudes decay exponentially inside the metal as well as dielectric medium as shown on the left side of Fig. 8.20. Using the Maxwell's equations the dispersion relation can be shown as

$$k_{\mathrm{x}} = \frac{\omega}{c} \left(\frac{\varepsilon_{\mathrm{m}} \varepsilon_{\mathrm{d}}}{\varepsilon_{\mathrm{m}} + \varepsilon_{\mathrm{d}}} \right)^{1/2} \tag{8.37}$$

with

$$k_{\mathrm{x}} = \frac{\omega}{c} \sqrt{\varepsilon}$$

ε_{m} and ε_{d} are dielectric constants of the metal and dielectric respectively.

The dispersion behaviour of surface plasmon polariton can be understood from Fig. 8.21.

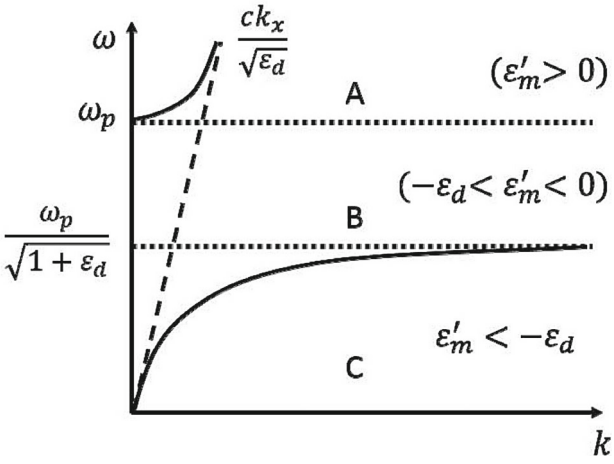


Fig. 8.21: Dispersion curve for the surface plasmon polariton.

The dotted line represents the dispersion of light in the dielectric medium, air with $\varepsilon_d = 1$ here. Obviously the dispersion is divided into three parts, A, B and C, depending upon the dielectric constant values of the metal. Here we consider the real part of the dielectric constant viz. ε'.

When $\omega > \omega_p$ (region A),

$$\varepsilon_m \text{ and } \varepsilon_m + \varepsilon_d \text{ are } > 0 \tag{8.38}$$

The modes are radiative. In this region, k_x and k_z are both real.

In the region B, we have

$$\frac{\omega_p}{\sqrt{1+\varepsilon_d}} < \omega < \omega_p \tag{8.39}$$

Here $\varepsilon_m < 0$ but $\varepsilon_m + \varepsilon_d > 0$ and k_x is imaginary but k_z is real and no propagating modes exist.

In the region C,

$$0 < \omega < \frac{\omega_p}{\sqrt{1+\varepsilon_d}} \tag{8.40}$$

The quantities ε_m and $\varepsilon_m + \varepsilon_d$ are < 0 and k_x is real but k_z is imaginary. When the values of k_x are small they approach the dotted line for dispersion of light in the dielectric.

In region C, we have bound modes and through the coupling of the electromagnetic waves and surface plasmons, surface plasmon polaritons can be propagated along the surface without radiative decay. This is what is expected in this branch of nanophotonics.

We shall not go into more details about surface plasmon polaritons here but just mention that surface plasmon modes and electromagnetic modes can be coupled also at the glass prism/dielectric and metal surface as illustrated in Fig. 8.22. Two different configurations can be set to investigate the surface plasmon polariton or to make its application as a sensor.

The light rays are internally reflected at glass prism and a dielectric between metal thin film, known as Otto configuration as in the left of Fig. 8.22 or by depositing the thin metal film directly on the prism as in the Kretschmann geometry. The silver or gold films are often used.

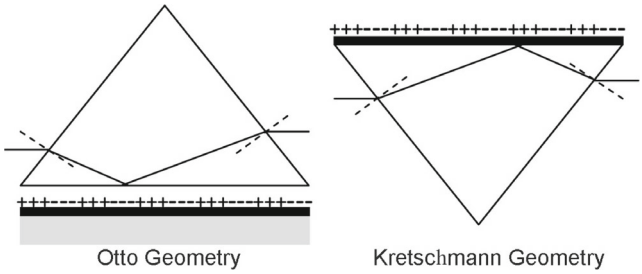


Fig. 8.22: The light rays passing through the prisms couple with the surface plasmons shown as positive–negative charge waves on the metal thin film (dark black). By keeping the analyte between the prism and the metal thin film, either in Otto or Kretschmann geometries, analytes can be studied by finding out the changes in the reflected rays.

8.5 NANOCOMPOSITES

Materials with uniform physico-chemical properties throughout their extent associated to structure and bonding are known as 'single phase' materials. A composite on the other hand is a multiphase, homogeneous dispersion of two or more components each one being a single phase in itself. Homogeneity of the resultant material is dictated by the microstructure, number of interfaces made and the interfacial bonding. The amount of each constituent is determined by the end properties desired and achieved with the constituting materials. Properties of composites are expected to be superior to the constituent phases on their own. However, this expectation is fulfilled provided the choice of constituents and processing has been carried out properly. The superiority can be in terms of not only physico-chemical properties but also their economical viability and large scale production.

When one of the constituent phases has at least one of the dimensions at nanoscale (1-100 nm), it is known as nanocomposite.

Nanomaterials like quantum dots (semiconductors), plasmonic materials (metals), metals oxides, carbon nanotubes, nanofibrils, Graphene, Graphene Quantum Dots, Carbon Quantum Dots, Graphdiyne, 2-D Metals sulphides, 2-D metal selenides, Graphitic Carbon Nitride, black phosphorus etc. are intermixed to make nanocomposite. They can be nonofibers or nanosheets occupying large volume so that they can be useful in large scale applications.

Nanocomposites are often made of a bulk polymer (matrix) in which nanoparticles are used as fillers. Sometimes there is another polymer used, which interfaces with the bulk polymer as well as nanoparticles. The properties and amount of interfacial polymer (bonding with matrix and filler) finally control whether nanocomposite has better properties than individual phases or not. The desired properties could be better strength, Young's modulus, heat control, gas permeability, antimicrobial activity, appearance etc.

There is another type of polymer nanocomposite. Here a polymer matrix is filled with nanoscale clay (nanoclay). Clay (< 4-5 μm grains) forms the crust of earth

(62.5% oxygen, 21% silicon, 6.5% aluminum and remaining other elements) and is used in making bricks, earthen pottery or ceramics. Silicates form the basic structural crystals, mostly layered. With oxygen ions at the apexes of a tetrahedron silicon occupies the central position. Other metal atoms like aluminum, magnesium etc. sometimes occupy some of the interstitial vacancies or substitute silicon sites. Consequently a variety of clays called as montmorillonite, kaolinite, illite, hectorite etc. emerge depending upon the structure and composition. Clays can be considered as layered materials often weakly bonded together. They may be phase separated, intercalated or exfoliated as schematically illustrated in Fig. 8.23 to make nanocomposites.

A large number of applications have been demonstrated using nanocomposites. Automotive, aviation, space, submarine, civil construction, electric and electronic industry, renewable energy sources, food package industry etc. has started experiencing the nanomaterials and nanoclay nanocomposites, extremely useful in making light weight, strong and economically better parts.

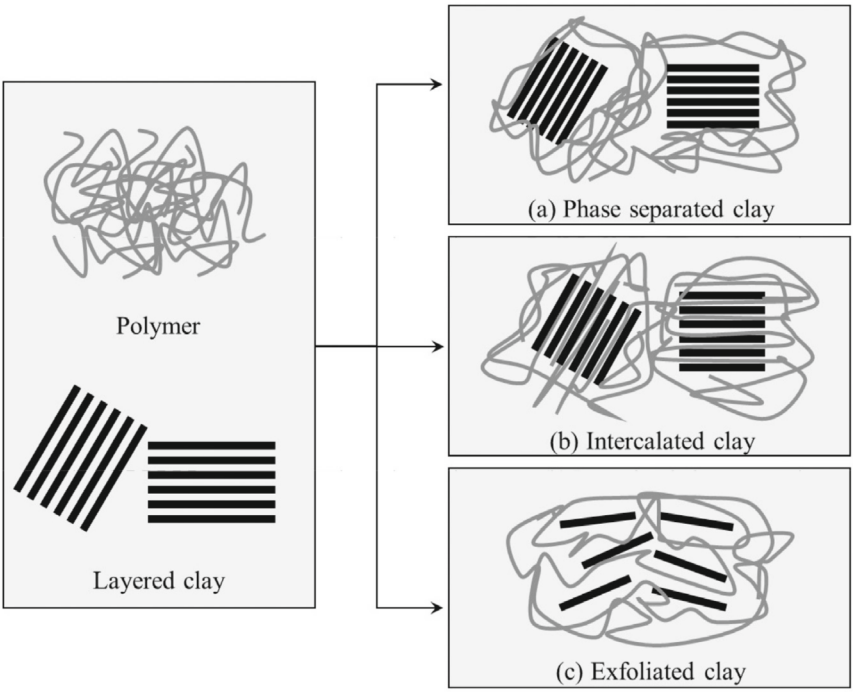


Fig. 8.23: Nanocomposite structures using clay.

8.6 NANOMAGNETISM

Magnetism in bulk materials is due to magnetic moments on constituting atoms, ions or molecules. Presence of a magnetic field around a moving charge is well known. In an atom, spin and orbital motion of electrons and change of orbital motion in the presence of external applied magnetic field **H**, gives rise to magnetic moment. Nuclear magnetic moment is usually very small and will be neglected in our discussion.

Total spin angular momentum due to spin of electrons in incomplete shells is given by

$$\mathbf{S} = \sum_{i}^{N} \mathbf{S}_{\mathrm{i}} \tag{8.41}$$

where $\mathbf{S}_{\mathrm{i}}$ is the spin on the ith shell and the sum is over N shells. Corresponding dipole moment is given by

$$\boldsymbol{\mu}_{\mathrm{s}} = -\frac{e}{m}\mathbf{S} \tag{8.42}$$

Total orbital angular momentum due to that of electrons in incomplete shells is given by

$$\mathbf{L} = \sum_{i}^{N} \mathbf{L}_{i} \tag{8.43}$$

where $\mathbf{L}_i$ is the orbital momentum of ith shell and the sum is over all the N shells. The magnetic orbital angular momentum is given by

$$\boldsymbol{\mu}_{\mathrm{L}} = \frac{e}{2m}\mathbf{L} \tag{8.44}$$

Total angular momentum **J** is

$$\mathbf{J} = \mathbf{L} + \mathbf{S} \tag{8.45}$$

Interaction between **L** and **S** is known as spin–orbit interaction. The total magnetic moment of an atom is given by

$$\boldsymbol{\mu} = \boldsymbol{\mu}_{\mathrm{L}} + \boldsymbol{\mu}_{\mathrm{S}} \tag{8.46}$$

It can be shown (see Fig. 8.24) that in free space magnetic moment on an atom is given by

$$\boldsymbol{\mu}_{\mathrm{avg}} = g\left(-\frac{e}{2m}\right)\mathbf{J} \tag{8.47}$$

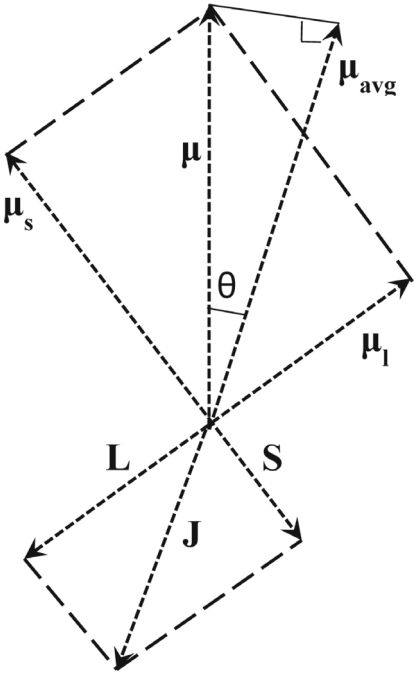


Fig. 8.24: Spin–orbit interaction.

With

$$g = 1 + \frac{J(J+1) + S(S+1) - L(L+1)}{2J(J+1)} \tag{8.48}$$

In the presence of an applied magnetic field, angular momentum can have $2J + 1$ orientations. One can determine S, L and J values if one knows the angular momentum of incomplete shells and number of electrons using Hund's rules as follows.

Hund's Rules

1. The maximum allowed value of S will be decided by Pauli's exclusion principle.
2. The maximum value of L is consistent with the value of S from L.
3. The value of J will be $|L - S|$ when the shell is less than half filled and $|L + S|$ when the shell is more than half filled.

The resultant magnetic moment of an atom could be defined as:

$$|\mu| = \mu_B g(J(J+1))^{1/2} \tag{8.49}$$

where μ_B is the Bohr magneton $\left(9.3 \times 10^{-24}\text{Joule/Tesla}\right)$ and g is Lánde factor having a value between 1 and 2.

Types of Magnetic Materials

Depending upon the response of a material to the external magnetic field, i.e. how much magnetization (M) is induced in the material by an external magnetic field most of the materials can be classified as diamagnetic, paramagnetic, ferromagnetic, antiferromagnetic and ferrimagnetic. The ratio of M to H is called the susceptibility:

$$\chi = \frac{M}{H} \tag{8.50}$$

$$\mu_0 = \frac{B}{H} \tag{8.51}$$

where B is magnetic induction and μ_o is the permeability of the free space.

Permeability of the medium μ_m is given by

$$\mu = 1 + 4\pi\chi \tag{8.52}$$

We shall now discuss these materials in brief.

Diamagnetic Materials

Diamagnetism can be considered as the atomic manifestation of Lenz's law. When a magnetic field is applied to a conducting loop, a current is induced which opposes the change of flux. Electrons in atom are equivalent to conducting loop. In the applied magnetic field, electrons precess about the magnetic field direction. The precession is known as Larmor precession. It induces a magnetic moment

$$\mu = -\left(\frac{\mu_0 e^2}{4m}\right) r^2 H \tag{8.53}$$

where r is the radius of the electron orbit. There is no diamagnetic component of magnetic moment in the absence of an external magnetic field. However, as soon as an external magnetic field is present, diamagnetic component would always be present.

The magnetic susceptibility for N atoms having atomic number Z is given by

$$\chi = -\left(\frac{N\mu_0 e^2 Z}{6m}\right) a^2 \tag{8.54}$$

where a is the average radius of electrons in an atom.

Thus diamagnetic materials have temperature independent negative susceptibility.

Paramagnetic Materials

Materials having some net magnetic moment present on atoms, molecules or ions even in the zero magnetic field are paramagnetic. In the zero magnetic field these magnetic moments would be randomly distributed and the material can show zero magnetization. In the presence of the magnetic field the magnetic moments try to

align themselves in the direction of the magnetic field. Thermal energy kT tries to disturb this alignment. Diamagnetic component also arises in the opposite direction of the magnetic field but paramagnetic response can be larger resulting into a net positive response. Magnetic susceptibility is given by

$$\chi = \frac{Np^2\mu_B^2 Z}{3kT} = \frac{C}{T} \tag{8.55}$$

where C is known as Curie constant and

$$p = g[J(J+1)]^{1/2} \tag{8.56}$$

p is known as effective Bohr magneton.

Equation (8.55) is known as Curie law. It shows that the magnetic susceptibility in case of paramagnetic substances is inversely proportional to the temperature. Figure 8.25 illustrates schematically the temperature dependence of paramagnetic susceptibility.

Ferromagnetic Materials

Some magnetic materials have regions or domains in which large number of atoms or ions or molecules having permanent magnetic moments, like in paramagnetic materials, are aligned in a particular direction. Different domains are separated by what is known as domain walls. The domains themselves may be randomly oriented in the absence of a magnetic field. Such materials are said to have spontaneous magnetization and are known as ferromagnetic materials. This is due to the magnetic interaction, known as exchange field, amongst the magnetic moments of atoms. It was proposed by Weiss that the effect of mean magnetic field due to all other moments on a given moment of an atom is given by

$$\mathbf{H} = \lambda \mathbf{M} \tag{8.57}$$

$$\mathbf{H}_{\text{total}} = \mathbf{H} + \lambda \mathbf{M} \tag{8.58}$$

Modified Curie law, known as Curie–Weiss law for the susceptibility of ferromagnetic materials, is given by the equation

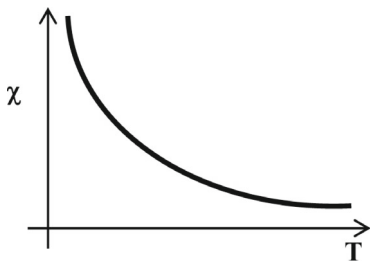


Fig. 8.25: Temperature variation of magnetic susceptibility in case of paramagnetic materials.

$$\chi = \frac{C}{(T - T_c)} \tag{8.59}$$

where T_c is the Curie temperature. In ferromagnetic materials below the characteristic temperature T_c of the material, spontaneous magnetization exists in zero magnetic field. At temperature higher than T_c, the material behaves like a paramagnetic material. Ferromagnetic materials exhibit a hysterisis behaviour (see Fig. 8.26) of magnetization when an external magnetic field is applied.

Heisenberg explained the spontaneous magnetization by considering that the exchange energy U_{exch} which is due to the spin moments say $\mathbf{S}_i$ and $\mathbf{S}_j$ on i^{th} and j^{th} atoms is given by

$$U_{exch} = -2J_{exch}\mathbf{S}_i \cdot \mathbf{S}_j \tag{8.60}$$

where J_{exch} is known as the exchange integral.

When J_{exch} is negative, antiparallel configuration of spins is favoured, whereas for positive J_{exch}, parallel configuration is favoured, as illustrated in Fig. 8.27.

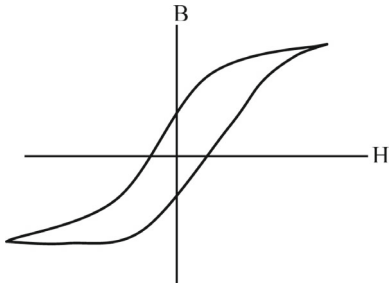


Fig. 8.26: Hysterisis loop of a ferromagnetic material.

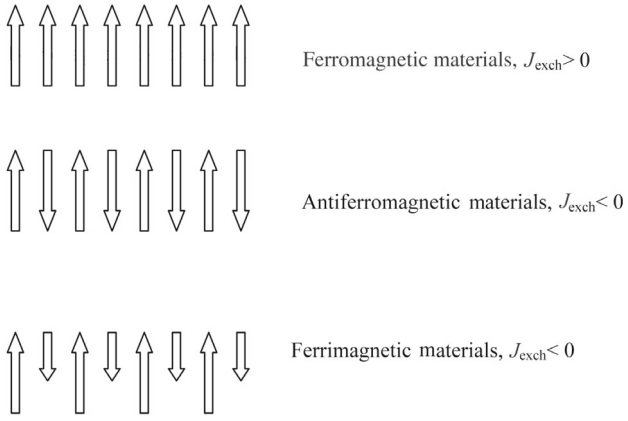


Fig. 8.27: Schematic of spin orientations on neighbouring atoms in case of ferromagnetic, antiferromagnetic and ferrimagnetic materials.

Slater showed that there exists a correlation between the interatomic distance and radius of incomplete filled d shell in case of transition elements, which is responsible for the ferromagnetism or antiferromagnetism, as shown in Fig. 8.28. It can be seen that this makes Fe, Co and Ni ($J_{exch} > 0$) ferromagnetic but Mn and Cr ($J_{exch} < 0$) antiferromagnetic materials.

Antiferromagnetic Materials

For antiferromagnetic materials the J_{exch} is negative and spins on neighbouring atoms are antiparallel to each other. Susceptibility in these materials is small but positive, as applied magnetic field tends to orient the spins in the direction of magnetic field overcoming the diamagnetism. Figure 8.29 illustrates schematically the susceptibility of antiferromagnetic materials in general.

Antiferromagnetic arrangement exists below a critical temperature known as Néel temperature T_N. Above T_N, susceptibility is paramagnetic and has the form

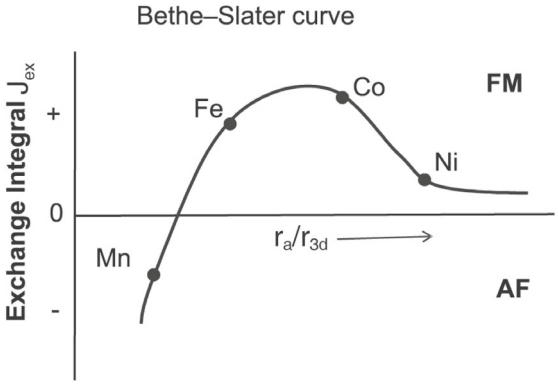


Fig. 8.28: A relation between J_{ex} and interatomic distance for transition elements; Here r_a is interatomic distance and r_{3d} is the radius of 3d shell.

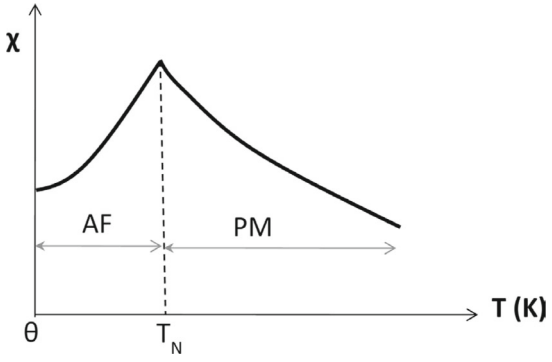


Fig. 8.29: Typical susceptibility curve in case of antiferromagnetic material.

$$\chi_\theta = \frac{2\chi}{T + T_N} \tag{8.61}$$

Ferrimagnetic Materials

As illustrated in Fig. 8.27, there is also ferrimagnetism which is similar to antiferromagnetism but alternate spins are of smaller magnitudes than the rest. In general the ferromagnetic, antiferromagnetic and ferrimagnetic materials are quite complex in their behaviour.

Nanomagnetic Materials

With this background of magnetic materials we now proceed to understand properties of some special types of nanomagnetic materials in which materials are reduced at least in one of the dimensions. Magnetic nanoparticles, assemblies of nanoparticles, magnetic nanowires, magnetic thin films or multilayer films and some metal oxide films, doped semiconductor particles or thin films have become the focus of attention due to their interesting magnetoresistive or magnetooptical properties they exhibit. The field of nanomagnetic materials is quite vast and still expanding. We shall discuss some of these materials.

8.6.1 Magnetic Nanoparticles

Ferromagnetic materials like Fe, Co, Ni, Fe_3O_4, γ-Fe_2O_3 and many others have very interesting behaviour below a critical size, characteristic of each material. Bulk ferromagnetic materials have spontaneously magnetized domains. Formation of domains occurs in order to minimize the total magnetostatic energy of the system. However below the critical size, domain formation is not energetically favoured and material prefers to be single domain. In such a situation all the spins of atoms are oriented in one direction. Typically, the particles with a size below 100 nm are likely to be single domain. Coercivity H_c in such particles is given by

$$H_c = \frac{2K}{M_s}\left[1 - \frac{T}{T_B}\right] \tag{8.62}$$

where M_s is the saturation magnetization due to applied magnetic field, K – the anisotropy constant and T_B is the blocking temperature. Anisotropy (preferred orientation of magnetic moments) can arise due to crystal structure, shape, stress and surface. Blocking temperature is the temperature, above which thermal energy is able to set the orientation of magnetic moments free. Below the blocking temperature, they are as if frozen. Figure 8.30 shows a common behaviour in materials of H_c as a size of particles.

Single domain particles of extremely small size which do not show coercivity or hysteresis (see Fig. 8.31) are known as superparamagnetic particles.

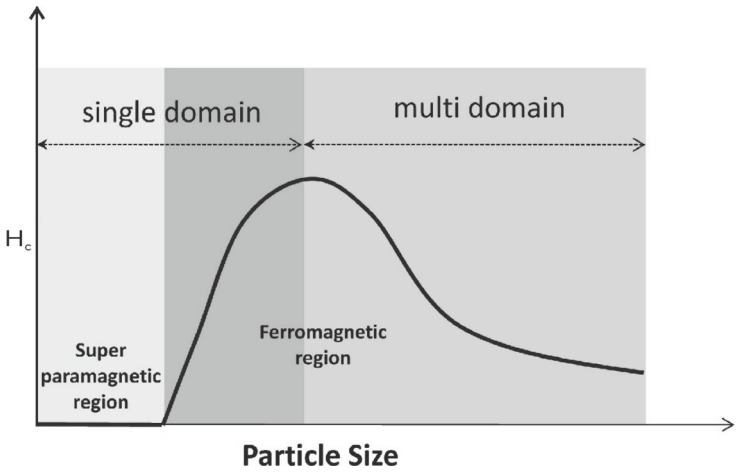


Fig. 8.30: A relation between particle coercivity and size.

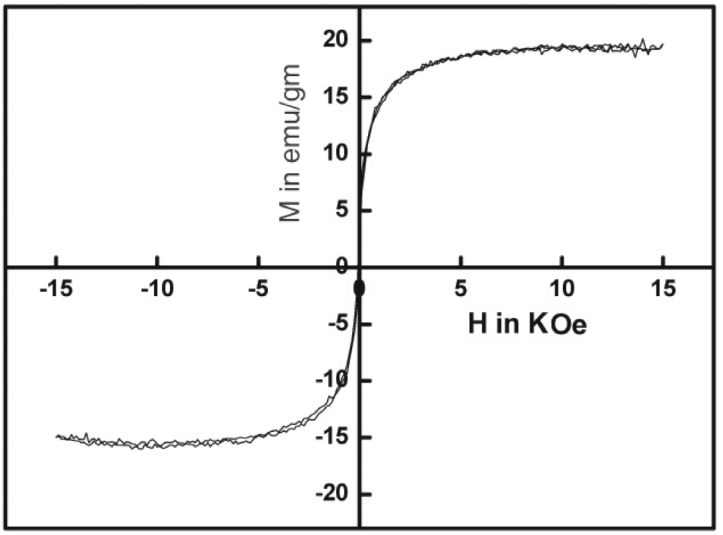


Fig. 8.31: Magnetization behaviour in case of a superparamagnetic particle.

In superparamagnetic particles, spins are oriented in one direction and switch coherently in the opposite direction. The switching time is of the order of few nanoseconds and is given by

$$t = \tau_0 e^{-KV/kT} \tag{8.63}$$

where KV is the barrier for total spin orientation. Typical measurements of magnetization require 10 to 100 s. Thus magnetization (M) versus applied field (H) would exhibit the curve as shown in Fig. 8.31 with no coercive field. If magnetization M versus (H/T) are plotted then all the points for a superparamagnetic material lie on a single curve.

Small particles are characterized by large surface to volume ratio. Therefore surfaces and interfaces play an important role in their magnetic properties of nanostrucures. At surfaces there is not only the symmetry breaking of the bulk crystal structure but there is a change in the coordination number as well as change in the lattice constant. Such effects can give rise to observation of ferromagnetic behaviour in materials which are not ferromagnetic in the bulk form. The coupling of magnetic nanoparticles spread at short distances also is an interesting area of current research.

8.6.2 Magnetic Oxide Materials

Magnetic multilayers are artificially obtained materials in which thin layers of magnetic layers separated by nonmagnetic layers are coupled and show Giant Magneto Resistance (GMR). In some metal oxides it has been observed that some planes are magnetically coupled to each other. This gives rise to huge change in the resistance when magnetic field is applied. The change in resistance is even larger than that in magnetic multilayers and is referred to as Collosal Magnetoresistance or CMR.

Colossal Magnetoresistance has been predominantly discovered in manganese-based perovskite oxides. In these crystals strong mutual coupling of spin and charges on atoms is observed. Hence not only high temperature superconductivity, but also new magnetoelectronic properties are increasingly discovered in materials with perovskite structures.

The 3d transition metal oxides, particularly the manganites are being exploited from the prospects of having improved device performance as compared to the GMR materials. These oxides display a diverse nature of properties such as paramagnetic to ferromagnetic transition accompanied by insulator to metal transition and realization of high magnetoresistance on application of comparatively low magnetic field. Historically magnetoresistance in manganites is known since nineteen fifties. But recent interest is developed in doped manganites for exploring further high magnetoresistance value at room temperature at low magnetic field value to interplay of spin, charge and lattice distortion.

For example, undoped $LaMnO_3$ is an antiferromagnetic charge transfer insulator wherein Mn is in +3 valence state. As shown in Fig. 8.32 it has a perovskite structure. If any divalent atom (such as Ca, Sr, Ba, Pb) is doped at La site for instance, $La_{1-x}Ca_xMnO_3$, it converts Mn^{+3} to Mn^{+4} state in equal proportion of the doping concentration. This is equivalent to the hole doping in the system. The hole doping introduces a number of dramatic changes in electric and magnetic properties from

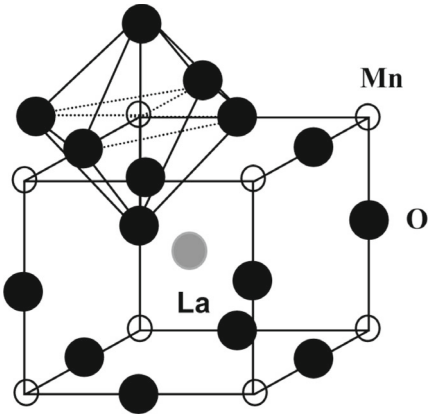


Fig. 8.32: Perovskite structure of $LaMnO_3$.

the parent $LaMnO_3$ compound, such as insulator–metal transition, paramagnetic-ferromagnetic transition, charge ordered state, phase separation etc. In the light of these rich physical properties, hole doped $LaMnO_3$ has a potential for promising device applications such as magnetic sensors, magnetic valves, read head technology and bolometric application. Thin films of these materials are therefore a topic of great current interest.

8.7 MECHANICAL PROPERTIES OF NANOMATERIALS

Mechanical properties of materials depend upon the composition and bonds between the atoms viz. covalent, ionic, metallic etc. As a result purest materials may be inherently weak or strong or brittle. Presence of impurities affects all these properties. Most of the materials have various impurities like C, O, N, P, S etc. present in them as well as point defects, grain boundaries, dislocations etc., which are responsible for the deviations of the properties expected from high purity and ordered materials.

When the size of materials is reduced to nanoscale, materials tend to be single crystals. However we need to consider different types of nanomaterials as schematically illustrated in Fig. 8.33 to specify the properties.

Indeed it is possible to determine various mechanical properties like elasticity, hardness, ductility etc. of different nanostructures. Techniques like bending measurement, velocity of sound measurement, nanoindentation etc. can be used. It should be, however, noted that measurements on single nanoparticles, rods, tubes etc. would inherently be difficult, though not impossible. However measurements on nanocrystalline solids, thin films etc. are possible using some conventional methods.

It has been shown in case of metallic nanocrystalline materials (as in Table 8.4) that elastic modulii reduce dramatically. For example in case of magnesium nanocrystalline material (grains ~12 nm size), Young's modulus was observed to be 3900 N/

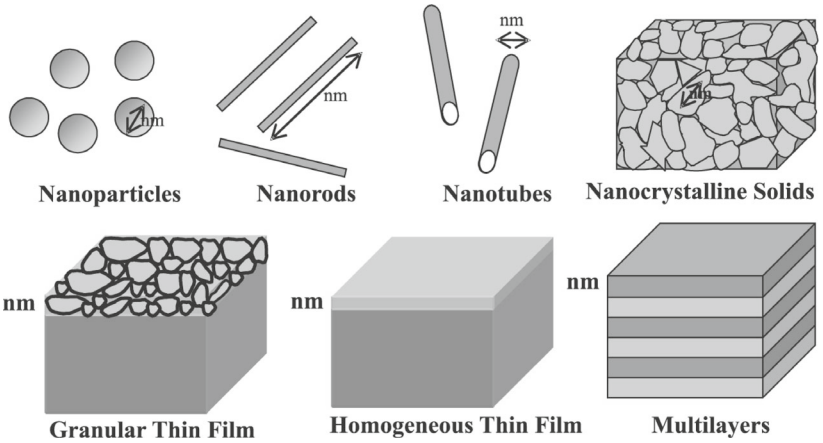


Fig. 8.33: Some common types of nanocrystalline materials.

mm^2 as against 4100 N/mm^2 for polycrystalline (size > 1 μm) magnesium. In case of palladium, nanocrystallites of ~ 8 nm size had Young's modulus 8800 N/mm^2 as against 12,300 N/mm^2 for polycrystalline palladium.

Ceramic materials are often compacted and sintered using powder material. This also increases the hardness of materials. It has been shown that in case of TiO_2 nanoparticles (~12 nm size) produced in powder form much less temperature was required to densify and achieve the hardness comparable to usual polycrystalline material.

In fact density of nanocrystalline pellet is often low due to some pores left when powders are compressed to form pellets. Nanocrystalline pellets densities approach those of bulk polycrystalline materials as the sintering at high temperature progresses.

Plastic deformation in nanocrystalline materials strongly differs from that of polycrystalline bulk counterpart as shown in Fig. 8.34. It was shown in case of nickel that stress removal results in more effective recovery of the material as compared to corresponding polycrystalline material.

Table 8.4: Elastic properties of some common materials

Material	*Density kg/m^3*	*Young's modulus 10 ^{9}N/m^2*
Steel	7860	200
Al	2710	70
Glass	2190	65
Concrete	2320	25–35
Bone	1900	9
Polystrene	1050	3
Diamond	3510	1035
CNT	2600	1280

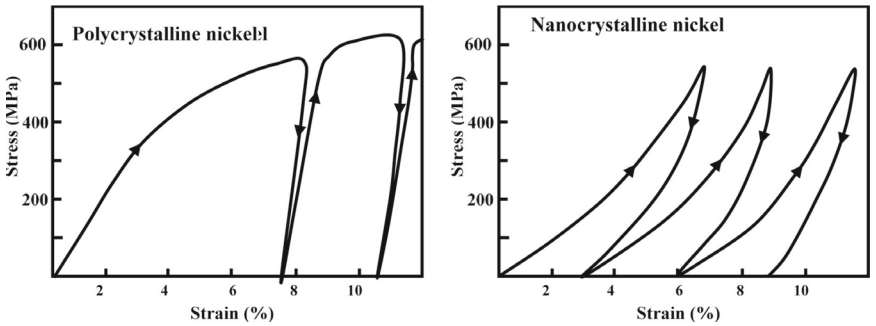


Fig. 8.34: Comparison of stress and strain relation in case of nanocrystalline and polycrystalline material.

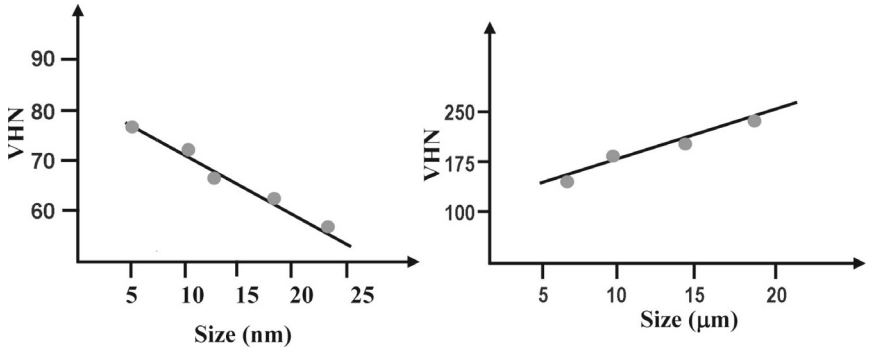


Fig. 8.35: Hardness variation for copper in nanometer and micrometre grain size range.

Hardness of materials is also related to the grain size. As illustrated in Fig. 8.35 for copper, even in micrometer grain size range there is a linear dependence of hardness on particle size. It decreases with increase of grain size. However in nanometer size range, the hardness increases with increase of particle size linearly. Similar results are found in case of palladium nanoparticles and microparticles.

8.8 STRUCTURAL PROPERTIES

Even though some nanomaterials with slightly large number of atoms (>50–60 atoms) may acquire bulk crystalline structure, it is found that the lattice parameters may not be the same as in the bulk material. For example it has been shown by rigorous analysis of X-ray diffraction patterns of ZnS that as small as 1.4 nm particles had liquid disorder. However larger crystals of ZnS indeed show same sphalerite (cubic) structure as in the bulk. It has been observed that there is a lattice contraction of ~ 1%

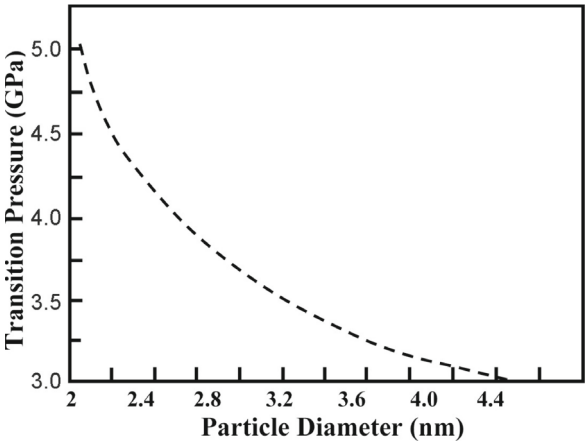


Fig. 8.36: Structural deformation of CdSe nanoparticles.

for 1.4 nm ZnS nanoparticles. Other small particles also show up to ~2-3% lattice constant deviations compared to bulk crystalline materials.

With increase in temperature the disordered structure of small particles of ZnS were found to transform to wurtzite (hexagonal) structure. Further the chemical capping, often used in the synthesis of nanoparticles, gets removed and particles tend to agglomerate or coalesce forming larger particles.

Effect of pressure on structural properties (using X-ray diffraction) has also been well investigated for some nanoparticles. It has been found that indeed the structural transformations do take place in case of nanoparticles with applied pressure. However the pressures required for this are larger for nanoparticles than for corresponding bulk material and depend upon the particle size, as illustrated in Fig. 8.36 for CdSe nanoparticles. Thus CdSe nanoparticles of 1 to 2.1 nm required 4.9 GPa to 3.6 GPa pressure to transform them from wurtzite to rock salt structure. Bulk CdSe needs just 2.0 GPa for the same transformation.

8.9 MELTING OF NANOPARTICLES

A variety of nanoparticles like Au, Ag, CdS etc. have been investigated for their thermal stability and melting. Melting begins at the surface. As the particle size decreases surface to bulk atoms ratio increases dramatically. In small particles or clusters the central atom may be considered as surrounded by 1st, 2nd, 3rd, … compact shells of atoms. Number of atoms in shells is given as $10\,n^2 + 2$. Thus 1st shell would have 12 atoms, 2nd shell would have 42 atoms and so on. It can be easily seen that number of surface atoms is quite large in nanoparticles and surface to bulk atoms ratio goes on increasing with decreasing particle size (or shells). Large surface is related to large surface energy which can be lowered by melting. Extensive experimental data on spherical as well as some non-spherical nanoparticles, thermodynamic model and

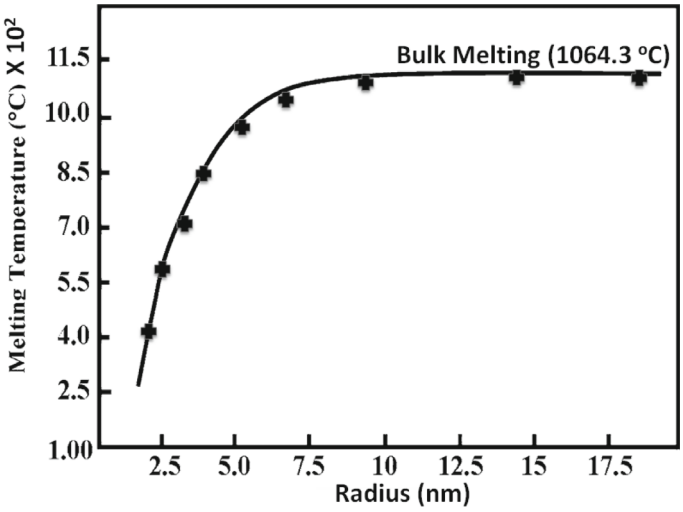


Fig. 8.37: Variation of melting point with radius of gold nanoparticles.

molecular dynamic simulations agree with each other regarding the size dependent changes. In general with decreasing size the melting temperature also decreases. The changes due to shape and exposed surfaces for the same material may show different melting temperatures. Figure 8.37 illustrates how nanoparticles of gold show dramatic variation of melting temperature with reduction in the size below around 6-7 nm radius. In case of metal nanoparticles embedded in a foreign host material, melting at higher temperature than its bulk are observed. This is called superheating.

Melting of nanoparticles is usually determined either by X-ray diffraction or electron diffraction. Heating increases the lattice parameter and at melting point long range order is lost.

FURTHER READINGS

Optical Properties of Solids by Mark Fox. Oxford University Press (2012).
Principles of Optics by Max Born and Emil Wolf. Cambridge University Press (2003).
Phosphor Handbook (Eds) Shigeo Shionoya and William M. Yen. CRC Press (1999).
Materials Science and Engineering: An Introduction, William D. Callister, Jr. 4th Edition. John Wiley & Sons, Inc. New York (1997).
Nanocrystalline Materials by H. Gleiter. *Progress in Materials Science*, **33:** 227 (1989).

Chapter 9
Carbon Nano Family

9.1 INTRODUCTION

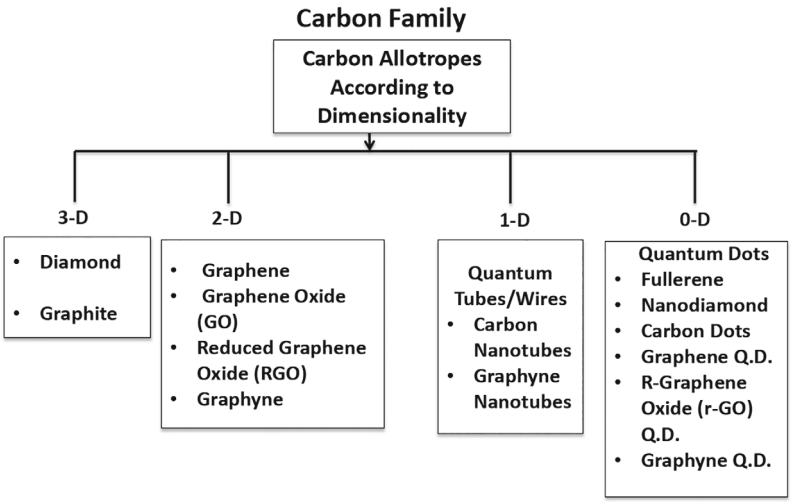
We discussed in last few chapters synthesis, characterization and properties of nanomaterials in general. Few examples were given at various places. In this chapter we shall discuss nanomaterials belonging to carbon family. Here nanomaterials like fullerenes, graphene, graphene oxide, carbon nanotubes, graphyne and various carbon family quantum dots and nanodiamonds will be discussed. Along with carbon nanomaterials transition metal dichalcogenides, and some other very interesting new developments in nanotechnology will be briefly outlined as they have become very important. As we will see the reason for a separate chapter on these materials is reasonable, considering various developments in carbon nanomaterials (and other fallouts) as many nanotechnologically important materials have been developed and their large scale production has been achieved over the years. Here we will try to see their discoveries and current understanding.

9.2 CARBON NANOMATERIALS

Carbon is one of the very interesting elements which constitutes a major part of the living as well as non-living world. It also is the backbone of an important branch of chemistry viz. polymer chemistry. But over a long time diamond and graphite were the only known crystalline carbon allotropes until the discovery of fullerenes. Since then carbon became a wonder element with many new discoveries of carbon based as well as similar materials. We can obtain carbon in various forms as 0-D (small clusters and fullerenes as well as nano diamonds), 1-D (carbon nanotubes), 2-D (graphene and graphayne) and 3-D (diamonds) materials. It can be said that carbon has grown in a large family as depicted schematically in Box 9.1.

Apart from scientific novelty of these materials, they have become important due to their applications in various fields like, nanoelectronics, nanosensors, medical field, environmental applications to name a few. In fact it is often said that half of the community in nanotechnology is working on carbon based materials. In the following sections we shall briefly try to understand these nanomaterials.

Box 9.1



9.3 FULLERENES

Crystalline carbon can exist in diamond, graphite, fullerene, carbon nanotubes and graphene forms. Out of these allotropes of carbon, fullerene, carbon nanotubes and graphene are relatively new. In 1985 a sixty-atom carbon molecule was experimentally discovered by Robert F. Curl, Harold W. Kroto and Richard E. Smalley. Although there were some theoretical predictions about the existence of such a hollow carbon molecule, no proof existed. Therefore Curl, Kroto and Smalley in 1996 received a Nobel prize for chemistry due to their groundbreaking discovery. It was a hollow molecule with carbon atoms on the surface. Discovery of this molecule was followed by observations of many such hollow carbon molecules, which have 20, 28, 32, 44, 50, 70, 72, 84, 240, 540, 960 and 1140 carbon atoms. These molecules are collectively known as 'Fullerenes' and written with element symbol of carbon 'C' with number of atoms as suffix, C_n where n is the number of carbon atoms of the fullerene. Each carbon atom at the apex position of a fullerene, is bonded to three nearest carbon

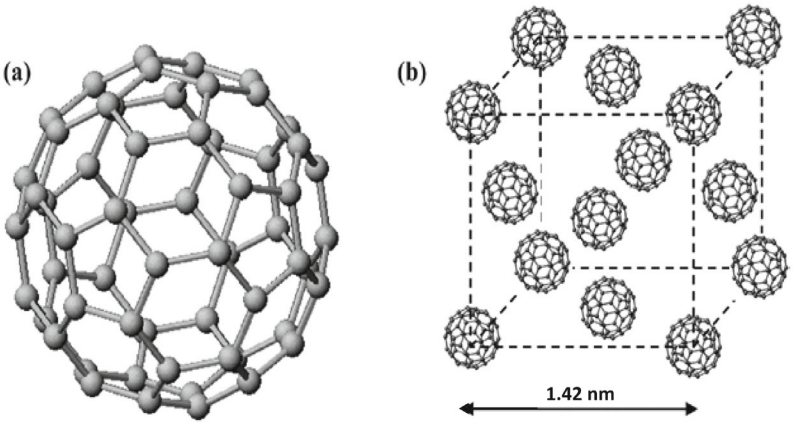


Fig. 9.1: (a) Fullerene C_{60} and (b) Crystalline form of C_{60}.

atoms by three sigma (σ) bonds. The pi (π) bond is dangling. The common feature of fullerene molecules, besides being hollow is that they have a distribution of atoms in five membered rings (pentagons) and six membered rings (hexagons). All Fullerenes above C_{60}, always have 12 pentagons distribution such that no two pentagons touch each other and are always surrounded by five hexagons. As the number of carbon atoms increases number of hexagons may increase and they can touch each other but number of pentagons is always 12. For example C_{60} fullerene, has 12 pentagons, 20 hexagons and 32 faces, C_{70} fullerene has 12 pentagons, 25 hexagons and 37 faces. Out of all the fullerenes most abundant are C_{60}, C_{70}. Metal atoms, ions or molecules such as La, K, ScN3 when doped in fullerenes are known as endohedral metallofullerenes or simply endofullerenes.

Pure as well doped fullerenes can be synthesized by arc deposition or laser ablation method. Fullerene crystal is a molecular crystal having Face Centered Cubic (FCC) structure as illustrated in Fig. 9.1. Besides opening the gateway for carbon family, fullerenes themselves have a variety of applications in the medical field, pharmaceuticals, water purification, organic photovoltaics, electronics etc.

9.4 CARBON NANOTUBES (CNTs)

Carbon Nanotubes (CNT) are 1-D form of carbon as the diameter of carbon nanotubes can be as small as less than a nanometer and length few tens of cm. In 1991 S. Iijima observed under a transmission electron microscope few CNTs. He was actually examining some sample of carbon clusters viz. 'fullerens' synthesized using electric arc discharge method. Carbon nanotubes were so less in this sample that Iijima himself said somewhere that it was like observing 'a needle in a hay stack'. However prior to this observation, Smalley, one of the Nobel laureates who made

the discovery of Fullerenes had speculated that like fullerenes which were spherical or nearly spherical molecules, even tubes should be possible. This was followed by some elegant theories predicting the nature of electron and phonon spectra of carbon nanotubes. Discovery of nanotubes was a breakthrough in that sense. His experiments suggested that even a simple set up as used in producing fullerenes should produce even nanotubes under certain conditions. It was quite clear as more and more scientists started reporting the nanotubes that the conditions favoured for fullerenes were unsuitable for producing the nanotubes. Novelty and later the potential applications of nanotubes created a wave of excitement amongst the scientists which led to unfolding of many unique properties carbon nanotubes possess. Their potential applications in electronics, optoelectronics and energy saving systems have been well realized. Just in one decade several groups all over the world have dedicated their research activities to synthesize and analyze the nanotubes. Following the original arc deposition method, it was found that chemical vapour deposition, laser ablation and some other methods could be employed to produce carbon nanotubes. Further it was found that not only carbon but many other materials like ZnO, TiO_2 and MoS_2 can have shape of nanotube. They have their own applications but carbon nanotubes still remain the most important ones due to their technological potential. A number of excellent reviews and books have been published on carbon nanotubes. Here we shall briefly discuss some of the unique features of carbon nanotubes, synthesis methods, properties and applications.

Box 9.2
Why the Name Fullerenes?

Why the name fullerene is given to such cage-like carbon clusters has an interesting story. H.N. Kroto of the Sussex University, U.K. and R. Smalley's group in the Rice University, U.S.A. were performing some experiments in the Rice University. Their objective was to simulate some interstellar molecules in the laboratory. They were trying to laser ablate a graphite rod and study its mass spectrum using a mass spectrometer. Evaporation from graphite was carried out in ultra high vacuum chamber in helium atmosphere. To their great surprise they repeatedly got a mass signal due to sixty carbon atom cluster. They tried to model it nearly for a week and suddenly Kroto remembered of an industrial exhibition he had once visited. There he had seen a dome-like steel structure constructed by a very famous architect Buckminster Fuller. This dome had some pentagons and some hexagons distributed on its surface. Kroto, Smalley and Curl succeeded in constructing sixty atoms cluster model with 12 pentagons and 20 hexagons with a restriction that no two pentagons touched each other and each pentagon shared an edge with hexagon. Later it was found that one could also have carbon clusters with more number of atoms having 12 pentagons but more number of hexagons. It was in the honour of Buckminster Fuller that the name Fullerene was given to the family of this newly found cage-like carbon cluster. It is normally written as carbon atom symbol C with suffix to denote the number of atoms present. For example C_{60}, C_{70}, C_{78} etc. are 60 atoms, 70 atoms and 78 atoms carbon fullerenes. Kroto, Smalley and Curl received the Nobel Prize in 1996 for their discovery.

However it was not until 1989 that fullerenes were widely investigated. The original apparatus of Smalley group at Rice University where the discovery of fullerenes was made was quite expensive and not available at many places. Besides one could not get any samples out of this apparatus, it was only detection by a mass spectrometer. Huffman and Krätschmer when they heard of the discovery of fullerenes, wondered if they too were producing this form of carbon. They had found that they too in their experiments had some strange form of carbon. Their further work showed that indeed they too were producing fullerenes. To their great surprise with a simple set up they were producing large quantities of fullerenes. When their work was published, there started a big activity all over the world to synthesize fullerenes. It was also found that it is possible to trap some ions inside the fullerene cage or attach some functional molecules to fullerene from outside. It was realized that due to small size (diameter of C_{60} fullerene is just 0.7 nm; that is why we can consider fullerenes as 0-D material) of fullerenes, they can be used for even drug delivery. The fullerenes could crystallize in a solid form. Properties of fullerenes and their crystals are well investigated now. Smaller clusters of carbon atoms (<60 atoms) also have been investigated. They are detected in mass spectrometers but are not stable and not possible to collect. But most important is that investigations on fullerenes also led to the discovery of Carbon Nanotubes which are found to have a great technological potential.

9.4.1 Structure of Carbon Nanotubes

Carbon nanotubes can be considered as cylinders made of graphite sheets, mostly closed at the ends, with carbon atoms on the apexes of the hexagons, just like on a graphite sheet. Thus, as shown in Fig. 9.2, one can consider carbon nanotube as folding of a graphite sheet (it is only an imaginary sheet, actual growth can be different), just like one rolls a piece of paper into a cylindrical form. The difference however is that, a paper is a two dimensional solid material (area much larger, few cm^2, as compared to thickness of ~ few micrometers) and graphite sheet we are talking about has an area of few μm^2 and thickness just the atomic size of carbon atom. If we consider the rolling of graphite sheet, we can imagine carbon atoms being spread in hexagonal arrangement with some lattice strain. The lines showed connecting the filled spheres (carbon) are the bonds that exist between the atoms. Besides, during their formation, nanotubes get capped with hemispheres of fullerenes. It is also possible that many concentric cylinders may be formed as a nanotube. Such concentric nanotubes are termed as Multi Wall Carbon Nanotubes (MWCNT). The distance between their walls is ~0.334 nm. This is similar to what one gets between two graphite layers in a single crystal. MWCNT are most common and easily formed. However under certain conditions, it is possible to obtain even Single Wall Carbon Nanotubes (SWCNT). Figure 9.3 illustrates the concept of both SWCNT and MWCNT.

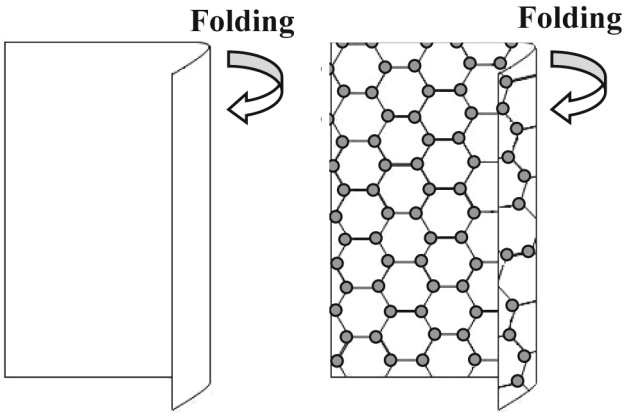
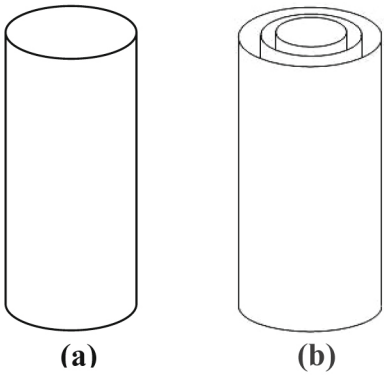


Fig. 9.2: Rolling the carbon sheet to obtain carbon nanotube.

Fig. 9.3: (a) SW and (b) MW tubes.



MWCNTs can be turned into SWCNTs using some etching methods. SWCNTs have diameters ranging from 1 to 2 nm. MWCNTs have outer diameters ranging from 2 to 25 nm. The concentrically formed MWCNTs are, however, rotationally disordered (turbotactic). Both MWCNTs and SWCNTs have their own range of applications and studied rigorously.

As the carbon nanotubes can be imagined as folding of a graphite sheet, two things are obvious: (1) carbon atoms on nanotubes are sp^2 bonded like in graphite, although some strain would be expected due to curvature and (2) there should be more than one way of folding the graphite sheet.

Indeed three types of carbon nanotubes (we will consider here only the SWCNT for the sake of simplicity) are possible viz. armchair, zigzag and helical, under appropriate conditions of growth. In order to understand the differences in these three types, consider a graphite sheet as shown in Fig. 9.4.

It is indeed possible to uniquely identify each hexagon as (a, b) with $a = 0, 1, 2, 3\ldots$ and $b = 0, 1, 2, 3\ldots$ only (b, a) is not allowed. Position of any hexagon would

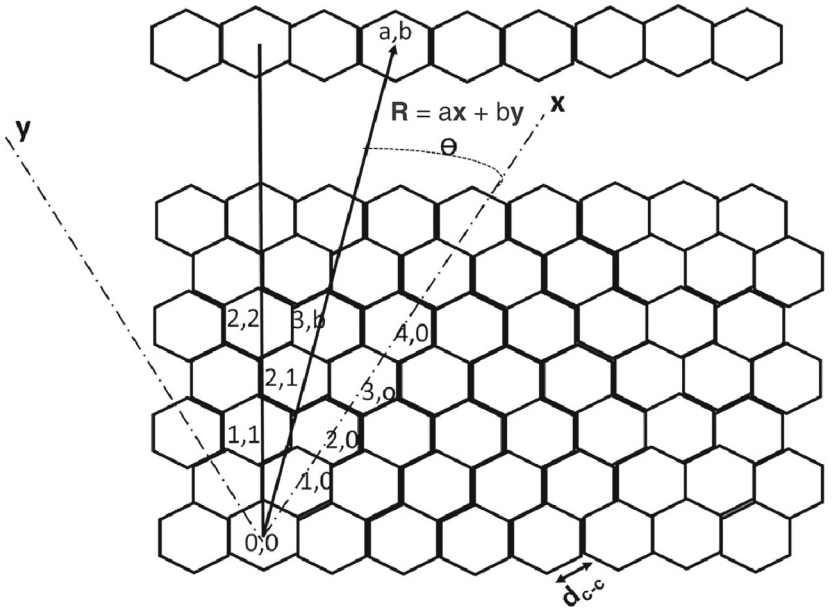


Fig. 9.4: Graphite sheet and generation of tubes.

be given by a vector **R** as

$$\mathbf{R} = a\mathbf{x} + b\mathbf{y} \tag{9.1}$$

Consider now the length of primitive vectors **x** and **y**, in terms of distance '*d*' ($d_{c\text{-}c}$) between two nearest carbon atoms. It is clear from the simple geometrical considerations that

$$\mathbf{x} = \sqrt{3}.d \text{ and } \mathbf{y} = \sqrt{3}.d \tag{9.2}$$

Vector **R** denoting the position of a hexagon is known as a 'chiral vector'. A tube obtained by folding the sheet along **R** (*a*, *b*) is called a chiral tube (*a*, *b*). An angle between x-axis and vector **R**, θ also can be used to denote the folding. It is observed that all the angles between $0 < \theta < \pi/6$ are sufficient to uniquely define different types of tubes except (*b*, *a*) is not possible. The tubes whose mirror image is identical with its own image is known as 'achiral' tube and 'chiral' otherwise. The tubes are normally terminated with the hemispheres of fullerenes (see Fig. 9.5), as was predicted by Smalley.

These are simply called as 'caps' or 'end caps'. Caps contain six pentagons (half the number in C_{60} fullerene) and different number of hexagons so that they can fit on the tubes properly. It should be remembered that in the notation used here, for a nanotube we need to have $a > b$. Diameter of the nanotube is obtained as follows

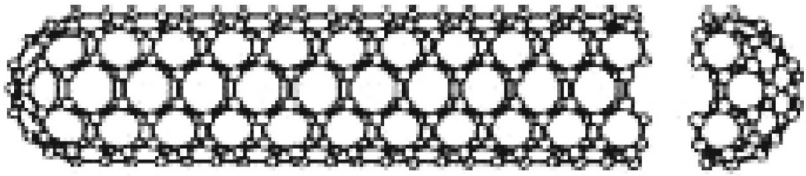


Fig. 9.5: Fullerene with end caps.

$$D = \frac{\text{circumference of the tube}}{\pi} \tag{9.3}$$

$$\begin{aligned} \text{Circumference of tube} &= |\mathbf{R}| \\ &= \sqrt{R.R} \\ &= \sqrt{(ax+by).(ax+by)} \\ &= \sqrt{3}.d_{c-c}\sqrt{\left(a^2+ab+b^2\right)} \end{aligned} \tag{9.4}$$

as

$$x \cdot x = y \cdot y = 3d_{\text{c-c}}^2 \tag{9.5}$$

$$D = \sqrt{3}.d_{c-c}\frac{\sqrt{\left(a^2+ab+b^2\right)}}{\pi} \tag{9.6}$$

where D is diameter, $d_{\text{c-c}}$ – distance between two nearest carbon atoms, and a and b are chiral lengths of vector $\mathbf{R}$.

Angle θ in terms of chiral lengths a and b is obtained as

$$\cos\theta = \frac{\mathbf{R} \cdot a\mathbf{x}}{|\mathbf{R}| \cdot |a\mathbf{x}|} \tag{9.7}$$

$$= \frac{(a\mathbf{x}+b\mathbf{y}) \cdot a\mathbf{x}}{\sqrt{3}.d_{c-c}\sqrt{\left(a^2+ab+b^2\right)}\sqrt{3d_{c-c} \cdot a}} \tag{9.8}$$

$$\theta = \cos^{-1}\left\{\frac{2a+b}{2\left(a^2+ab+b^2\right)}\right\} \tag{9.9}$$

For the angles $0 < \theta^\circ < \pi/6$ (Table 9.1).

Table 9.1: SWCNT properties compared with copper

Property	*Comparison with Copper*	*SWNT*
Density	8.9 g/cm^3	1.3–1.4 g/cm^3
Young's Modulus	130 GPa	500 – 700 GPa
Tensile Strength	262 MPa	3.6 - 4.0 GPa
Poisson ratio	0.34	0.16 – 0.29
Thermal Conductivity	401 W/m.K	2000 W/m.K
Electrical Resistivity	1.7×10^{-6} Ω.cm	~10^{-4} Ω.cm
Energy Gap	nil	~1.1 eV, Tunable

9.4.2 Types of Carbon Nanotubes

Depending upon their chirality or the way of folding as discussed above for a SWCNT, basically three types arise viz. zigzag, armchair and helical CNT.

Zigzag CNT: These are formed for $\theta = 0$ and chirality $(a, 0)$. i.e. by folding parallel to x-axis. The name zigzag has been given due to zig zag arrangement of carbon atoms that can be seen (Fig. 9.6) if cross section of the tube as shown in the figure is taken. This type of tubes are 'achiral' tubes i.e. their mirror images are similar as the original structure.

Armchair CNT: These are formed for the angle $\theta = \pi/6$ and chirality (a, a). These also are 'achiral in nature'.

Armchair structure with arrangement of carbon atoms can be seen in Fig. 9.7.

Helical CNT: These are obtained when angle θ is anywhere between 0 and $\pi/6$ and chirality is (a, b). Helical structure of CNT is shown in Fig. 9.8.

Such tubes are 'chiral' and their mirror images appear to differ from their original structure. Table 9.2 shows the comparison of three types of CNT.

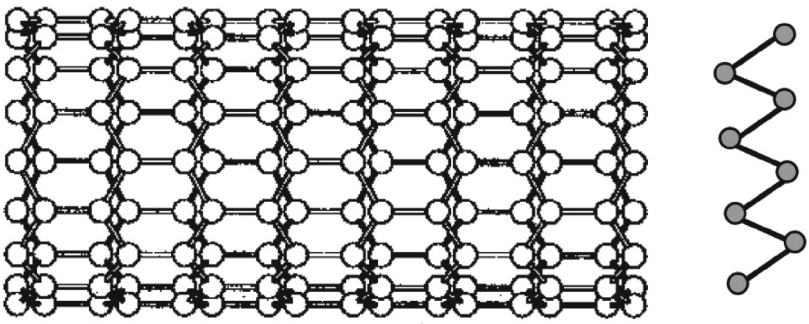


Fig. 9.6: Zigzag CNT.

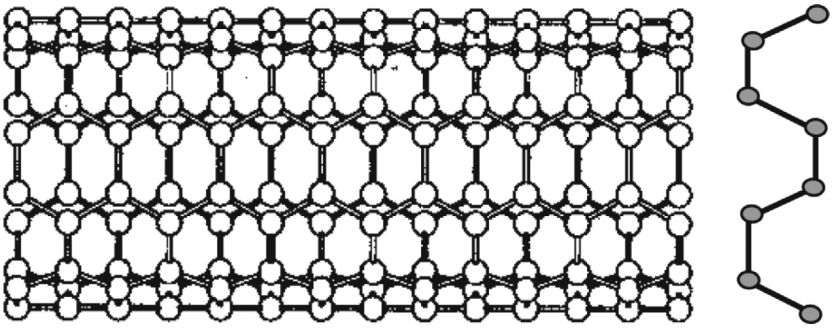


Fig. 9.7: Armchair CNT.

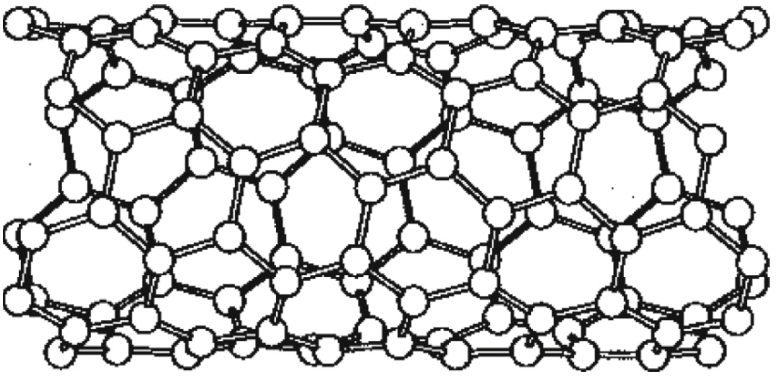


Fig. 9.8: Helical CNT.

Table 9.2: Comparison between different types of CNTs

Type of CNT	*R*	$\theta°$	*Cross Section*
Zigzag (achiral)	a, 0	0	trans
Armchair (achiral)	a, a	30	cis
Helical (chiral)	a, b	$0 < \theta < 30$	mixture of tran and cis

Table 9.3: Carbon polymorphs and their properties

Dimension	*3D*	*2D*	*1D*	*0D*
Polymorph	Diamond	Graphite/graphene	Nanotube	Fullerene
Bonding type	sp^3	sp^2	sp^2	sp^2
Bond length, Å	1.54	1.42	1.44	1.40
E_g, eV	5.47	0	1 to 0	1.9
Conductor type	Insulator	Metal/semimetal	Semicond. to metal	Semicond
Density, gm/cc	3.52	2.26	~1.2–2.02	1.7 (for C_{60})

Besides these basic types, a variety of shapes like ropes, springs, stripes, bamboo structure, conical shapes etc. are observed to have been formed of CNTs under different experimental conditions.

9.4.3 Synthesis of Carbon Nanotubes

Iijima had detected carbon nanotubes in the sample obtained from an electric arc discharge set up for synthesizing fullerenes. His nanotubes were multiwall type. The yield of nanotubes was very low compared to the fullerene content. Due to tremendous interest the scientific community took in CNTs, soon arc discharge and other methods like laser ablation and chemical vapour deposition were optimized to increase the yield and even to get SWCNTs. It is now understood that electric arc discharge mostly can produce MWCNT and laser ablation can be used for the production of SWCNT. There are also some attempts to use ion beams to obtain CNTs. However such methods are uncommon. In the following we briefly discuss the parameters and few points regarding the CNT synthesis using electric arc discharge, chemical vapour deposition and laser ablation techniques. More details about these techniques are discussed in Chapter 3.

(a) **Electric Arc Discharge**

Carbon nanotubes, in an electric arc discharge set up (see Fig. 9.9) between two graphite rods as electrodes, are formed under certain conditions as follows.

Electrodes: Graphite.
Diameter of electrodes: 5 to 20 mm.
Gap between the electrodes: ~1 mm.
Helium gas pressure: 100 to 500 torr (no CNT formed if $p < 100$ torr).
Current: 50 to 120 amperes.
Voltage: 20 to 25 V.
Cooling of the chamber is preferred.
Yield of MWCNT: ~30 to 50%

Nanotubes are deposited within few minutes in the central region of cathode. In this deposition of MWCNT it is not necessary to use any catalyst. Central region of cathode reaches a temperature close to 3000 °C and nanotubes are aligned in the direction of the current between the two electrodes. The central region of cathode is surrounded by a region in which nanoparticles, fullerenes and amorphous carbon are formed. It, therefore, becomes necessary to purify the MWCNTs to get rid of these other particles. No tubes are found to be deposited on the walls of the experimental chamber in contrast to the case with fullerenes.

Diameters of nanotubes are in the range of 0.7 nm to 1.5 nm. Incidentally, 0.7 nm is the diameter of C_{60} fullerene. Length of the tubes is typically one micrometre. However lengths in the few μm to mm range also have been possible.

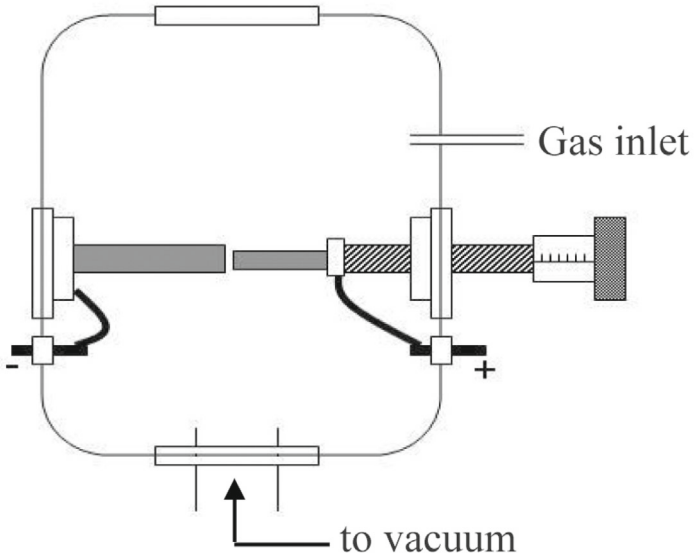


Fig. 9.9: Electric arc discharge set up.

(b) **Chemical Vapour Deposition (CVD)**

For large scale production of the nanotubes, this method is most commonly used. Here a hydrocarbon gas is cracked under certain conditions. A schematic of typical set up is shown in Fig. 9.10. Modifications are possible in the actual set up.

As there are no graphitic hexagons present in some of the precursors used to deposit carbon nanotubes, catalysts play a very important role in carbon nanotubes formation. Typical catalysts are less than 3 nm metal particles of Co, Ni, Fe, Mn, Pt and Pd. Iron-aluminum or magnesium oxide coated silicon substrates also have been used to grow high quality CNTs. 'Jungles' of vertically aligned CNTs have

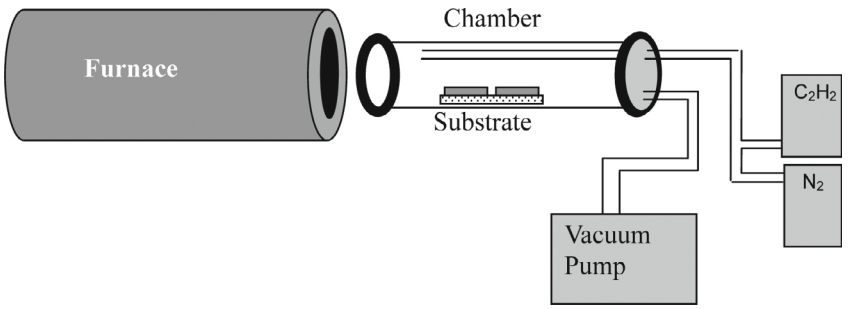


Fig. 9.10: CVD set up (schematic). The chamber and the substrates are shown outside the furnace just for the illustration.

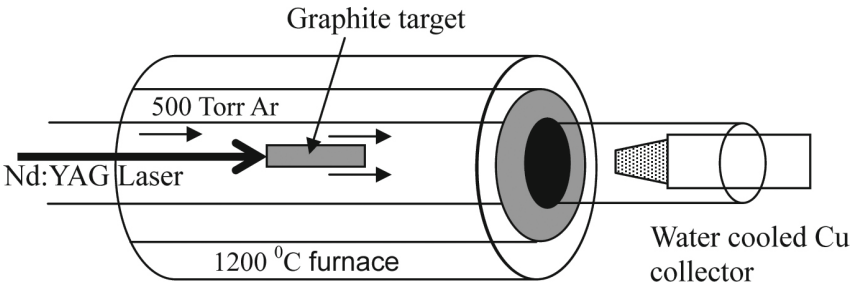


Fig. 9.11: Laser set up (schematic) to deposit CNTs.

been obtained in various types of CVD like Hot wall CVD, microwave plasma CVD, Hot Filament CVD (HFCVD) and so on. Advantage of some of these methods are: aligned nanotubes are deposited on some solid substrates (sometimes lithographically patterned substrates) which can be used for some electronics or other applications. Both MWCNT and SWCNT are possible to obtain by this method. No nanoparticles or amorphous carbon formation takes place, making high purity carbon nanotubes with high yield.

Gases: CH_4, C_2H_2, C_6H_6 (or any other hydrocarbon gas) or alcohols.
Pressure in the chamber: ~10^4 Pa.
Catalyst: Fe, Co, Ni or Pt.
Furnace Temperature: ~1000 °C can be optimized from 550 to 1200 °C.

(c) **Laser Ablation**

Schematic sketch of the laser ablation set up for CNT deposition is given in Fig. 9.11. Advantage of using laser in the synthesis of carbon nanotubes is that the nanotubes are invariably SWCNTs. Ropes of 10–20 nm diameter and lengths ~100 μm also are observed. Narrow size distribution of diameters, long lengths and high yield are some of the attractive features of this technique for synthesizing SWCNTs. However, compared to electric arc discharge and CVD techniques laser ablation is an expensive technique. Along with CNTs other forms of carbon viz. fullerenes and other carbon clusters or catalysts may be present in the samples and need to be removed. Purification techniques using magnetic field, microfiltration, heat and chemical treatments have been devised.

9.4.4 Growth Mechanism

One may wonder at this stage as to how do the nanotubes grow at all. At the synthesis temperatures as high as used in CVD (>800 °C) and electric arc deposition (~3500 °C) or laser ablation, we really don't think that long sheets of graphite are really released and folded. It is probably *atom by atom* or *molecule by molecule addition* that takes place and under certain favourable conditions nanotubes are formed. Some scientists

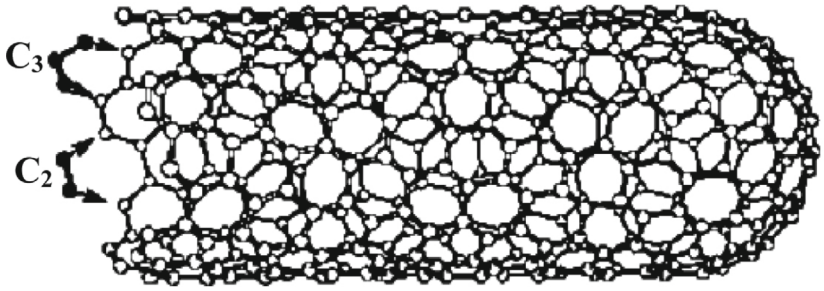


Fig. 9.12: Growth of CNT by absorption of C_2 dimers and C_3 trimers.

think that an initially formed fullerene with 12 pentagons and hexagons would open up its cage to accommodate a dimer (C_2) and/or a trimer (C_3) as shown in Fig. 9.12.

Pentagons on a fullerene are more strained and can break to accommodate a C_2 dimer or C_3 trimer. This can continue and long nanotubes be formed. Some others are of the opinion that during the growth process, nanotubes are not capped but ultimately get end caps (Fig. 9.13). Other structures with concentric tubes and different shapes also have been explained. As there are a variety of methods, temperature, catalysts and pressure which dictate the growth, it is rather impossible that a single mechanism would be responsible for the nanotube growth.

Experiments to determine the conductor type of CNT includes Scanning Tunnelling Spectroscopy, Electron Spin Resonance and ^{13}C nuclear magnetic resonance. Measurements are in agreement with the theoretical prediction that both

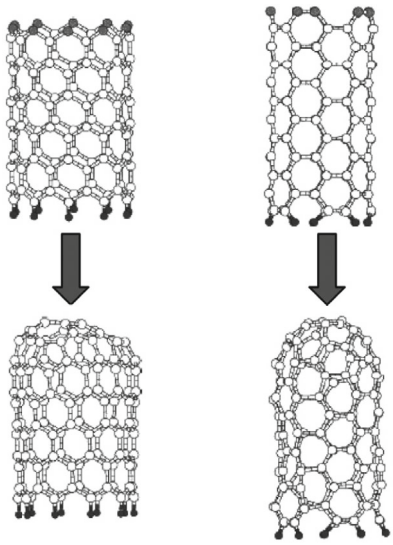


Fig. 9.13: Carbon without end cap, dimer, trimer addition for growth and spontaneous closure.

semiconductor and metallic nanotubes should be formed with above mentioned conditions.

Superconducting transition temperature T_C in CNT is as low as ~1.5×10^{-4} K. Doping can alter T_C in superconductors as well as affect metallic or semiconductor behaviour.

Carbon nanotubes, MWCNTs and SWCNTs find diverse fields of applications. Environment, nanoelectronics, optoelectronics, nanocomposites, catalysis, energy storage and medical field are some of the few areas where CNTs make an impact due to their various electrical, electronic and mechanical properties.

9.5 GRAPHENE

Until, 2004, Graphene, which is an isolated 2-D sheet of carbon atoms with honeycomb structure (see Fig. 9.14) was considered to exist only as a part of graphite or when grown epitaxially on a suitable 3-D material. This was due to theoretical predictions that 2-D nanocrystals were thermodynamically instable. There were some experimental evidences also which showed dramatic depression of melting point with reduction in the thickness of materials at nanoscale. This convinced the scientist that it would not be possible to get isolated sheet of carbon atoms. It is possible that some scientists suspected that they obtained few layers of carbon atoms or even single layers of carbon atoms in their experiments, but without much conviction and confirmation. In fact three scientists Boehm, Setton and Stumpp had coined the term 'Graphene' in 1986. However, Andre Geim and Konstantin Novoselov from the University of Manchester, U.K. were the ones who were successful in isolating single carbon sheets (monolayer thick) by simply peeling them off (mechanical exfoliation) from a graphite single crystal, using a scotch tape. They were also able to show that one could study the graphene properties as well as use them in novel applications. This was a major breakthrough in nanoscience because of the occurrence of some exotic properties in graphene and its family! Geim, Novoselov and their research group showed that a 2-D isolated sheet of graphene has an electron gas of massless Dirac Fermions. Electron transport was better explained by charge particles (electrons) with mass zero, showing properties of relativistic particles. The unusual properties of electrons could be explained using relativistic Dirac equation rather than the Schrödinger equation used otherwise (see Eq. 1.16) to explain the transport properties in solids. As shown in Fig. 9.14 (a), graphene has two sets of nonequivalent carbon atoms (represented by A and B in the figure). Honeycomb lattice of graphene has six carbon atoms and central position of hexagon lacks a carbon atom. The unit cell of graphene has two atoms per unit cell. This is also reflected in the reciprocal lattice of graphene (rhombus in Fig. 9.14 (b)), which also has a hexagonal lattice. In Fig. 9.14 (c) the hexagonal reciprocal lattice of graphene is depicted. As a consequence of graphene with in plane covalent σ bonding of carbon atoms in plane, each carbon atom is left with an electron in the π bond perpendicular to the plane of carbon atoms. These electrons are free to move in the plane of graphene.

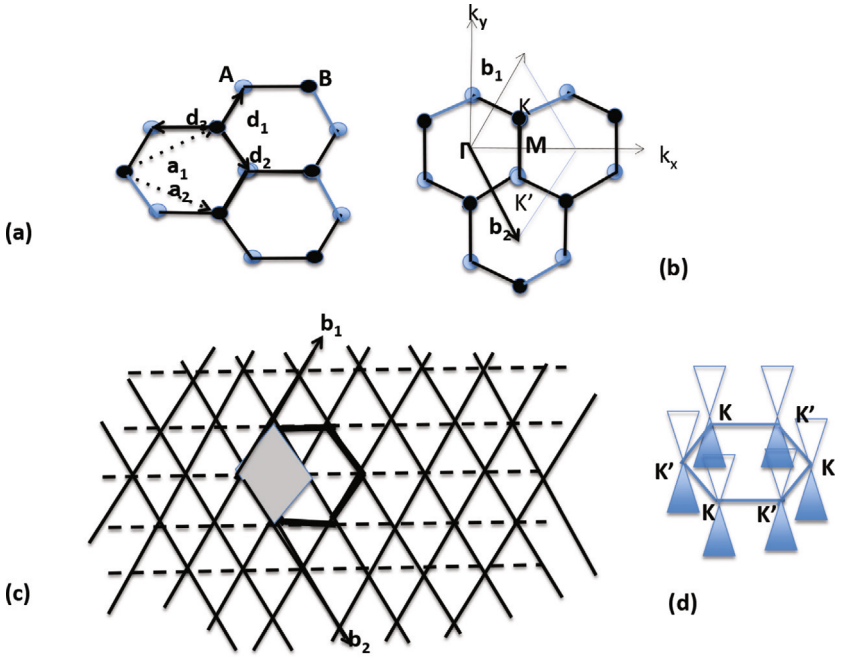


Fig. 9.14: (a) Graphene in honeycomb Structure. Note that there are two non-equivalent carbon atoms A (light) and B (dark) (b) reciprocal lattice of carbon atoms in (a). (c) A rhombus in (b) when repeated results in hexagonal structure (d) Dirac cones appear at points K and K′ at each corner of reciprocal lattice point.

Theoretical considerations (Dirac equation, not discussed here) lead to Dirac cones at six vertices of honeycomb lattice in the first Brillouin zone (schematically shown in Fig. 9.14(d). Actual shape of cone would be different with overlap of cones). The portion of the cones below the plane of the graphene (shaded) is the valence band and that above (hollow) the plane is conduction band. Interesting point to be noted here is that at room temperature the valence band is completely filled and conduction band completely empty and they touch each other. Therefore cross over of electrons is smooth. This in turn leads to ultra high conductivity (ultra low resistivity) as well as Quantum Hall Effect (QHE) observation at room temperature in graphene (See Box 9.3).

Occurrence of QHE at room temperature was unusal as many attempts so far to obtain it using other materials had failed. Moreover with a single layer of graphene, fractional QHE was demonstrated which did not exist even for two layers of carbon layers. The existence of fractional QHE lead to consider that electrons in graphene need to be treated collectively as correlated particles and not as individual particles. Graphene is thus one of the truly ‘Quantum Material’ discussed later. In 2010 Andre Geim and Konstantin Novoselov were awarded the Nobel Prize in Physics.

Box 9.3
Hall Effect & Quantum Hall Effect

The Hall Effect was discovered by E. H. Hall in 1879. Consider that electric and magnetic fields are simultaneously applied to a rectangular solid (e.g. in X and Z directions) then depending on the charge carriers in the rectangular solid a voltage appears on the faces in Y direction (as shown in Fig. 9.15 (a)). This can be understood with Lorentz force equation as

$$\mathbf{F} = e(\mathbf{E} + \mathbf{V} \times \mathbf{B})$$

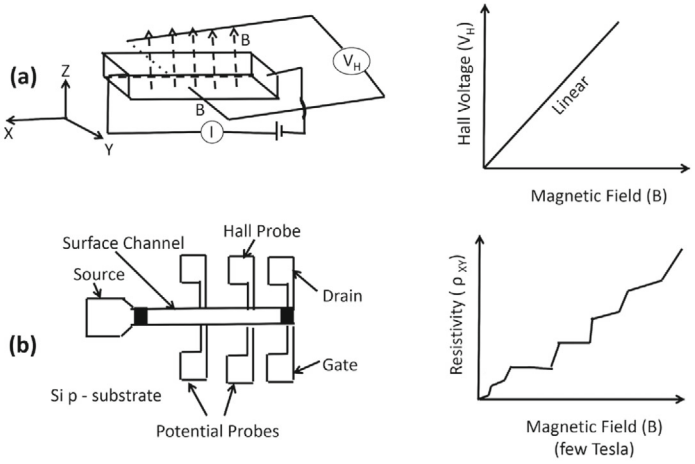


Fig. 9.15: Arrangement to measure (a) Hall effect (b) QHE/FQHE.

where $\mathbf{F}$ = Force on electric charge carrier, e = electric charge of electron, $\mathbf{E}$ = electric field, $\mathbf{V}$ = velocity of electron and $\mathbf{B}$ = Magnetic field. Charge carriers would accumulate on two surfaces in Y direction. Measuring Hall voltage across the surfaces in Y direction, it is possible to determine effective charge carrier density and the type of material viz. metal, semiconductor (n or p type) or insulator. Hall voltage is given by

$$R_H = -\frac{1}{ne}$$

where R_H = Hall coefficient, n = charge density (charge per unit volume) and e = electron charge. Obviously Hall effect measurements are very important in study of solids.

Quantum Hall Effect

Klaus von Klitzing discovered Quantum Hall Effect (QHE) in 1980, who got the Nobel Prize in Physics for the same in 1985. He found that when the thickness of the solid was very very small (2D solid), temperature is low and applied magnetic field B is vey high, then in plane resistance (R_{xy}) against magnetic field shows steps (Fig. 9.15 (b)). In plane resistance R_{xy} is given by

$$R_{xy} = h/e^2 v$$

where, h-Planck's constant, e-electron charge and $v = 1, 2\ldots$.

At each Plateau the quantized resistance is observed. In Graphene after the first plateau the Quantum Hall Effect occurs at $v + 1/2$.

Fractional Quantum Hall Effect

Robert Lughlin, Daniel Tsui and Horst Störmer observed in 1982 Fractional Quantum Hall Effect (FQHE). In very high purity thin solids (2D) with low concentration of electrons, temperature and at specific magnetic fields the 2D electron gas condenses into liquid states, which have.

$v = p/q$, p and q are integers.

but $p/q = 1/3, 2/5, 3/7$ etc.

as well as $p/q = 2/3, 3/5, 4/7$ etc.

This is a very interesting state of electrons which is only due to collective movement of charge particles and precisely controlled material as well as experimental conditions. In this situation charge may be smaller than charge of an electron or 'fractional'.

Properties of Graphene (Table 9.4, properties are compared to those of copper because of its wide use) are extremely phenomenal, not found in any other known material so far. It has extremely high thermal conductivity and electrical conductivity. Large flexibility associated to Graphene makes it suitable in applications like wearable electronics. Theoretically, Graphene has as high as 2630 m^2/g surface area making it suitable in many applications like electrodes in batteries, transparent displays, variety of sensors or even wearable electronics. This can be achieved often by making graphene nanocomposites.

It is found that graphene can be obtained using methods other than mechanical exfoliation, like chemical exfoliation and chemical vapour deposition as well as by many others which are summarized in Fig. 9.16.

9.6 GRAPHENE OXIDE (GO) AND REDUCED GRAPHENE OXIDE (r-GO)

Graphene oxide (GO) was known since 1859 when Benjamin Brody developed a process to synthesize graphite oxide, in which at certain temperature, layers of carbon in graphite had intercalated oxygen ions. Graphene, a single layer of carbon atoms

Table 9.4: Graphene properties compared with copper

Property	*Copper*	*Graphene*
Density	2.267 g/cm^3	8.9 g/cm^3
Specific Surface Area (theoretical)	~ 75 m^2/g	2630 m^2/g
Young's Modulus	130 GPa	2.4 TeraPa
Tensile Strength	262 MPa	130 GPa
Poisson ratio	0.34	0.19
Thermal Conductivity	401 W/m.K	4000 W/m.K
Electrical Resistance	1.7×10^{-6} Ω.cm	~ 0.2×10^{-6} Ω.cm
Energy Gap	nil	nil

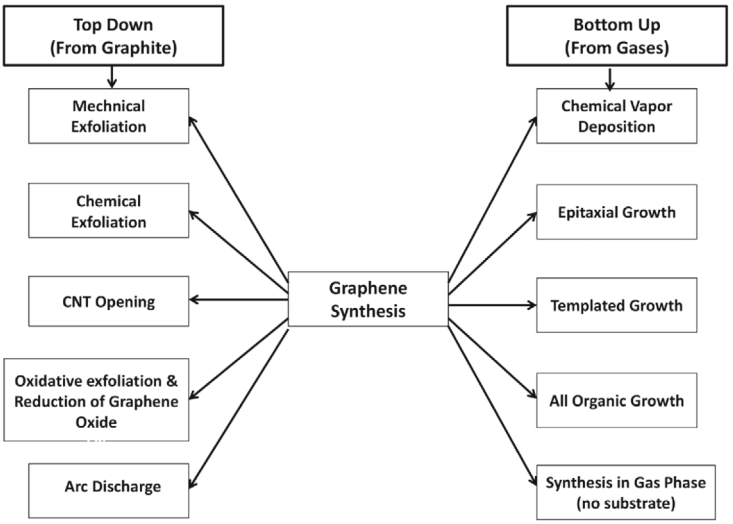


Fig. 9.16: Graphene synthesis with top down and bottom up methods.

first exfoliated in 2004 from graphite using mechanical exfoliation method and other methods was known much later. As schematically illustrated in Fig. 9.17 graphene oxide, as is known now, is a single layer of carbon atoms which has functional molecules containing oxygen, are attached on both the faces as well as edges. The number of such molecules and attached oxygen is variable. Therefore stoichiometry and exact properties of graphene oxide are difficult to specify.

Graphene oxide (GO) is obtained by treating flakes of graphite powder with potassium chlorate and strong acids of HNO_3 or H_2SO_4. This is a method proposed by Brodie, Staudenmaier and Hoffman. However, most common and quicker procedure giving better yield is due to Hummars-Hoffeman, which employs a mixture of H_2SO_4, $NaNO_3$, and $KMnO_4$. Graphene oxide is a precursor to graphene. By

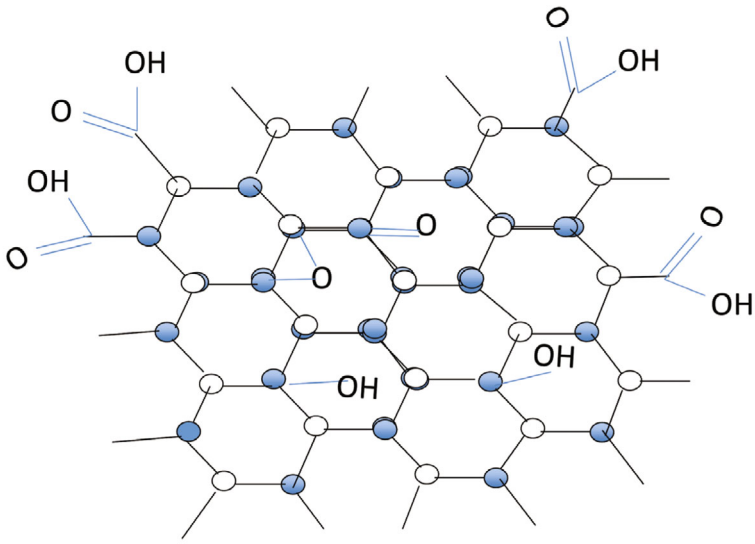


Fig. 9.17: Typical sheet of graphene oxide (number of oxygen or OH groups can change and so would the properties).

removing oxygen or oxygen containing groups one gets graphene. Main advantage of using graphene oxide is that it can be obtained using inexpensive raw materials and turns out to be useful precursor to get graphene, single layers of two dimensional layers of carbon atoms in large quantities.

Graphene oxide is soluble in water (hydrophilic) as well as highly dispersible in many organic solvents, which makes it easy to incorporate in other materials or making thin films increasing its range of applications. Graphene oxide is an insulator but by reducing it to graphene i.e. removing a number of attached oxygen or oxygen containing molecules in graphene oxide layer it is possible to reduce the band gap in graphene oxide. Indeed it is possible to remove all oxygen atoms and get pure carbon sheet, single layer, or graphene. However, such a graphene layer can be inferior to graphene synthesized using methods meant for graphene synthesis (as illustrated in Fig. 9.16). This is because during reduction process it is likely that a number of defects are produced in the carbon layer. However, reduced graphene oxide or r-GO has many applications in electronics as transparent conducting coating, energy material and many other fields. The conventional Indium-Tin-Oxide (ITO) is very expensive coating compared to r-GO due to process simplicity. Besides GO and r-GO are flexible, electrically tunable and chemically very active. Therefore they find applications in solar cells as well as flexible electronics. The application range of GO and r-GO range from electronics, photo catalysis, membranes, sensors, supercapacitors, EMI shielding to drug delivery and so on.

Like in graphene, there can be few or more layers of graphene oxide and reduced graphene oxide making either multilayers or ‘dots’ or Quantum Dots. Such particles

are found to be highly luminescent due to surface effects as well as lattice defects and have applications in optoelectronics.

9.7 GRAPHYNE

Graphyne is an allotrope of carbon with sp and sp^2 bonding of carbon atoms. Similar to graphene it can be obtained as a 2-D network of carbon atoms, rolled like carbon nanotubes or obtained as Quantum Dots less than 3–4 nm thick having few sheets of graphyne.

Structure and properties of Graphyne were first predicted by Baughman, Eckhardt and Kertesz in 1987. They named the new family of carbon atoms as 'Graphyne' because it has similarity to carbon hexagonal network connected by acetylene like bonded carbon atoms as shown in Fig. 9.18. Here we show single unit for each type that can be repeated to make a two dimensional network. The experimental evidences of this material along with properties measurements started coming around 2010 when a group of scientists G. Li, Y. Li, H. Liu, Y. Guo, Y. Li and D. Zhu reported deposition of nanoscale graphdiyne (with diacetylene link). Since then many attempts are made to obtain various graphyne films. One can use as illustrated in Fig. 9.18 (b) different acetylenic linking molecules that give rise to various graphyne 2-D materials. In 2019 Q. Li, C. Yang, L. Wu, H. Wang and X. Cui developed a mechanochemical method to obtain large quantities of graphyne. Still considerable work is going on to synthesize graphyne and various graphdiynes having good crystallinity Fig. 9.18 (a).

Mechanical stiffness of graphyne is isotropic (hexagonal symmetry) but other allotropes like 6, 6, 12 graphyne have all the properties directional. In graphyne the expected band gap is 0.48 eV. In other graphyne there are two Dirac cones at two points. Advantage of graphyne over silicon is that its band gap is direct while that of silicon is indirect. This makes it suitable as optoelectronic material.

9.8 GRAPHENE AND GRAPHENE OXIDE QUANTUM DOTS

Graphene oxide quantum dots are nanoparticles (~2 to 20 nm) of graphene with size dependent properties. As the name suggests they have few graphene sheets and are carbon analogue of semiconductor quantum dots. However properties of graphene quantum dots differ from semiconductor quantum dots as graphene quantum dots are dictated by not just the size, but edges and oxygen, which make bonds with graphene sheets. The difference in the synthesis method can decide, as depicted in Fig. 9.19 whether Quantum Dots will be of graphene or graphene oxide (or reduced graphene oxide).

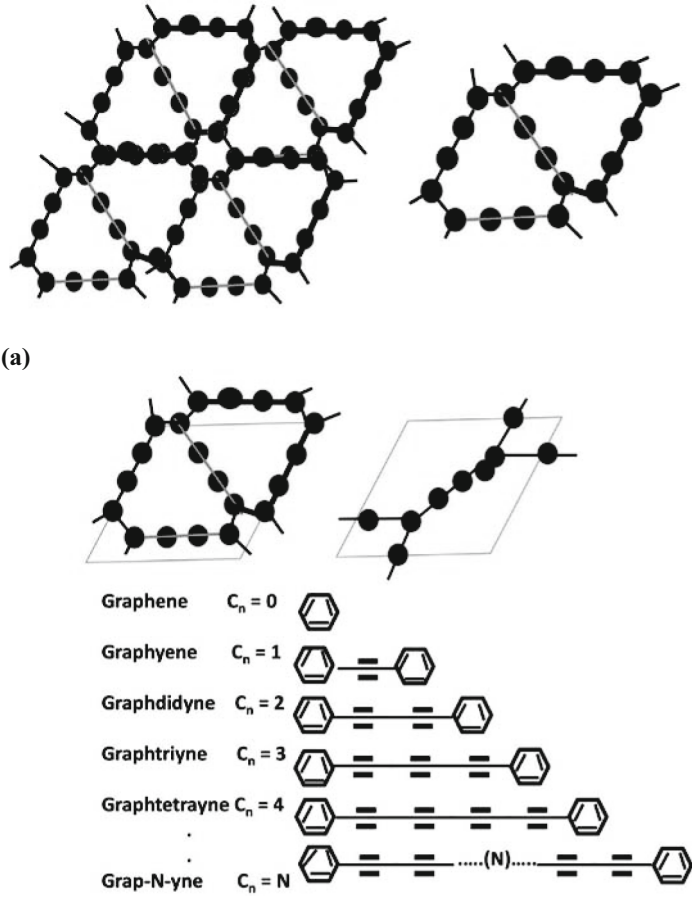


Fig. 9.18 (a): Graphyne 2-D along with the unit cell of the same. (b): Two unit cells and different molecular chains to obtain variety of graphynes.

Oxygen groups attached to edges and on surfaces are often responsible for the solubility and optical properties of the GQDs. Luminescence of GQDs is tunable from blue to red side of the visible spectrum with large quantum efficiency. Easy solubility of GQDs makes them valuable in a variety of applications. Graphene and graphene oxide quantum dots are often used in nanocomposites to achieve desirable optical or electrical properties.

There is also interest in amorphous carbon material viz. carbon dots and crystalline carbon form viz. nanodiamonds of sizes smaller than ~100 nm.

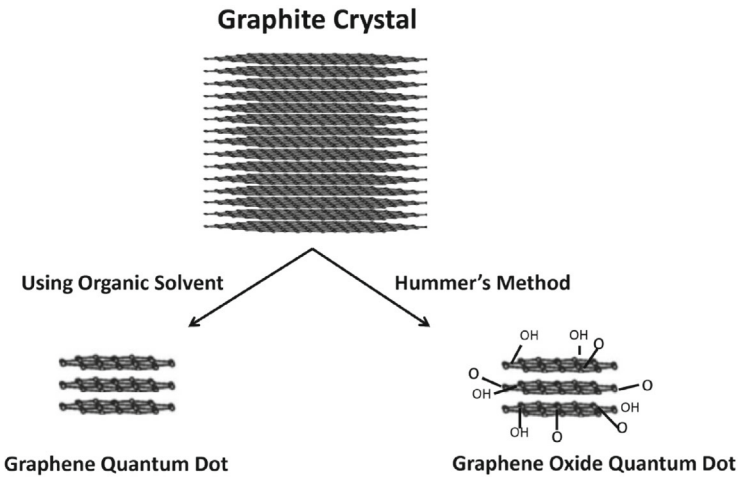


Fig. 9.19. Graphene, graphene oxide and graphyne quantum dots.

9.9 BEYOND GRAPHENE (OTHER TWO DIMENSIONAL (2-D) NANOMATERIALS)

The rise of graphene as an important and rather inexpensive material with unusually superior electronic, thermal as well as mechanical properties has led the scientists to search for other materials with the properties similar or even superior to graphene. Graphene, an atom thick layer, is metallic in nature. The question was, can one get 2-D semiconductors with tunable band gap? There are many naturally occurring layered materials besides graphite. Several layered materials are known. Can they be mechanically or chemically exfoliated or even deposited so that single or few layers are obtained?

Search for 2-D, single or few layers of materials has shown that indeed there are several possibilities. The plethora of 2-D materials (thickness of single to couple of layers) exists. Figure 9.20 illustrates how these 2-D materials are classified. It also lists type of the materials investigated so far. Some of them exhibit properties close to graphene such as presence of Dirac cone, large surface area, mechanical strength, high metallic conductivity and so on. But some show tunable band gap and are semiconductors. Therefore they have attracted lot of attention in potential energy production (solar cells, batteries etc.) and optoelectronics. Some 2-D materials find applications in medical field. It is beyond the scope of this book to discuss the properties of each and every 2-D material here. For further reading see the references given at the end of this chapter.

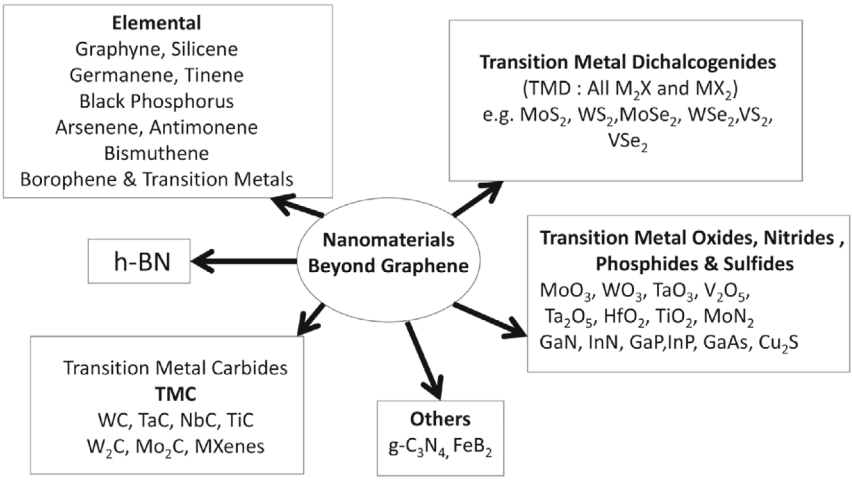


Fig. 9.20: Two dimensional nanomaterials beyond graphene.

9.10 NANODIAMONDS

Diamonds an allotrope of carbon have highest hardness and stiffness amongst all the known materials. Diamonds are scratch resistant, and microcrystal powder of diamonds is useful for grinding and polishing of softer materials. Another interesting and important property of diamonds is that they are not affected by any chemicals, hence insoluble even in very strong acids. Diamonds appear transparent and colourless in the purest form but if there are any vacancies in their crystals they can trap atoms producing beautiful fluorescent colours. The exotic extreme properties of diamonds can be explained on the basis of the sp^3 bonding of carbon atoms and face center cubic (fcc) crystalline structure. In the natural form they are found deep in the earth's mantle as they need extremely high pressure and temperature to get formed. Figure 9.21 depicts the phase diagram of graphite, diamond and nanodiamonds, materials of large interest now. Plasma Enhanced Chemical Vapour Deposition (PECVD), Catalytic High Pressure High Temperature (HPHT) or simple HPHT and Detonation (in water) are some of the techniques used to produce nanodiamond as shown in the figure. HPHT ball milling of microdiamonds also is a useful technique to obtain nanodiamonds. Though somewhat expensive Laser ablation of graphite also can produce nanodiamonds. Properties, specially surface properties of nanodiamonds are largely dependent on the method of synthesis (environment).

Large and microdiamonds are not soluble in water or any other liquid, nanodiamonds, diamonds in the size range ~5-100 nm have structure, mechanical, thermal and electrical properties close to bulk diamond. However their surfaces are rather different i.e. there can be adsorption of hydroxyl, SO^{4-}, Cl^- and others which makes them soluble in various liquids. These properties make them useful in drug delivery

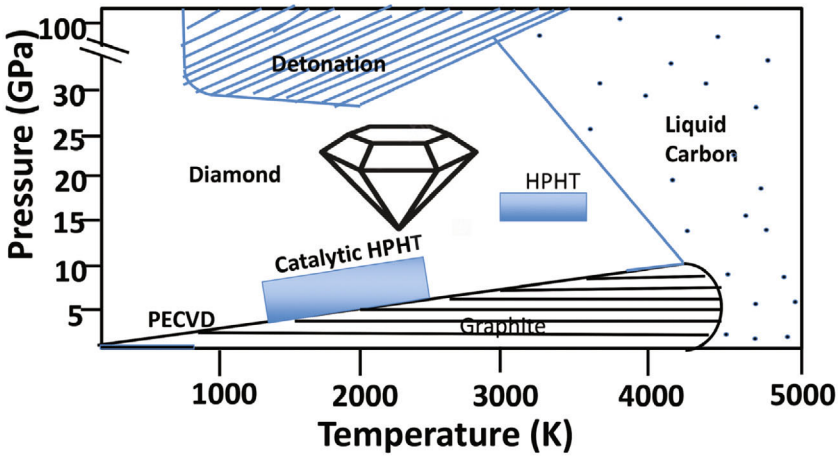


Fig. 9.21. Phase diagram of graphite, diamond and nanodiamonds (Plasma enhanced chemical vapour deposition (PECVD), Catalytic High Pressure High Temperature (HPHT), HPHT and Detonation).

and clinical diagnosis. Nanodiamonds are biocompatible and nontoxic which makes them useful in various biotechnological applications. Surface to volume ratio being large like any other nanomaterials, surface defects along with any point defects (vacancies) make nanodiamonds highly fluorescent. Adsorption of nitrogen atom near a vacancy in nanodiamonds (denoted as NV) is responsible for producing strong fluorescence. This enables applications of nanodiamonds even for sensing electrical field, magnetic field and Optically Detected Magnetic Resonance (ODMR).

FURTHER READING

The Science and Technology of Carbon Nanotubes by K. Tanaka, T. Yamabe and K. Fukui. Elsevier (1999).

Carbon Dots, Junjun Liu, Rui Li and Bai Yang. *ACS Cent. Sci.*, **6(12):** 2179-2195 (2020).

Exploring Two-Dimensional Materials toward the Next-Generation Circuits: From Monomer Design to Assembly Control, Mengqi Zeng, Yao Xiao, Jinxin Liu, Kena Yang and Lei Fu. *Chem. Rev.*, **118(13):** 6236-6296 (2018).

Graphene synthesis, Keith E.Whitener Jr and Paul E. Sheehan, *Diamond and Related Materials*, **46:** 25–34 (2014).

Chapter 10
Novel Nanomaterials

10.1 INTRODUCTION

In Chapter 9 we discussed a rich variety of carbon nanomaterials. We have also considered the theory of Quantum Dots and Plasmonic nanomaterials with some examples in chapter 8. Yet there remains a huge variety of nanomaterials which have been the result of the efforts by many scientists. In this chapter we discuss some Porous Materials, Core–Shell particles, Metamaterials, Quantum Materials and Bioinspired Materials.

10.2 POROUS MATERIALS

There is a large variety of porous materials with pore size of few nanometers to micrometres. Porous materials are classified as microporous, mesoporous and macroporous materials based on the pore sizes. Materials with pore diameters less than about 2 nm are called microporous materials and those with larger than about 50 nm size are known as macroporous. The pores of intermediate sizes i.e. between 2 to 50 nm are referred to as mesoporous materials. There exist some natural porous materials like soil, snow, sandstone, corals, bones, wood and so on. Several examples of manmade porous materials around us are bread, biscuits, sponge, brick, chalk etc.

In this section we shall discuss some of the artificially synthesized porous materials. These materials have emerged over a long period and have considerable applications from drug delivery, energy storage to space science. Due to their porous nature in general, all types of porous materials inherently have lower density and larger surface area (includes surface area of pore walls) than the corresponding bulk materials.

10.2.1 Porous Silicon

Silicon is the most widely used semiconductor material by electronics industry. It is abundant in nature and techniques to purify, grow single crystals economically and on large scale are well developed. The methods of doping and polishing also have been perfected. Oxide of silicon is stable and metal-silicon contacts can be made and understood to good extent. Thus microelectronics industry is well established to use silicon in making various components and systems like diodes, transistors, Field Effect Transistors (FETs), Metal Oxide Semiconductor FETs (MOSFETs), Integrated Circuits (ICs), Very Large Scale Integrated Circuits (VLSIs) and Mechano-Electrical Machines (MEMS). However, there is a basic drawback with silicon viz. unlike some other semiconductors like ZnS, CdS, GaAs, InAs and InP, light emission capability of silicon is extremely poor. Therefore it is not possible to make light emitting diodes (LED) or lasers using silicon wafers that are usually used in electronics. It was thus considered for a long time that silicon cannot be an optoelectronics material.

However in 1990, Canham showed that porous silicon can be very easily and routinely formed on silicon wafers, used by electronics industry, to emit photoluminescence in the visible range at room temperature. He further showed that silicon nanocrystallites formed, as a result of pores formation, were responsible for light emission and as the crystallite size reduced, the emission wavelength also decreased. This triggered the wave of enthusiasm in the scientists and technologists, as it was a major breakthrough for silicon industry to use silicon as an optoelectronic material. As a result, considerable work has been carried out since 1990 to make porous silicon by number of ways, use different substrates, etchants etc. and investigate the various structural, optical and electrical properties and demonstrate new applications. Even other semiconductors like GaAs also have been made porous and investigated. It should be remembered that scientists were using for quite a long time the procedures, specially the electrochemical polishing, similar to that used in porous silicon formation and some interesting properties were already noticed. However, prior to Canham's publication, emphasis of early work was on studying the properties of thin film observed during etching or electropolishing of silicon. Detailed investigations showed that such films were porous and since 1970 the word 'porous silicon' was being used. Canham showed the nanocrystalline nature of the pore walls and its role in luminescence that became interesting and decisive to consider porous silicon as an optoelectronic nanocrystalline material. Here we will discuss, how porous silicon can be made, its properties and briefly some applications.

Box 10.1
History of Silicon Etching

- 1956 – A. Uhlir at Bell Labs. USA [ref. *Bell Syst. Tech. J.* **35** (1956) 333] observed while polishing silicon by electrochemical method that at high current density polishing occurred but at low current density surface looked as if there was some blackish, brownish or sometimes reddish deposit. He attributed this to the formation of some silicon suboxides
- 1958 – D.R. Turner [ref. *J. Electrochem. Soc.* **105** (1958) 402; Surface Chemistry of Metals & Semiconductors (H.C. Gatos, ed.), Wiley, New York (1960); The Electrochemistry of Semiconductors (P.J. Holmes, ed.) Academic Press, New York, Vol. 155 (1962)] attributed anodic film formation to $(SiF_2)_n$ polymeric network with predominance of silicon. He considered that the formation of anodic regions depends upon whether oxidizing or reducing species attack a certain region. When HNO_3 attacks, it removes an electron from the surface making it a local cathode. The positively charged hole is created as a result of loss of electron. Hole can wander around and towards Si where HF attacks. It is then ready to form an anodic region. Silicon complexed with fluorine, dissolves into the solution forming a pit.
- 1959–1956 – Robins and Schwartz used HNO_3 and HF solution for etching silicon. In a series of papers [*J. Electrochem. Soc.* **106** (1959) 505; ibid. **107** (1960) 108; ibid. **108** (1961) 365; ibid **123** (1976) 1903], they discussed that process of etching first involved the formation of silicon oxide and then its removal by HF. They, however, did not mention the formation of any porous film.
- 1960 – R.J. Archer [*J. Phys. Chem. Solids* **14** (1960) 104] mentions porous silicon formation and observation of silicon hydride formation.
- 1965 – K.H. Beckman [*Surf. Sci.* **3** (1965) 314] studied infra red spectrum of porous silicon and showed that polymeric network of $–SiH_2$ existed on the etched silicon surface. If electrochemically prepared SiH_2 gets connected through Si–Si bonds and if chemically prepared, the network is connected through Si–O-Si bonds.
- 1972 – M.J.J. Theunissen [*J. Electrochem. Soc.* **119** (1972) 351] clearly showed for the first time that porous silicon had crystalline nature.
- 1983 – G. Bomchil, R. Herino, K. Barla and J.C. Pfister [*J. Electrochem. Soc.* **130** (1983) 1611] measured surface area and sizes of pores in various lightly and heavily doped p and n type silicon samples.
- 1984 – C. Pickering, M.I.J. Beale, D.J. Robbinson, P.J. Pearson and R. Greef [*J. Phys.* **C 17** (1984) 6535] carried out optical studies of etched Si. Refractive index of p^+-Si varied from 3.47 towards 1.0 with increasing porosity. They observed the presence of complex amorphous alloy of Si, H and O along with crystalline Si in the porous film. They also mentioned the observation of an intense band gap luminescence at 4.2 K as well as orange red luminescence. They attributed it to amorphous SiHx formation. Further by changing H concentration, room temperature photoluminescence also was observed.
- Rutherford Back Scattering (RBS) and nuclear measurements of porous silicon showed that it was crystalline and surface was covered with 50% H, 5% O, 1% F and 3% C (mostly as contaminant). SiH formation was concluded.

- 1988 – Work on porous silicon was reviewed by G. Bomchil, A. Halimaoui and R. Herino [*Microelectron. Eng.* **8** (1988) 293]. Even some applications of porous silicon for semiconductor on insulator (SOI) appear but optical studies were neglected.
- 1990 – Canham [*Appl. Phys. Lett.* **57** (1990) 1046] proposed that quantum confinement took place as a result of nanocrystallites formation in porous silicon. Photoluminescence was observed from porous silicon at room temperature and its wavelength depended upon the size of nanocrystallites produced by etching. This observation of Canham can be considered as the most important step in porous silicon research

How to Make Silicon Porous?

Surface of solid silicon can be turned into porous layers by a variety of methods like ion irradiation, spark erosion, chemical etching or electrochemical etching. However, many of the investigations show that most effective method of making porous silicon is electrochemical etching. Chemical etching involves dipping the crystalline silicon in some strong etchant solution like HF mixed with HNO_3. The reaction responsible for making pores begins autocatalytically. Reproducible and controlled porosity in relatively short time can be obtained using electrochemical method and is described here.

Electrochemical Etching

Porous silicon can be easily formed with appropriate conditions at room temperature in a simple electrochemical cell. There are various designs that can be used to obtain anodic etching of silicon but a schematic one is shown in Fig. 10.1. A crystalline piece from a silicon wafer is dipped in an electrolyte, usually a solution of HF, water and ethanol. By making silicon a working anode and a piece of graphite or platinum as cathode a current is passed through the solution. Typical parameters used to etch a *p*-Si are shown in Box 10.2. As HF (hydrofluoric acid) is used as an etchant, precaution has to be taken to use Teflon or polyethylene container as HF severely attacks glass. *It also should be remembered that although the experimental procedure is very simple, HF is extremely dangerous to use. If HF comes in contact with the skin, fluorine makes entry through the skin to the bones affecting the limbs.* CaF_2 *is formed which mixes with the blood, poisoning it and can turn out to be fatal. Even the waste electrolyte needs to be handled properly, otherwise due to contact with some chemicals it can even explode.*

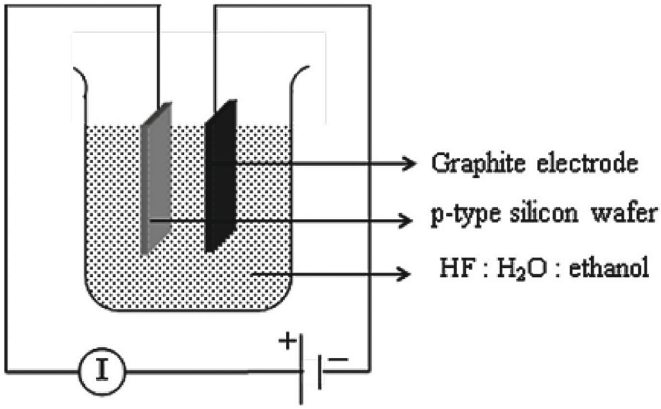


Fig. 10.1: Electrochemical cell for making silicon porous.

Box 10.2
Typical parameters in electrochemical reaction used for making porous silicon

Substrate: *p*-Si(111)(B doped) Voltage: 24 V
Current: 30 mA Solution: HF + ethanol + water

Mechanism of Pores Formation

Mechanism of pores formation on silicon surface is a complex process. Attempts are made to understand them from the observed species formed on the surface.

For the formation of a porous network on silicon surface, it is necessary that local cathode and anode regions be formed on its surface. When an electron is extracted into the electrolyte from certain region of silicon, usually some sharp point acts like a local cathode. Release of an electron is equivalent to the production of a positively charged hole. The hole can migrate to some other region on the surface, which becomes an anode. If such a region is also attacked by HF present in the electrolyte, following reaction becomes feasible.

$$2h^+ + Si + 2F^- \rightarrow SiF_2 + \left(2h^+ + 2e^-\right) \rightarrow SiF_2 \tag{10.1}$$

$$\left(h^+ - \text{hole charge}, e^- - \text{electron charge}\right)$$

$$SiF_2 + 2HF \rightarrow SiF_4 + H_2 \tag{10.2}$$

$$SiF_4 + 2HF \rightarrow H_2SiF_6 \tag{10.3}$$

H_2SiF_6 dissolves in the electrolyte forming the initiation of pits. It can be seen from above equations that two holes are necessary for the formation of pores. At some low current value and HF concentration such conditions are created.

As the current density is increased beyond a certain value, it is possible to generate more holes at a time and chances of forming surface oxide are increased. If following reaction becomes possible then it is expected that polishing would occur rather than porous structure.

$$4h^+ + Si + 4(OH)^- \rightarrow (SiOH)_4 + \left(4h^+ + 4e^-\right) \tag{10.4}$$

$$(SiOH)_4 \rightarrow SiO_2 + 2H_2O \tag{10.5}$$

$$SiO_2 + 6HF \rightarrow H_2SiF_6 + 2H_2O \tag{10.6}$$

H_2SiF_6 and water dissolve in the electrolyte. Obviously the h^+ and e^-, already present in the silicon substrate as a result of doping, would greatly influence the pores formation. Specially, it would matter considerably, whether it is a p-type (excess h^+) or n-type (excess e^-) silicon substrate being used for pores formation. It should not also be surprising if doping level—whether low concentration, moderate concentration or extremely high concentration—would alter the nature of pores formed. Depending upon the concentrations of excess charges, either thermionic or tunnelling of charge carriers would be effective.

High porosity of a material means large area of the exposed material to the ambient. Therefore porous silicon can easily get oxidized. The optical properties of porous silicon are found to be unstable after their formation. However they are reversible too.

Factors Affecting the Porous Structure of Porous Silicon

From the vast literature available now on porous silicon, few trends are noticed as follows.

The morphology, optical and structural properties of porous silicon depend upon (1) Substrate type (n or p), (2) Doping level (low or high), (3) Concentration of HF in the electrolyte, (4) Current density used to etch silicon, (5) Effect of light, and (6) Duration of etching.

Box 10.3
Porous Material

Materials become porous when some solid portion is removed from them leaving voids in them. The volume of the original material does not alter. Therefore density of a porous material is lower than the original material. Materials can be made even 98% porous. There is a large variety of porous materials, natural (e.g. wood) as well as synthetic (e.g. thermocole). The pores can be of regular, irregular and different sizes

and shapes. Pores can be closed, open, penetrating or ink bottle type. Internal area increases with pores (Fig. 10.2).

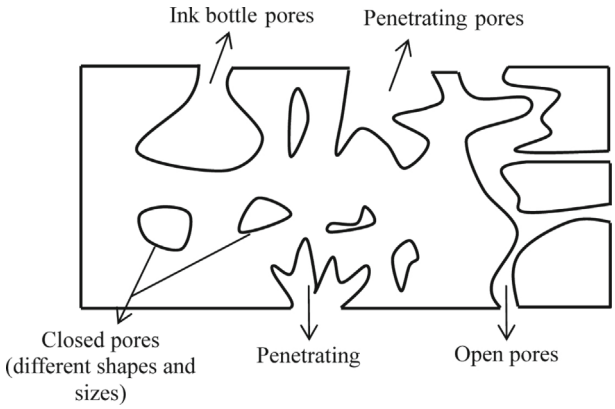


Fig. 10.2: Various types of pores.

In general, properties of porous materials are quite different than those of corresponding bulk materials. Porous materials are often divided into microscopic (less than 2 nm), mesoscopic (2 to 50 nm) and macroscopic (larger than 50 nm) depending upon the diameters or size of the pores.

1. **Substrate Type:** Pore structure is normally formed due to holes and silicon atoms interacting together with fluorine. As was discussed in the mechanism of pore formation, it is necessary to consider the substrate type viz. p or n type as they already have excess holes or electrons due to doping. For example it has been found in case of moderately doped n or p type silicon that columnar or sponge-like structure is respectively formed.
2. **Doping Level:** One can identify four types of silicon wafers for forming pores on their surfaces by electrochemical etching. They are: p^+-Si (very high concentration of holes), p-Si (moderate or low concentration), n^+-Si (high concentration of electrons) and n-Si (moderate or low concentration of electrons). It is found that p, p^+ and n^+-Si substrates have similar trends in their pores formation, but n-Si needs large voltage to form pores.
3. **Concentration of HF:** If the current density is held constant, concentration of HF in the electrolyte is an important factor in determining pore formation. Although porosity has been found to increase with decrease of HF concentration, it also is a function of current density. At large current density, usually electropolishing takes place as schematically shown in Fig. 10.3. In the transition region, a rough surface is produced. At lower concentration of HF, the pH of the solution increases, which promotes pore formation.

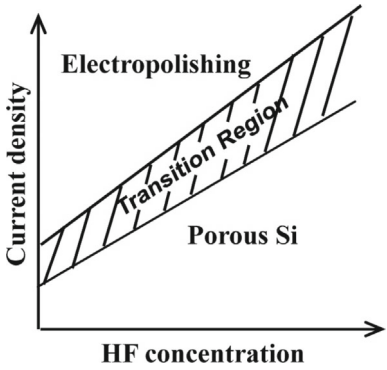


Fig. 10.3: Different regions of silicon surface morphology, which depend upon current and HF concentration in electrochemical etching of Si.

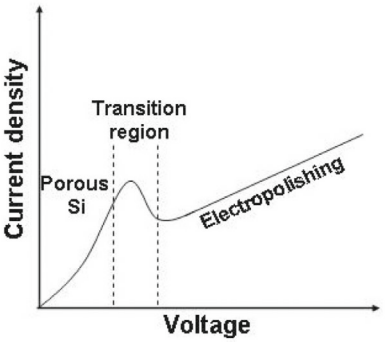


Fig. 10.4: Region of silicon morphology which depends upon current density and the applied voltage in the process of electrochemical etching of silicon.

4. **Current Density:** Fig. 10.4 shows three different regions viz. a region of pores formation, an intermediate region and electropolishing region, similar to that in Fig. 10.3.

For a given electrolyte, it can be seen that porous structure can be formed only at low voltages and low current densities. The curve may differ for n-Si substrate quite considerably.

5. **Effect of Light:** Chemical reactions can be often controlled using light. When light of appropriate wavelength falls on a semiconductor, it generates electron–hole pairs. If intensity of light is adequate, a large number of charge carriers necessary for the pores formation can be generated using light. Thus the illumination of the silicon anode during electrochemical reaction can affect pores formation. It is found that p-Si can etch out in dark but for n-Si illumination by photons is necessary for pores formation. Sometimes, ambient light can be sufficient.
6. **Duration of Etching:** Etching creates pores which are surrounded by the silicon walls or pillars. These walls contain nanocrystallites of silicon as was first postulated by Canham. Longer the etching time, smaller are these crystallites. The

photoluminescence which is observed in porous silicon is due to the formation of these nanocrystals. Therefore properties of the porous silicon are affected by the crystal size and the etching time.

Properties of Porous Silicon

Morphology

Depending upon the substrates used for etching, the morphology of porous silicon can change. In general, it is found that 'nanopores' produced with sizes less than ~10 nm (difficult to image) when p-Si with low doping level are used. On the other hand if n-Si substrates are etched using large current, pores even larger than ~1 μm size are easily formed. Medium sized (mesopores) pores ~10 to 100 nm can be formed in heaviliy doped p^+ or n^+ type silicon. Porosity, of course, will go on increasing with etching time and typically 50 to 90% porosity can be achieved. Shapes of pores also can be altered using different substrates. Often circular, square, triangular, star or random shapes and non-homogeneous are observed. Heavily doped p-Si substrate has columnar structure but lightly doped p-Si has spongy surface structure. Heavily doped n-Si substrate also has columnar porous structure. Figure 10.5 shows an SEM image of porous silicon.

Structure

Structure of porous silicon is quite complex. Nanocrystallites of silicon are usually embedded in amorphous silicon. However, in order to produce porous structure, it is necessary to use crystalline silicon. Attempts to use amorphous silicon substrates have failed to produce porous silicon.

Chemical Nature

Surface of porous silicon alters chemically during the etching process. It is non-homogeneous on microscopic level and different researchers often obtain conflicting

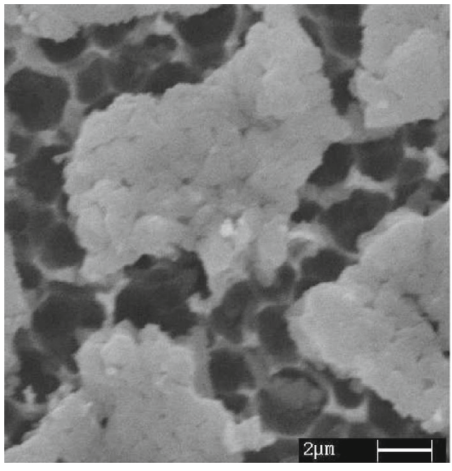


Fig. 10.5: SEM image of porous silicon.

results. This is because of the use of different substrates and variations in electrochemical parameters. In general, along with Si, one finds F, H and O to be present. The relative amounts, however, differ from experiment to experiment. Sometimes carbon also can be detected which arises as a contaminant from the solution or ambient. Samples left in the laboratory environment changes in H or O content with time, also alter the electrical or optical properties. Networks of SiH_2, SiO_2 or complex Si-O-H are often reported.

Electronic Structure

Electronic structure of porous silicon differs considerably from that of the bulk, crystalline silicon. For the bulk silicon crystal the band gap is 1.17 eV at absolute zero and is 1.125 eV at room temperature. This band gap is an indirect band gap as schematically shown in Fig. 10.6.

This is because the maximum in the valence band occurs at $\mathbf{k} = 0$ or Γ (gamma) point in Brillouin zone. The minimum in the band is closer to Brillouin zone boundary. An electron from valence band, even if excited using a photon with sufficient energy, cannot go to conduction band minimum, as both energy and momentum must be conserved in an absorption or emission process. Thus an electron in conduction band minimum at $\mathbf{k} \neq 0$ also cannot make a transition to valence band state at $\mathbf{k} = 0$. This is why Si is not an optoelectronic material. However, as soon as Si nanocrystallites are formed, the electronic structure alters dramatically.

As there is no long range periodic arrangement, the small momentum conservation rule breaks down and transitions become allowed. According to the Effective Mass Approximation (EMA) theory, energy gap increases compared to that in the bulk and depends upon the magnitude of the nanocrystallite size.

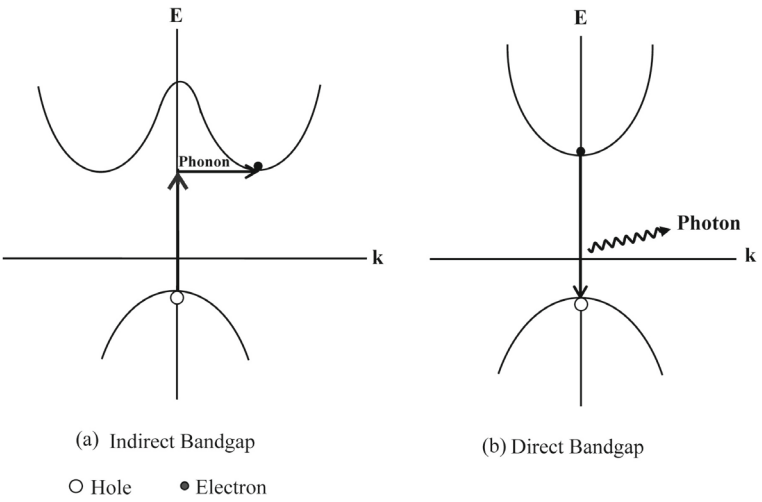


Fig. 10.6: Schematic band diagrams (a) silicon bulk and (b) Nanocrystalline silicon.

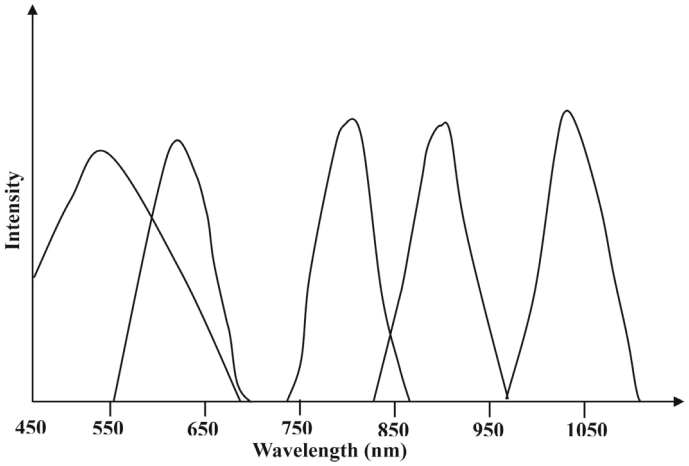


Fig. 10.7: Photoluminescence spectra for different porous silicon samples at room temperature.

Photoluminescence

Depending upon the preparation conditions and the nanocrystallite size porous silicon can exhibit photoluminescence over the entire visible range as illustrated in Fig. 10.7.

Due to wide distribution of nanocrystallites, luminescence peak widths can be large.

10.2.2 Aerogels

Aerogels constitute another class of highly porous materials. These ultra low density, monolithic materials are formed by interconnection of particles of nanometer size, to form a nanoporous solid. The pores themselves are usually nonuniform with sizes from ~10 to 100 nm. These materials are synthesized by sol–gel method (discussed in Chapter 4) and dried by special procedures to retain their porous structure. Mixture of reactants forming colloidal particles, which are dispersed in liquid, is known as 'sol'.

Box 10.4
Brief History of Aerogels

Aerogels were first synthesized by an American scientist Samual Kistler in 1930s. He was working in Stanford University on puzzle of cracking of gels while drying. It was found that gels were made up of network of interconnected solids dispersed in some solvent (water). Drying of gels led to evaporation of this solvent and surface tension exerted by evaporating liquid collapsed the solid network. It was realized that while drying gels, they shrink and develop cracks. Kistler developed, therefore, crack-free drying method by removing the solvent above its critical point. Dried materials thus obtained were crack-free and retained their original size and shape. Kistler replaced water in the gel (which has very high critical temperature and pressure) with alcohol and removed it by putting it in an autoclave (a vessel able to withstand high temperature and pressure, just like a pressure cooker, but equipped with temperature, pressure measurement gauges) and raising its temperature and pressure above critical point of the alcohol. Above critical point, alcohol is a gas-like fluid which can be removed from the solid network without any surface tension. He named that transparent, low density solid 'aerogel'. These first aerogels synthesized by Kistler required very long (weeks) time consuming process. In 1970s, S. Teichner at University of Lyon, France, wanted to use these highly porous materials for storing rocket fuel. Looking at lengthy and laborious process he tried to develop a simpler and faster process to synthesize aerogels. He used tetra methylorthosilicate (TMOS) and hydrolyzed it in methanol, in presence of some catalyst. He could get gel in one step and exchange of solvent with alcohol was not required. He could synthesize aerogels in one day (Fig. 10.8).

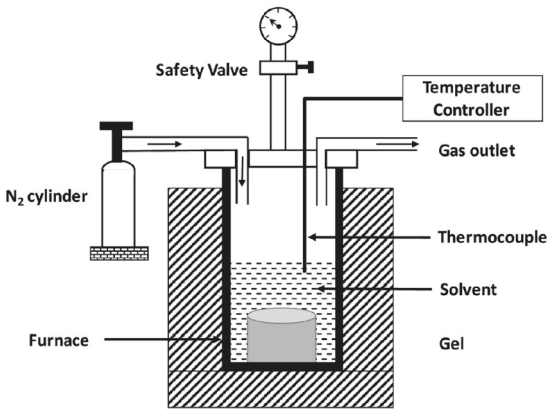


Fig. 10.8: Schematic diagram of an Autoclave (for the photograph, see Fig. 4.33).

These colloidal particles aggregate to form a continuous three dimensional network, which occupies total volume of the solution. The viscous semi-fluidic solid hence formed is known as 'gel'. Formation of gel from solution is dependent on various parameters such as reactant concentration, temperature, pH value etc. In such

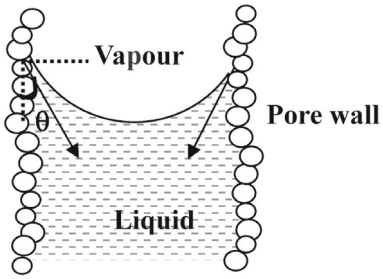


Fig. 10.9: Surface tension at liquid vapour interface.

gels, liquid is filled in the pores (empty space between the networks of solid particles). In simple evaporation, liquid and vapour coexist within the pore and surface tension of liquid at liquid vapour interface causes collapse of the network (as shown in Fig. 10.9). Magnitude of this interfacial force is given by Eq. (10.7).

$$P_s = \rho g h = \frac{2\sigma \cos \theta}{r} \tag{10.7}$$

where ρ is density of liquid, g – acceleration due to gravity, h – height of the capillary, σ – surface tension of the liquid, r – radius of the pore, θ is contact angle of the liquid with the pore wall.

Hence liquid has to be evaporated in such a fashion that network of particles does not collapse. One way to achieve this is by supercritical drying in which drying of gels is carried out above the critical point of liquid present in pores. Above critical point surface tension is zero. Whole liquid is present in vapour state and can be removed from gels with out rupturing the network. In recent years it was realized that it is also possible to dry gels in ambient condition. This method is known as sub-critical drying. A comparison of sub-critical and super-critical drying procedure is shown in Fig. 10.10. In this case to overcome the interfacial pressure (surface tension), various techniques such as strengthening of gel by aging it in suitable solvent or use of some polymers or template to produce uniform size pores (to avoid pressure gradient while drying) etc. are used. Only drawback of this process is that it is time consuming and aerogels obtained by this method have higher density as compared to those dried by supercritical drying. Another way of drying is by freezing the liquid in the pores and sublimating by vacuum pumping. This method is known as freeze drying.

Types of Aerogels

Kistler had successfully demonstrated aerogel formation of various materials such as silica, alumina, nickel tartarate, stannic oxide, tungstic oxide, gelatin agar, cellulose egg albumin etc. Inorganic aerogels such as SiO_2, TiO_2, ZrO_2 and Al_2O_3 have interesting properties and applications. It is also possible to synthesize mixed oxide aerogels such as SiO_2:TiO_2, Al_2O_3:SiO_2, SiO_2-ZrO_2 etc. Fascinating properties and applications of inorganic aerogels motivated also the development of organic aerogels such as RF (Resorcinol Formaldehyde) and MF (Melamin Furfural). They were synthesized by using different organic precursors such as Phenol-Furfural, Polyurethane,

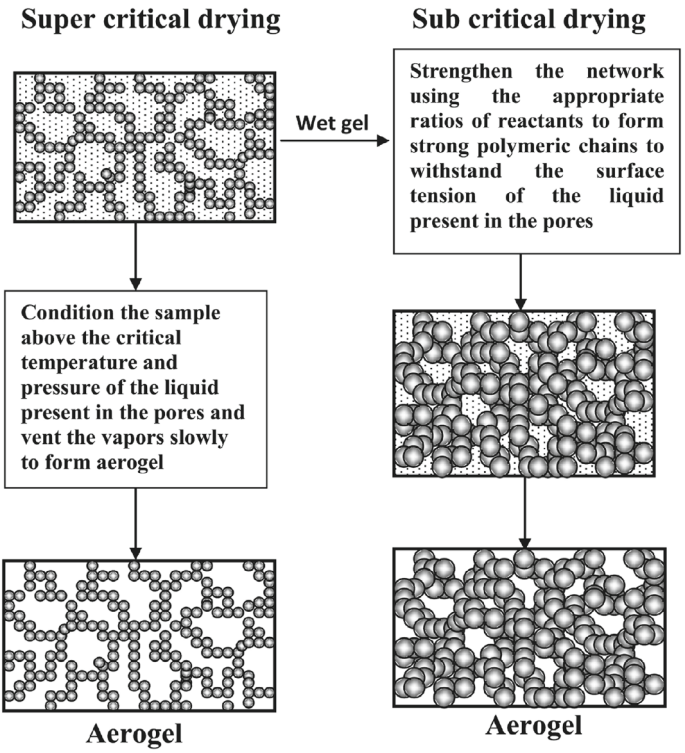


Fig. 10.10: Comparison of supercritical and sub critical drying.

cellulose-toluene 2, 4-diisocyanate etc. RF aerogels were first synthesized by R.W. Pekala in United States in 1989. Organic aerogels can be heat treated with some specific heat cycle in inert atmosphere to convert them into carbon aerogels. It is indeed possible to dope aerogels with various metals such as gold, silver, manganese and even organic dyes for variety of applications. Among these materials silica aerogels are very commonly synthesized in many laboratories as well as commercially. Aerogels can be treated by various chemicals to make them hydrophilic or hydrophobic.

Silica Aerogel

Silica aerogels are historically the first ones that scientists have synthesized. They have been explored a lot due to their immense applications in various fields. Tetraethylorthosilicate (TEOS) or Tetramethylorthosilicate (TMOS) are typically used as primary reactants with ethanol or methanol used as solvents. Acid (HCl, HNO_3) or base (NH_4OH, NH_4F) catalyst or combination of both is used for accelerating the reaction. Reaction consists of two steps viz. hydrolysis and polycondensation. For example, TEOS is first hydrolyzed using water. Two hydrolyzed molecules get

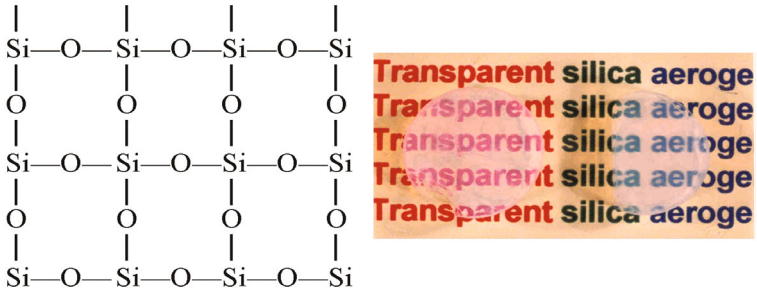


Fig. 10.11: Formation of 3D network of silica aerogel and a photograph showing transparent silica aerogel (diameter 2.5 cm and thickness 1.0 cm).

condensed to form a dimer. The condensation process continues to form polycondensed silica gel, with Si-O-Si linkage as shown in Fig. 10.11. These gels can be further dried using supercritical drying. For carrying out this type of drying, autoclave is used. Ethanol or methanol can be used as solvent. Critical temperature for ethanol is 249 °C and critical pressure is 78 bars. Depending upon reaction conditions aerogels can be transparent or opaque and density can typically vary between 0.01 and 0.8 gm/cc.

$$\mathrm{Si(OR)_4 + H_2O \longrightarrow Si(OH)_4 + 4C_2H_5OH} \tag{9.17}$$

$$\equiv \mathrm{Si - OR + HO\text{-}Si} \equiv \ \leftrightarrows \ \equiv \mathrm{Si - O - Si} \equiv + \mathrm{ROH} \tag{9.18}$$

$$\equiv \mathrm{Si - OH + HO\text{-}Si} \equiv \ \leftrightarrows \ \equiv \mathrm{Si - O - Si} \equiv + \mathrm{HOH} \tag{9.19}$$

RF Aerogels and Carbon Aerogels

Inorganic aerogels such as silica have been explored a lot but organic aerogels such as RF (Resorcinol Formaldehyde) have been relatively less investigated. These aerogels are interesting, particularly because of the possibility to convert them into carbon aerogel by pyrolysis (i.e. by heat treatment). Unlike silica aerogels they need high temperature (85 °C) for gelation. For synthesis of these aerogels, resorcinol ($C_6H_5(OH)_2$) and formaldehyde (CH_2O) are reacted in presence of sodium carbonate in aqueous medium. The reaction is shown in Fig. 10.12. The chemicals are mixed in a flask and stirred vigorously for three hours and then left to themselves for one day at room temperature. The solution is then kept at 85 °C for six days to form a gel. This gel can be dried supercritically or subcritically. To convert RF aerogel into carbon aerogel, it is necessary to heat RF aerogel under inert atmosphere at ~1100 °C. A photograph showing RF and carbon aerogels along with SEM image of RF aerogel showing porous structure are as in Fig. 10.12.

Aerogels, however, due to their weak network of nanoparticles (bonded by Van der Waals interaction) have inferior mechanical properties. Therefore they can be

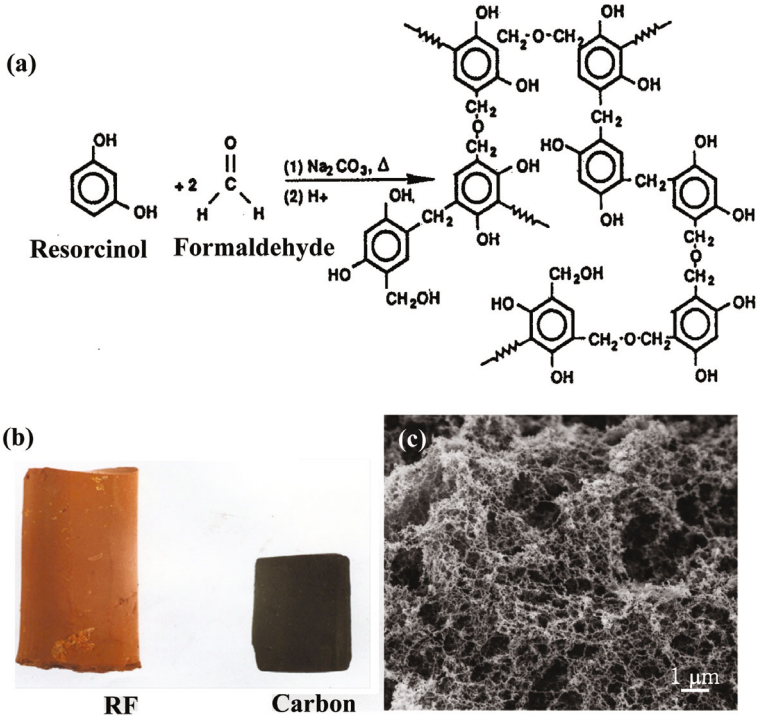


Fig. 10.12: (a) Formation of three dimensional network of RF aerogel; (b) photograph of RF and carbon aerogel and (c) SEM of RF aerogel.

easily broken with as small pressure as exerted by fingers. There are now attempts to increase the strength of aerogels by incorporating high strength nanomaterials like graphene or carbon nanotubes during gel formation process as strengths of graphene or carbon nanotubes are even superior than the steel of same dimension.

Properties of Aerogels

Aerogels possess many interesting properties because of their highly porous structure. They are the lightest materials (comparison is made with densities of other materials in Table 10.1) man has ever synthesized having pore diameters of few tens of nanometers with 2–5 nm size particles. It is possible to have very high porosity (80–90%), very low density and high surface area (500–1500 m^2/gm) for these aerogels. They have very low index of refraction (1–1.05) and very low speed of sound through them (~20 m/s). Young's modulus in these materials is quite low (10^6–10^7 N/m^2). Very low value of thermal conductivity (0.003 W/m.K) which is lower than most commonly used insulators make them best available thermal insulator. See Fig. 10.13.

Table 10.1: Densities in g/cc of some commonly known materials

Iron	*Diamond*	*Glass*	*Graphite*	C_{60}	*Paper*	*Charcoal*	*Aerogel*	*Air*
7.87	3.5	2.4	2.3	1.7	0.7	0.57	0.003 to 0.8	0.0013



Fig. 10.13: Photograph of silica aerogel as best thermal insulator.

Applications of Aerogels

For the decades since their first formation it was considered to be quite difficult to use aerogels commercially because of number of reasons like risky supercritical drying process, expensive synthesis procedure and their low fracture toughness etc. With the development of cost effective synthesis routes and technological developments and demands, interest in aerogels is increasing. They are nontoxic and biodegradable making them eco-friendly. Their unique properties such as very low thermal conductivity makes them best thermal insulators for various applications such as insulation of space vehicles, automobile engines etc. Aerogels have been commercially used in jackets and blankets to be used under extreme low temperature conditions. Large windows of aerogels are being used in houses and buildings in some places to control the temperature in the interior. The transparency and heat insulation properties are used. Indeed it is possible to make aerogel-textile composites so as to make blankets, coats etc. by making them rollable. They can be obtained in the form of thin films, molds and tiny balls. Various applications of aerogels depending on their properties are shown in Fig. 10.14.

10.2.3 Zeolites

We discussed in the previous two sections two different types of porous materials viz. porous silicon and porous aerogels. Both the materials have disordered pores

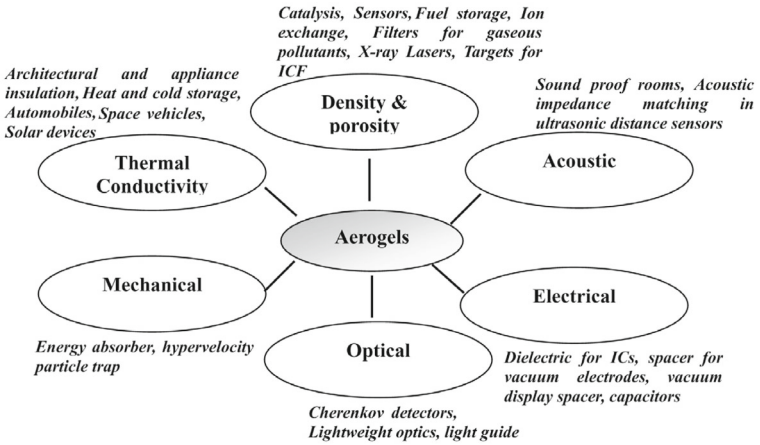


Fig. 10.14: Various applications of aerogels depending upon their properties.

of irregular shapes and sizes. There is another class of materials known as zeolites, which has pores usually smaller (<2 nm) than aerogels or porous silicon but the pores are highly ordered. The zeolites have not only the pores of uniform size but are periodically arranged to have long range order. The material is crystalline. Some zeolites occur naturally. However due to their technological importance as catalysts and highly sorbent material, they are also synthesized on large scale.

The word 'zeolites' origins from 'zeo' and 'lithos' meaning 'boil' and 'stone' respectively. Boiling away water from some materials makes them porous zeolites. They have a crystalline structure with translational long range order for their unit cells in three dimensions. Unit cell itself is often the result of some quite complicated subunits. Most common zeolites have the building blocks of tetrahedral units of Si and Al connected together by Si-O-Al types of bonds. There can be Si-O-Si bonds also in zeolites but no Al–O–Al bonds exist. This implies that the number of silicon atoms will be always equal or larger than Al. The network of zeolites can be viewed as shown in Fig. 10.15.

However, valence of silicon is 4 and that of aluminum is 3. This would mean that with Al substitution, there would be excess negative charge on the network, which needs to be balanced. This is done by adding some cations or electron acceptors in the material. A general formula of a zeolite can then be written as

$$\left[Al_xSi_yO_{2(x+y)}\right]M_xzH_2O \tag{10.8}$$

known as aluminosilicate. Besides this there can be some other tetrahedral units such as (GeO_4), (ZnO_4), (PO_4) etc. which can be used instead or in addition to Al. There are around 40 naturally occurring zeolites and much larger variety can be synthesized.

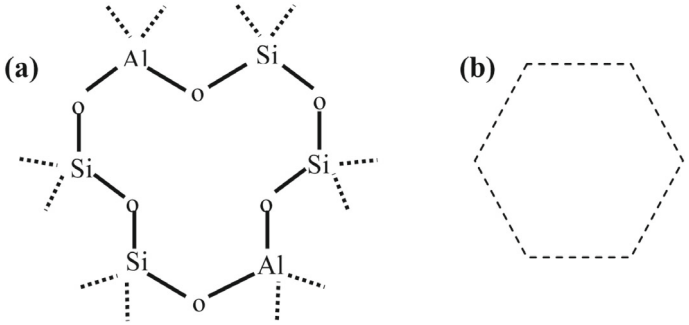


Fig. 10.15: (a) Al-Si–O subunit and (b) its schematic subunit.

Synthesis of Zeolites

Synthesis of zeolites, however, is quite a delicate balance of additives as well as processing conditions. Often some organic additives of some ammonium or amines have to be added in some suitable quantities. Both metal cations and organic additives are believed to act as templates for building various structures. Variation of Si/Al, Si/H_2O, cation/Si and organic additives can give rise to numerous zeolites exceeding the number of natural zeolites. Some of the chemical species are responsible for achieving particular structure in zeolites. Such chemical species play the role of so called molecular templates. However particular structure can be obtained using more than one organic molecule and one molecule can create different structures for different precursors forming zeolites. Silicon and aluminium precursor solutions are hydrolyzed and polycondensation takes place similar to that in sol–gel process. Organic and inorganic reactants are added and a gel thus formed is hydrolyzed in an autoclave at high pressure (several tens or hundreds of bars) and temperature (usually 100 to 350 °C).

Pore diameters can be controlled with a narrow size distribution in the size range of 0.3 nm to 2.0 nm. The pores can be connected in 1-D, 2-D or 3-D. Such porous materials find numerous applications. A widely used ZSM-5 zeolite structure is as shown in Fig. 10.16.

Properties of Zeolites

Zeolites are crystalline, highly ordered porous materials. The pore sizes can be precisely controlled. As illustrated in Fig. 10.17, there can be a large number of subunits forming zeolites of different constituents.

Pores in zeolites are useful to carry out different reactions. Zeolites are useful catalysts or sorption material. By controlling the pore sizes to desired size, it is possible to synthesize and organize nanoparticles inside zeolites.

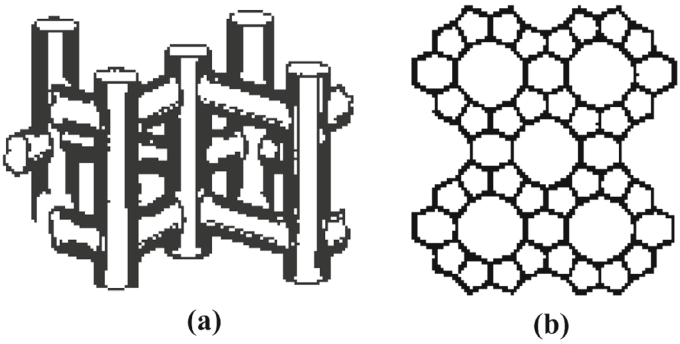


Fig. 10.16: Zeolite ZSM-5 (a) side view and (b) top view.

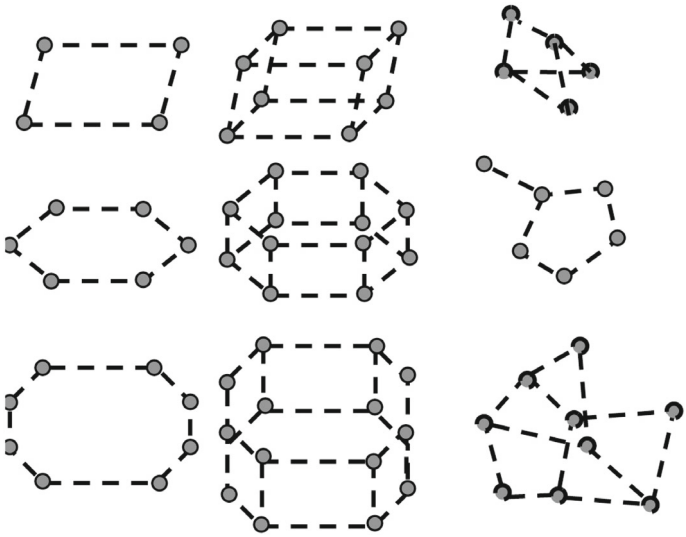


Fig. 10.17: Structural sub units in zeolites.

10.2.4 Porosity Through Templates

Porosity in nanomaterials can be obtained through various procedures. Here we shall discuss the method in which micelles are used as the templates. Another method involves using core–shell particles in which core is partially or completely removed and the shell becomes porous in the whole process. This part we shall discuss in the core–shell particles.

Micelles as Templates

This also is a good example of self assembled materials. Large (2–10 nm) and ordered pores in three dimensional solids can be obtained using micelles as the templates. However major difference between zeolites and M41S and its family as they are known, is that the pore walls are amorphous in this case. These are only synthetic materials with no naturally occurring material of this type. They were first synthesized in 1992 by some scientists in Mobil company. These are synthesized as 1-D ordered (MCM41), 2-D ordered lamellar (MCM-50) and 3-D ordered (MCM48) arrangement of pores.

Synthesis of Ordered Porous Materials Using Micelles

Very interesting, novel route of synthesis has been adopted to synthesize these materials. Amphiphilic organic chain molecules, water, inorganic silica precursor and some catalyst are used in making these materials. Reaction temperature can vary over a wide range from –14 °C to 180 °C depending upon the materials of interest. However the underlying idea is as follows. As discussed in Chapter 3, when amphiphilic molecules are mixed in water, depending upon the molecules above the critical concentration, one can spontaneously obtain micelles. With increasing concentration, one can even obtain different structures like hexagonal, cubic or lamellar at certain temperature. If silicates are condensed on preformed structures of micelles, it is possible to obtain ordered composites as shown in Fig. 10.18.

Alternatively, the formation of observed periodic pore structure can be viewed as follows. Before the self assembly of micelles takes place forming some ordered structures, each micelle may acquire a silicate shell around it and then form the ordered structure. The assembly always occurs irrespective of the stage at which the

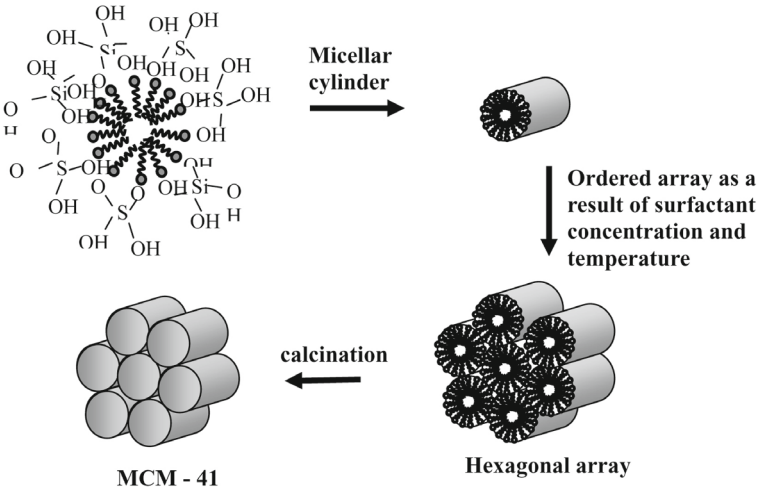


Fig. 10.18: Micelles covered with silica.

silicate precursor is added to the micellar solution. Once the reaction is over, micelles can be removed by plasma etching, calcination or solvent exchange processes.

Such ordered pore structures themselves are nanomaterials and can be also used as templates to synthesize nanoparticles in their cavities.

10.2.5 Metal Organic Frameworks (MOF)

Metal organic frameworks are under intense investigations from the point of view of finding new MOFs and their applications. MOF frameworks are co-ordination polymers in which metal ions make a one-dimensional, two-dimensional or three-dimensional pore network through organic molecules. They resemble zeolites to some extent but they are different from zeolites. Zeolites are mainly alumino silicates. The densities of zeolites are quite high 10.18 and can be ~50 g/cm^3 as against the densities of MOFs which can be ~0.5 g/cm^3 viz. two orders of magnitude smaller than zeolites. The pore sizes in MOFs can be as low as ~5 nm diameter. This results into extremely large internal surface area (~6000 m^2/g). In MOFs this large surface area is the most important parameter as this is important in deciding the choice of a material as the potential candidate for H_2 storage application or drug delivery. Some of the organic molecules which have been used in the synthesis of MOFs are oxalic acid (HOOC-COOH), malonic acid (HOOC-CH_2-COOH), succinic acid (HOOC-$(CH_2)_2$-COOH), gltanic acid (HOOC-$(CH_2)_3$-COOH), pthalic acid ($C_6H_4(COOH)_2$), citric acid (HOOC)CH_2C(OH)(COOH)CH_2(COOH)), trimeric acid ($C_9H_6O_6$), 1,2,3 tiazole ($C_2H_3N_3$) and squaric acid ($C_2H_2O_4$). Some of the metal ions used are Mg, Fe, Zn, Mn and Cu.

The synthesis of MOFs is carried out mostly by hydrothermal or solvothermal and mechanochemical techniques. Hydro or solvothermal reaction of precursor chemicals is carried out in an autoclave at high temperature and high pressure similar to that in the synthesis of zeolites. In mechanochemical synthesis metal acetates are mixed with organic molecules and ground in a ball mill.

The biodegradability of MOFs is excellent. Their functionality makes them useful in H_2 and other gases like CO_2, O_2 storage. Additionally MOFs can be useful in catalysis, drug delivery, data communication and replacement of liquid crystals which are under investigations.

10.3 CORE–SHELL PARTICLES

Core–shell particles form a novel class of nanocomposite materials in which a thin layer of nanometer size is coated on another material by some specialized procedure. The core can be just a nanoparticle (few nanometers to tens of nanometers) with a nanometer thick coating or it can be a large core (few tens to hundreds of nanometer diameter) with nanometer thick coating as schematically shown in Fig. 10.19. The

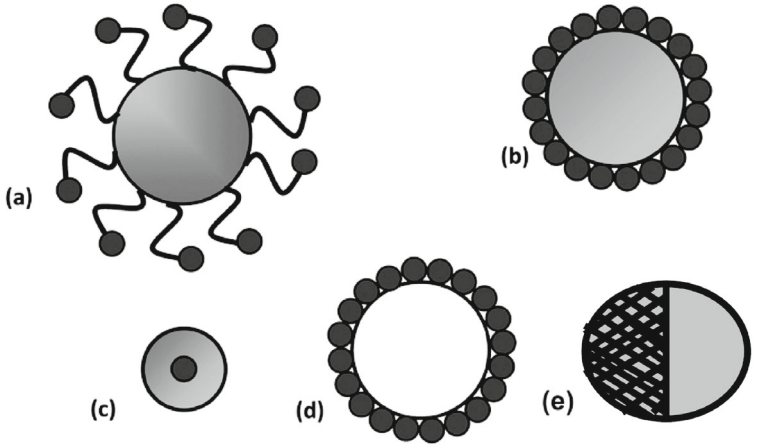


Fig. 10.19: Variety of spherical core–shell particles: (a) Surface modified core particles anchored with shell particles, (b) Smooth coating of dielectric core with shell, (c) Encapsulation of very small particles with dielectric material, (d) Quantum bubble and (e) Janus particle, half coated shell.

properties of core–shell particles are different from core or shell material. Their *properties depend usually upon core to shell ratio.* These particles are synthesized for a variety of purposes like providing chemical stability to colloids, enhancing luminescent properties, engineering band structures, sensors, drug delivery etc. These materials can be of economic interest also as precious materials can be deposited on inexpensive cores. Core particles of different morphologies such as rods, wires, tubes, rings, cubes etc. also can be coated with thin shell to get desired morphology in core shell structures.

Core–shell materials can be synthesized practically with all the materials, like semiconductors, metals and insulators. Dielectric materials such as silica and polystyrene are popular materials to use as core because they are soluble in water and hence can be useful in biological applications. Core–shell particles can be synthesized using variety of combinations such as dielectric-metal, dielectric-semiconductor, dielectric-dielectric, semiconductor–metal, metal–metal, semiconductor–semiconductor, semiconductor-dielectric, metal-dielectric, dye-dielectric, dielectric-biomolecules etc. Although core shell particles have novel properties, these can be further assembled and utilized for creation of another class of novel materials like colloidal crystal or quantum bubbles (i.e. hollow spheres with thin shells). It is indeed possible to create novel core shell structures having multishells and tuning optical properties from visible to infrared region of the electromagnetic spectrum.

Synthesis of core–shell particles requires highly controlled and sensitive synthesis protocols to ensure complete coverage of core particles with shell. There are various methods to fabricate core–shell structures which involve precipitation, polymerization, micro emulsion, reverse micelle sol–gel condensation etc. Although these methods themselves may appear to be simple, it is rather difficult to control the thickness and homogeneity of the coating. If reaction is not controlled properly,

eventually it leads to aggregation of core particles, formation of separate particles of shell material or incomplete coverage.

Here we shall discuss some silica based core–shell particles as an example. Preparation of core–shell particles is a multi-step synthesis procedure. One can make coating of silica on nanoparticles or grow silica particles of large size and then anchor nanoparticles or coat thin shell around, of few nm thickness. As mentioned earlier one can use metals, semiconductors or any other dielectric material as shell or core with silica.

Synthesis of Silica Cores

Silica is a very widely used material to form core–shell particles because of its stability against coagulation. It is also chemically inert and optically transparent. For various purposes it is desirable that particles remain well dispersed in the medium which can be achieved by coating silica on them to form an encapsulating shell.

Silica particles having very good monodispersity can be synthesized by what is known as Stöber method. This method involves hydrolysis and successive condensation of TEOS (tetraethylorthosilicate $Si(C_2H_5O)_4$) in alcoholic medium in presence of ammonium hydroxide (NH_4OH) as a catalyst. By varying relative concentration of TEOS to solvent (dilution) and amount of catalyst, one can synthesize these particles in various sizes ranging from 50 nm to 1 μm.

Synthesis of Metal, Semiconductor or Insulator Particles (<20 nm)

Synthesis of metal, semiconductor and insulators with very small size (<20 nm) and small size distribution can be performed using some of the chemical techniques described in Chapter 4 (as well as Chapter 15).

Core–shell Assemblies

Core–shell assembly is a multi-step process in which core and shell particles are separately synthesized and then shell particles are anchored on cores by specialized procedures or shell is directly synthesized on preformed cores.

In the first method, surface of core particles is often modified with a surfactant or bifunctional molecules to enhance the coverage of shell material on its surface. Surface of core particles such as silica can be modified using bifunctional organic molecules such as APS (3-aminopropyltriethoxysilane). APS molecule has an -OH group at one end while-NH group at the other end. APS forms a covalent bond with silica particle through -OH group and -NH group is available for interaction. Some nanoparticles having affinity for nitrogen can be attached with silica particles through NH group. There are several other modifiers such as APTMS (3-aminopropyltrimethoxysilane) and AEAPTMS (N-(2-aminoethyl)-3-aminopropyltrimethoxysilane) which produces surface terminated with amine group, MPTMS (3-mercaptopropyltrimethoxysilane) which produces surface terminated with thiols, DPPETES (2-(Diphenylphosphino)ethyltriethoxysilane) leaves the surface terminated by diphenylphosphine group and PTMS (propyltrimethoxysilane) gives surface terminated with methyl group. It is thus possible to make variety of

core–shell materials with dielectric cores such as silica, titania and polystyrene which can be anchored with different materials.

In another approach, synthesis of shell particles is carried out in presence of preformed cores (may be single nanoparticles or many nanoparticles). These core particles act as nuclei and shell material hydrolyses and gets condensed on cores forming core–shell particles. Reactant concentrations and amount of added core particles play important role in deciding shell thickness.

In other technique, which is known as layer by layer technique, alternate layers of anionic particles and cationic polymer are deposited on surface-modified template molecule by heterocoagulation.

Successive removal of core material (either by calcination or dissolution in suitable solvents such as toluene or HF) yields hollow particles consisting of shell material, popularly known as quantum bubble. This is known as template based synthesis of porous or hollow particles. For example SiO_2 nanoparticles are coated with SnO_2 thin film and etched mildly in controlled way using HF. Then with repeated deposition of SnO_2 and mild etching, core SiO_2 which mixes with SnO_2 layers is slowly etched out. This leaves SnO_2 particles with hollow core as well as porous SnO_2 shell. The process is known as galvanic replacement (of SiO_2 by SnO_2 in this case).

Selection of suitable pair for core and shell assemblies requires understanding of individual properties of core and shell materials. Core particle should withstand the process used for the coating of shell material. Core and shell particles should not interdiffuse and surface energies of core and shell particles must be similar so that probability of heterogeneous nucleation is more than homogeneous nucleation.

Properties of Core-shell Particles

Coating of colloidal particles with shell (in case of Janus particles, only half of the core is exposed and other half is covered with some inert material like wax permanently or till the coating of the desired material is carried out. The name Janus is that of a Greek god with two heads.) offers most simple and versatile way of modifying its surface chemical, reactive, optical, magnetic and catalytic properties. Silica particles coated with gold shell have been studied, which show changes in the Surface Plasmon Resonance (SPR) position of which depends upon core/shell optical properties. CdSe nanoparticles (<5 nm) coated with ZnO, CdS/ZnTe and CdTe nanoparticles coated with CdSe also have been studied for enhancement in luminescence of core nanoparticle. Use of higher band gap material as a shell for lower band gap material as a core increases the probability of photoexcited electron of core particle being trapped and increases the photoluminescence of core particle. Magnetic particles of iron oxide also have been coated with dye incorporated silica shell. Such particles show magnetic properties arising from core as well as luminescent optical properties arising from shell. Thus functional materials with novel properties can be synthesized by using various combinations of core shell materials.

10.4 METAMATERIALS

The literary meaning of 'metamaterials' is 'beyond Nature or Supernatural'. These materials have been fabricated artificially as a consequence of advances in the fabrication of nanomaterials in the desired shapes and sizes as well as improved understanding of materials, particularly their interaction with the electromagnetic waves. Metamaterials are also known as 'negative refractive index materials' or 'left handed materials'.

These materials are subwavelength periodic artificial structures capable of producing negative refraction. It has been realized after about year 2000 that having such materials would help making 'superlenses', band pass filters, novel photonic devices and even cloaking (invisibility of objects).

The concept of negative refractive index is not really new in physics. Lamb and Schuster had come across it in their theoretical work. However they had dismissed the idea thinking that it would not have any physical meaning or application. In 1967, Victor Veselago revisited the concept and showed that negative refractive index was possible provided dielectric constant ε and magnetic permeability μ were simultaneously negative. Dielectric constant ε is frequency dependent and becomes negative at certain frequencies (this was seen in the discussion on plasmonic materials, Sect. 8.4). However, permeability μ can be achieved negative with some artificial structures (usually some resonating circuits are required). Around 2002, several structures were theoretically suggested and experimentalists proved that indeed it is possible to obtain negative refraction, as shown in Fig. 10.20.

One such resonator arrangement section is illustrated in Fig. 10.21. Usually these are lithographically constructed by electron or ion beams in gold as gold does not get easily oxidized and has negative dielectric constant in the large frequency range.

With considerable research progress in this branch, it has been found that negative refractive index materials can be Double Negative (DNG) Refractive Index materials in which both dielectric constant ε and permeability μ are negative or only ε (epsilon) is negative (ENG) or only μ (mu) is negative (MNG).

Superlenses and cloaking also have been demonstrated. In superlenses the focussing can be achieved without the usual distortions due to curvature and other aberrations of the optical lenses. Cloaking has been demonstrated in the microwave region. In object kept inside the 'cloak' is invisible to microwaves as they are completely reflected. Based on this idea even 'plasmonic' cloaking has been proposed

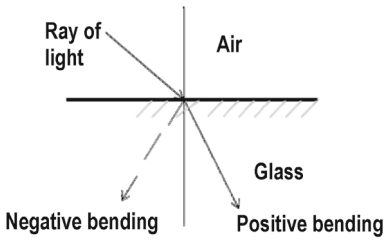


Fig. 10.20: Illustration of usual (positive) refraction of a ray and negative refraction of the ray when ray of light traverses from medium of low density into higher density medium.

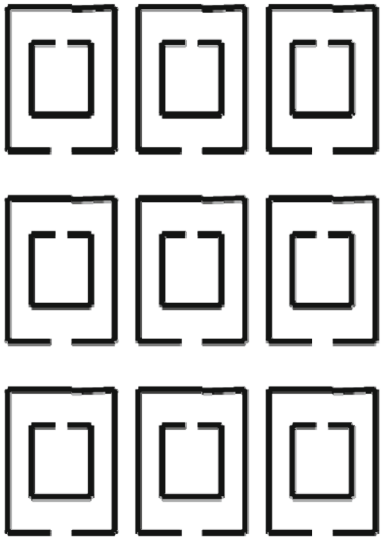


Fig. 10.21: Split resonators. Such structures are obtained in gold and form three dimensional arrays.

which would make the object invisible to particular wavelength but seen at other wavelengths.

10.5 QUANTUM MATERIALS

In Chapter 1 we have briefly discussed the scenario in science Pre-Quantum, Old Quantum and Modern Quantum era. In Box 1.1 milestones in these era were shown. From the contents of the Chapters 1 and 2 it is clear that even simple classification of macroscopic solids as metals, semiconductors and insulators is manifestation of Quantum Theory. The quantized electron states (energy) is the basis of Quantum theory as applied to solids. The use of Kronig-Penny model using Bloch wave function for obtaining electron energy states in a periodic lattice along with the use of Fermi–Dirac statistical distribution of electrons also correctly explains many special properties of materials like Hall effect, Field emission, Tunneling of electrons, Magnetism, and so on. These effects are abundantly used in various technologies like nanoelectronics, optoelectronics and many other currently used application areas. They are integral part directly or indirectly in today's science and technology. However, there remain some questions unanswered. For example, Fermi–Dirac distribution shows that at any given temperature the electrons only in the vicinity of Fermi level contribute to the transport properties in solids. What is the role of remaining electrons?

When one says 'Quantum Materials' then additional meaning is imperative. The term does not refer to individual charges in solids, dictated by Fermi–Dirac distribution and consequent properties mentioned above. The main features of Quantum

Materials are embedded in topology and entanglement of non-local particles. The prerequisite quantum confinement in one or more directions entails that the Quantum Material properties are not described by individual electrons but by their collective behaviour. The collective excitations are quantized and are known as 'Quasiparticles'. Lattice, charge, spin and orbit are interlinked at microscopic level. This results into unique electronic states and properties. Such materials with exotic properties are known as Quantum Materials.

Some examples of Quantum Materials are graphene as well as 2-D materials, superconductors, topological insulators, semimetals, spin ice, quantum spin liquids and plasmonic materials.

Although nanomaterials due to their quantum confinement attribute a major part of Quantum Materials, one cannot say that all Quantum Materials are nanomaterials.

10.6 BIOINSPIRED MATERIALS

We had discussed in Chapter 5 that the nature has mostly taken the 'bottom up' approach in building the animal and plant kingdom. Once the nanostructures are formed they may assemble (self assembly) into hiearchial structures to give functionalities. The scientists after getting the powerful microscopes at hand have gone into the details and found that many interesting phenomenon like cleanliness of lotus leaves, crawling of lizards on walls without falling or beautiful colours of peacock feather or butterflies or fish, birds are the results of 'Nature's Nanotechnology'. In this section we will try to understand some of the effects and the reasons behind them.

10.6.1 Lotus Effect (Self Cleaning)

In recent times there has been new understanding about how the hydrophobic (water hating) and hydrophilic (water loving) surfaces work. This effect has been there for millions of years and is now recognized as the 'lotus effect'. Lotus is a beautiful flower known to all civilizations and is being praised over generations in many cultures, religions and languages. How it appears beautiful and its leaves do not become dirty even by staying in muddy water without letting water stick to it has been a point of wonder. However, it has been investigated and found that this is due to the hierarchical micro-nano structure of the lotus leaf and many other leaves as well as body of animals.

A lotus leaf when observed under the microscope exhibits some bumps of micrometre size. The bumps are decorated with nanometer sized structure. This can help a water drop stay on the lotus leaf without spreading on it. The waxy substance on the lotus leaf also helps the water drop to form the droplet and not spread on the

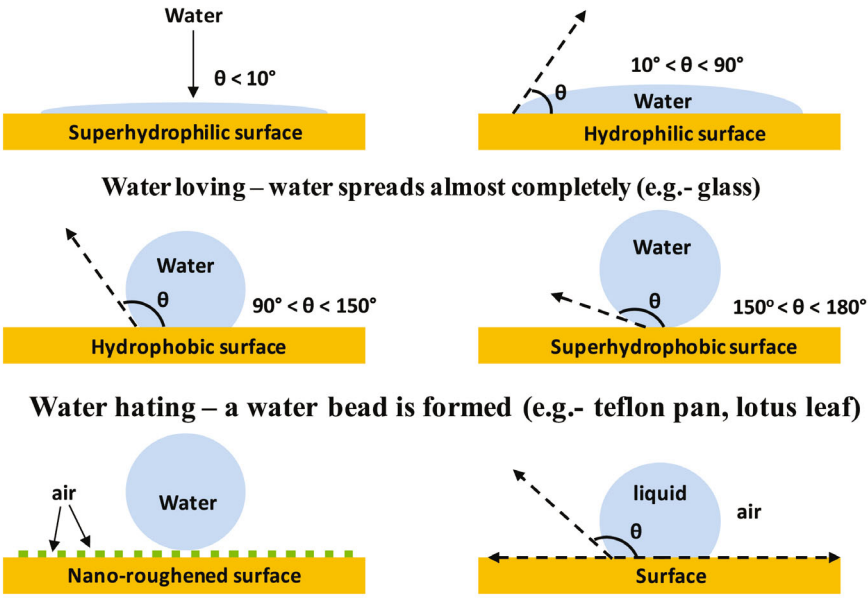


Fig. 10.22: Formation of water drop on the surface.

surface. When the water drop rolls on the leaf, it collects the dirt on the surface and makes the surface clean. Lotus effect can be understood from Fig. 10.22.

The phenomenon of wettability in lotus leaf is also common in many other plant leaves, butterfly wings, skin of fish and many others in plant and animal kingdom. It is understood in terms of hydrophibicity/hydrophilicity and the contact angle a water drop makes with the surface. A hydrophilic surface is water loving and hydrophobic surface repels water. Depending upon the contact angle of water drop, one can quantify the amount of hydrophilicity or hydrophobicity. As illustrated in Fig. 10.22, water contact angle is defined as the angle between solid surface and the liquid-surface. Contact angle depends upon surface roughness, surface condition and surface material. Interaction between liquid and solid surface is very important. When the contact angle is less than 90° the surface is known as the hydrophilic and if more than 90° it would be hydrophobic. If the angle is above ~150° then it is superhydrophobic and if less than ~20° then it is superhydrophilic. The angles which distinguish between superhydrophilic and hydrophilic or hydrophobic and superhydrophobic are not very precise and could vary within 5–10 degrees.

Scientists have tried now to make use of this phenomenon to artificially obtain the hydrophilic/hydrophobic surfaces in various materials of various shapes and sizes. One can even make coatings which would not only make a surface just hydrophobic or hydrophilic but also can change the nature in the controlled way using external stimuli such as application of an electric field or irradiation with UV light or X-rays.

Another interesting and more dramatic natural phenomena, which often goes unnoticed is known as 'petal effect'. It can be easily seen in common flowers e.g. a rose. If water is sprinkled on flowers, very tiny droplets can be observed. These droplets cling to the petals so strongly that even if the flower is turned upside down, the droplets do not fall under gravity. This effect is due to micropapillae of ~16 μm diameter and ~7 μm height with micropapillae folds ~730 nm size.

10.6.2 Gecko Effect (Adhesive Materials)

Somewhat similar interesting phenomenon observed in nature over centuries and wondered about is so-called 'Geko (family name for lizard) effect' or crawling of a lizard on wall. How does a lizard balance its weight? Does it secrete any fluid to stick or what else? Now the scientists believe that the peculiar construction of the tips of its feet on which millions of nano hair exist which help lizard in crawling. Interestingly the force that each hair strand exerts between wall and itself is just weak Van der Waals force. However, due to millions of hairs total force is sufficient to balance its weight and remove the force in controlled way to crawl. This effect may be useful to make robots which would climb the walls or do some scientific operations without manual aid. Some simple application of geko effect would be a stick tape which can be used number of times without falling like a lizard. One may also be perhaps able to make seals without nuts and bolts.

Basically understanding even natural phenomenon around us can lead to new discoveries and progress in science and technology.

FURTHER READINGS

Biomimetic Superhydrophobic Materials Construct from Binary Structure by Ji. Li, Z. Guo and W. Liu. *Adv. Mater. and Interfaces*, **10:** 2201847 (2023).

Synthesis of Inorganic Materials by Ulrich Schubert and Nicola Hüsing. Wiley-VCH. 2nd Edition (2005).

Reversing Light with Negative Refraction. *Physics Today*, **37** (June 2004).

Nanostructured Materials, Micelles and Colloids by L.M. Liz-Marzan, M.A. Correa-Duarte, I. Pastoriza-Santos, P. Mulvaney, T. Ung, M. Giersig and N.A. Kotov. *In:* Hand Book of Surfaces and Interfaces of Materials edited by H.S. Nalwa, Vol. 3. Academic Press (2001).

F. Caruso, *Adv. Mater*., **13:** 11 (2001).

Y. Xia, B. Gates, Y. Yin and Y. Lu. *Adv. Mater.*, **12:** 693 (2000).

Porous Silicon: From Luminescence to LEDs by Ruben T. Collins, Philippe M. Faushet and Michael A Tischler. *Physics Today*, 24 (1997).

Chapter 11
Nanolithography

11.1 INTRODUCTION

We have seen in chapters three to five a variety of methods to make nanoparticles (including spherical and other particle shapes) and thin films (or multilayers). These methods are popularly known as *bottom up* approach. In bottom up approach atoms and molecules are assembled so as to have nanomaterials of required size and shape by controlled deposition or reaction parameters. In the *top down* approach reverse is the case. Atoms and molecules are removed from a bulk material or sometimes thin films so as to obtain desired nanostructure.

Conventional *lithography* is a top down approach. The word lithography has its origin in the Greek word 'litho' which means stone. Lithography, therefore, literally means carving a stone or writing on a stone. It is used now to mean a process in which a sample is patterned by removing some part of it or sometimes even organizing some material on a suitable substrate. Lithography is very intensively used in electronics industry so as to obtain integrated circuits (IC) or very large scale integration (VLSI) on small piece of semiconductor substrate often called a 'chip'.

Immediately after the discovery of solid state transistor by Bardeen, Brattin and Schockley in 1947, the quest for making smaller and smaller electronic components began. First transistor was fabricated in germanium but soon it was found that silicon was a better material and was used commercially. Texas Instruments, U.S.A. marketed first transistor. Scientists found that speed of a device, system or an instrument finally depends on how fast a transistor can switch on and off. Speed of a computer also depends upon transistors. All electronic devices including cell phones, ATMs, etc. need fast transistors. It was further realized that smaller the device (or transistor), faster is the switching. Advantages of making smaller devices are manifold. They require smaller amount of material, space and consume less power for their performance making the resulting product cheaper. Initially the solid state transistors were assembled together along with different components like capacitors and resistors and wired to fabricate desired circuits.